

Mod 5: Hereditary

5.1 Reproduction		
IQ: How does reproduction ensure the continuity of a species?		
Syllabus Outcomes	Summary of information	
5.1.1 explains the mechanisms of reproduction that ensure the continuity of a species, by analysing sexual and asexual methods of reproduction in a variety of organisms, including but not limited to 1. animals: advantages of external and internal fertilisation 2. plants: asexual and sexual reproduction 3. fungi: budding, spores 4. bacteria: binary fission (ACSBL075) 5. protists: binary fission, budding	Term	Definition
	Asexual reproduction	Requires only one organism to reproduce and form genetically identical offspring
	Sexual reproduction	Requires the gametes of two organisms, to form a genetically varied offspring
	Continuity of a species	In the process of cell division, the continuity of a species is dependent on the fact that the daughter cells should have the same type and number of genes as the parent cell, despite having genetic variation.
	External reproduction	is Sexual reproduction in which the gametes combine and fertilisation occurs outside the female’s body
	Internal reproduction	is Sexual reproduction, in which the gametes combine and fertilisation occurs inside the female’s body
	Budding	Type of asexual reproduction in which the fungus can grow a genetically identical organism of itself, which can then separate and form another organism
	Binary fission	Type of asexual reproduction, in which a single cell organism splits itself into two, forming new organisms
	Bacteria	Single-celled prokaryotic organism
	Fungi	Eukaryotic cells that are classed separately from plants and animals, which can be unicellular or multicellular
★ Sexual Reproduction in Animals ➤ Requires the joining of the haploid gametes from a male and a female; these gametes are produced inside of the gonads (ovaries and testes). When these combine, it produces a diploid zygote that will develop into a fully formed organism, with half of each parent's genetic information, which guarantees genetic variation. ○ Advantages: ■ Increased diversity of species ● Meiosis: crossing over and independent segregation ● Fertilisation: the random joining of gametes ○ Increases variation within a species allowing them to survive the changing selective pressures ○ Disadvantages: ■ More time consuming ■ Takes more energy (gamete formation, finding a mate, mating) ➤ External Reproduction: male and female gametes combine and fertilisation occurs outside of the body. (more likely to be seen and take place in aquatic environments) ○ Broadcast spawning: multiple males and females release their gametes in a small space which results in higher genetic variety ■ Leads to a higher chance of survival of a species		

Advantages	Disadvantages
- Requires no parental care - Female is able to continue to reproduce without the need to wait for their young to develop - Greater genetic diversity as the gametes from one organism can fertilise the gametes from multiple other organisms from the same species - High frequency of breeding - Higher rate of fertilisation	- There is no guarantee that fertilisation of the gametes will occur - High energy expenditure (loss of gametes) - Must take place in a wet environment - Or else there is a high chance of desiccation of the gametes (they will dry out) - No parental care = high exposure to predation and infection

- **Internal reproduction:** gametes combine and fertilisation occurs within the female's body. (occurs in land animals/humans as the eggs need a moist environment to develop.)
- This takes more energy and more time to develop the egg, and there is a higher chance of survival as the offspring is a protected mother

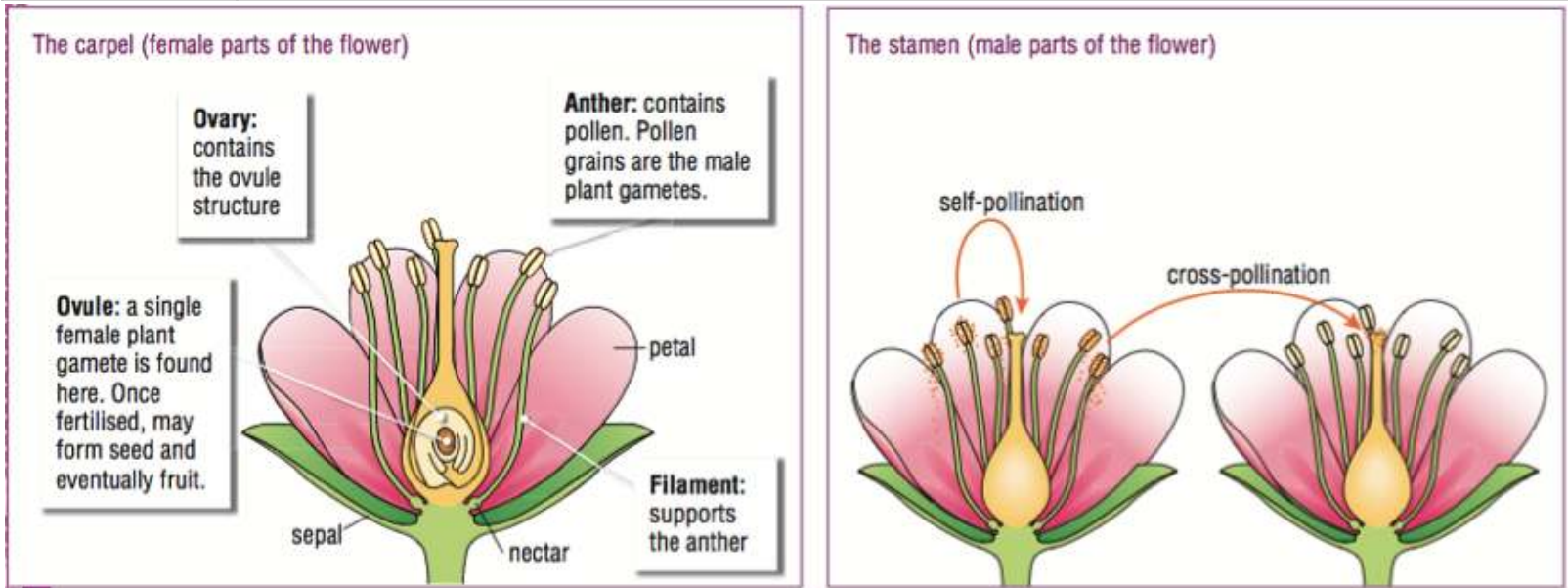
Advantages	Disadvantages
- Higher chance of survival of the offspring - Higher chance of fertilisation as the gametes are in a closed vicinity - Parental care allows a higher chance of survival of offspring. - Zygote is protected until birth. - Wider selection of mates	- Fewer offspring - Organisms require specialised reproductive structures (penis etc.). - Energy needed for mating and giving birth. - Parent care is lengthy. - They are constantly reproducing

★ Sexual Reproduction

- Reproduction in plants**
 - A flower/plant is able to reproduce through the male and female parts:
 - Carpel (female): made up of the ovary at the base of the flower, with the style extending from the ovary to the stigma at the top
 - The ovary holds the ovules, which hold female gametes of the plants
 - Stamen (male): made up of the filament and the anther, which hold the male gametes (pollen)
- Pollination**
 - Pollen (the male gamete) is produced in the stamen, and help up at the anther
 - The pollen is removed from the anther by the wind or by insects. It is transferred to the pistil of the same flower (self-pollination or another flower (cross-pollination)
 - Same or different
- Fertilisation**
 - The pollen falls on the stigma, and a pollen tube grows through the still of the ovary
 - Pollen migrates through pollen tubes, to the ovules
 - The pollen fertilises the ovules
- Development**
 - The fertilised ovule develops into a seed, containing the embryo
 - The seed protects the embryo
 - The fruit around it provides it with nutrients
- Germination**
 - When the fruit is removed, the seed within it grows separate from the parent plant(s)
 - The embryo lies dormant
 - If on suitable soil, germination begins
 - The embryo puts out its:
 - Radical (first root) for absorption of water and nutrients
 - Plumule (young stem) which develops leaves for photosynthesis

- The seed germinates into a seedling

Parts of the plant	Function
Petal	They surround the stigma and stamen, attracts pollinators
Stamen	Male reproductive part made up of the anther and filament, produces and distributes pollen
Filament	Long thin tubes/arm, holds up the anther and supports it
Anther	Male reproductive, produces the pollen grains
Carpel	Female reproductive part, ensure protection and efficient fertilisation
Stigma	The receptive part of the carpel, pollen grains will germinate only of the land here
Style	The stick up to the stigma, has a pollen tube; allows the male gamete to travel down into the ovary
Ovary	Holds the ovule structure, inside the ovules are the female gametes
Ovule	A single female plant gamete is found here, holds the eggs, once fertilised may form seed and eventually fruit
Receptacle	The swollen base of the flower, sometimes it forms the succulent tissue of the fruit.



★ Asexual Reproduction

- **Reproduction of Fungi**
 - Fungus: eukaryotic cells that are classed separately from plants and animals, which can be unicellular or multicellular. They are heterotrophs (cannot make their own food) and can reproduce both sexually and asexually)
 - **Budding:** an organism, that grows another organism on the surface that is genetically identical until it is big enough to break away and be a separate organism
 - A bud forms on the fungal cell; the bud cells then undergo mitosis in which the parent cells make a complete copy of the parent DNA
 - This is mostly a form used by yeasts
 - **Spores:** a fungal spore is a haploid cell that uses mitosis to replicate, without the need for a second haploid cell. They are genetically identical to the parent cells and are very light to enable them to travel in the wind, in order to germinate and disperse easier.
- **Reproduction of Bacteria**
 - **Binary Fission:** the splitting of a single parent cell into two genetically identical daughter cells.

	● Reproduction of Protists ○ Binary Fission: the splitting of a single parent cell into two genetically identical daughter cells. ○ Budding: an organism, that grows another organism on the surface that is genetically identical until it is big enough to break away and be a separate organism												
5.1.2 analyse the features of fertilisation, implantation and hormonal control of pregnancy and birth in mammals	★ Sex hormones → Chemical messenger that coordinates bodily actions ● These hormones control the development of the reproductive system. Produced in the ○ Ovaries, testes and pituitary glands (located at the base of the brain, and regulates the release of hormones for growth, metabolism and reproduction) ● They are responsible for ○ Development of reproductive organs ○ The development of secondary sexual characteristics <table><thead><tr><th>Males</th><th>Females</th></tr></thead><tbody><tr><td> ● Deepened voice ● Thicker body hair and more of it ● Increased size & thickness of muscles and bones </td><td> ● Larger breasts ● Widening hips ● Growth of pubic hair </td></tr></tbody></table> Types of Hormones <table><tbody><tr><td>Androgens</td><td> - Male hormones that regulate male sex organs. Testosterone is released for spermatogenesis - Responsible for the development of sperm and secondary sexual characteristics in males - Produced by the testis </td></tr><tr><td>Oestrogen</td><td> - The main female hormone but is also present in males - Is involved with the development of the female reproductive system - Stimulates the development of uterine smooth muscle layers ready for birth; this counter effect of the progesterone block that suppresses uterine contractions late in pregnancy. </td></tr><tr><td>Progesterone</td><td> - Female hormone - Ensures that the thickness of the endometrium is maintained during pregnancy - Without it, the endometrium would not be maintained or support the implantation of the embryo </td></tr><tr><td>Oxytocin</td><td> - Responsible for uterine contractions, the release of milk and maternal and pair bonding </td></tr></tbody></table> ★ Menstruation + Ovulation → Process of maturing an ovum, and preparing the uterus for fertilisation, but if that doesn't occur, then the lining sheds Process 1. Follicular phase (before the release of an egg) ● Oestrogen and progesterone are low, so lining breaks down and sheds ● Follicle stimulating hormone (FSH) increases, which stimulates the development of multiple follicles in the ovaries, of which each follicle contains an egg ● As FSH decreases, only one follicle develops to maturity ● The developing follicle releases oestrogen 2. Ovulatory phase (release of an egg) ● High levels of luteinizing hormone (LH) and FSH ● LH stimulates the release of the egg Complete Menstrual Cycle	Males	Females	● Deepened voice ● Thicker body hair and more of it ● Increased size & thickness of muscles and bones	● Larger breasts ● Widening hips ● Growth of pubic hair	Androgens	- Male hormones that regulate male sex organs. Testosterone is released for spermatogenesis - Responsible for the development of sperm and secondary sexual characteristics in males - Produced by the testis	Oestrogen	- The main female hormone but is also present in males - Is involved with the development of the female reproductive system - Stimulates the development of uterine smooth muscle layers ready for birth; this counter effect of the progesterone block that suppresses uterine contractions late in pregnancy.	Progesterone	- Female hormone - Ensures that the thickness of the endometrium is maintained during pregnancy - Without it, the endometrium would not be maintained or support the implantation of the embryo	Oxytocin	- Responsible for uterine contractions, the release of milk and maternal and pair bonding
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- Oestrogen levels to decrease and progesterone levels increase
3. **Luteal phase (after the release of an egg)**
- FSH and LH levels decrease
 - Ruptured follicle forms the corpus luteum, which produces progesterone, and the oestrogen levels rise again
 - Progesterone and oestrogen cause the lining of the uterus to thicken more, to prepare for possible fertilisation
 - If the eggs are not fertilised, the corpus lithium degenerates an no longer produces progesterone, the oestrogen level decreases
 - If the eggs are fertilised, the corpus luteum continues to function during early pregnancy. It helps maintain the pregnancy.

Hormones	Gonadotropin-releasing hormone (GnRH)	Regulates the production and release of FSH and LH
	Follicle stimulating hormone (FSH)	Stimulates the development of multiple follicles
	Luteinizing hormone (LH)	Stimulates the release of an egg from follicle
	Oestrogen	Made from developing follicle, and it thickens the lining
	Progesterone	Made from corpus luteum and thickens lining

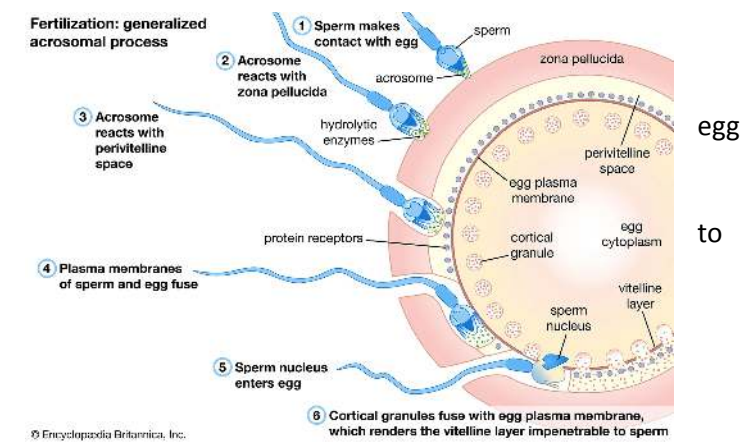
★ Fertilisation → **Gamete fusion and the production of a zygote**

- Process**
1. Sperm are attracted to the egg by chemical signals
 2. Sperm pushes through the follicle layer around the egg
 3. The first sperm to break through binds to the zona pellucida
 4. Acrosome reaction triggered
 5. Acrosome bursts and digests the zona pellucida, allowing the sperm to enter the
 6. Cell membranes of the sperm and egg fuse
 7. Cortical reaction triggered
 8. Enzymes from the cortical granule are released, which harden the zona pellucida stop more sperm from entering
 9. The two nuclei of the egg and the sperm undergo mitosis
 10. The entire cell divides, moving ½ of the egg's chromosomes and ½ of the sperm's chromosomes into separate cells
 11. Zygote-created

Hormones	Progesterone	Thickens lining, and allows maturity of sperm, for fusion
	Follicle stimulating, hormone (FSH)	Helps develop the sperm and the egg for fertilisation
	Luteinizing hormone (LH)	Helps release the sperm and the egg for fertilisation
	Testosterone	Spermatogenesis

★ Implantation → **the embryo is embedded into the uterus**

- Process**
- Oestrogen and progesterone, produced by the ovaries, prepares the uterus for the implantation of a fertilised egg. Progesterone then prevents

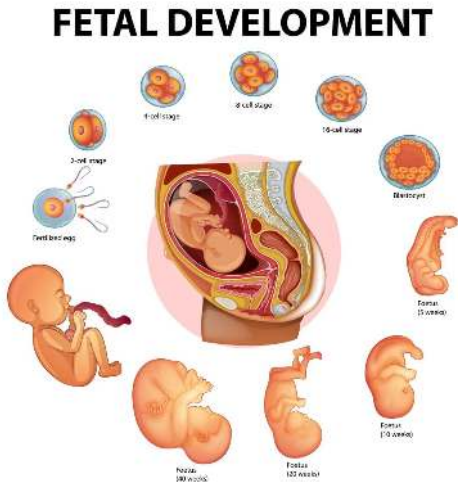


- uterine activity, as it inhibits muscular contractions and suppresses the immune response from attacking the embryo.
- The placenta
 - The foetus develops and the umbilical cord and placenta
 - The placenta is a disk-shaped tissue that gets embedded into the uterus wall by the placental villi. It connects the blood systems of the mother and the foetus without them fusing.

Hormones	Progesterone	Suppresses uterine activity
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★Pregnancy → Development of offspring from implantation to birth

Process	Corpus luteum in the ovary secretes hormones for the first three months of pregnancy. Once it begins to degenerate, the placenta develops and takes over the role of producing hormones to maintain pregnancy
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Hormones	Oestrogen	Promotes the growth of endometrium
	Progesterone	Stimulates the secretion of mucus by cells lining endometrium and the growth of blood vessels. Reduces the mother's immune response to foetal antigens. This is essential in maintaining pregnancy until the placenta takes over at 7-9 gestational weeks/

★Birth → pushing out of the body, via a birth canal

Process	• Softening the cervix tissue, to allow the baby to come out, and the contractions to push the baby and the placenta out.
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Hormones	Prostaglandins	Softening cervix tissue
	Oxytocin	Contractions of muscles
	Prolactin	Enlargement of mammary glands of milk production

5.1.3 evaluate the impact of scientific knowledge on the manipulation of plant and animal reproduction in agriculture

- ★Selective breeding (SB)
 → *process used to produce animals and plants with more useful or more attractive characteristics. Humans decide which individuals may breed and leave offspring to the next generation.*
- Breeding can either be:
 - Intraspecific (within a species)
 - Interspecific (between different species)
 - **SB in plants and animals**
 - Mostly done to produce higher-quality food (meat, grains etc)
 - Seeds are collected from the individuals with the largest, most attractive or more numerous grains, fruits etc.
 - Used to produce desirable traits so that at least one of these traits appears in the offspring

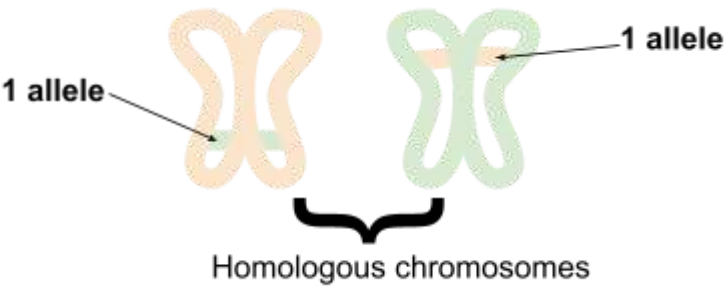
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★Artificial Semination (animals) → <i>the collection, storage and transport of semen from male animals with desirable characteristics so that more female can be impregnated and larger numbers of offspring with the desirable characteristics produced.</i>					
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★Artificial pollination → <i>choose which pollen is used to fertilise which plants to produce offspring with desired characteristics</i> • 2 methods ◦ Fertile stigmas are dusted with pollen from favourable male plants ◦ Pollinated flower is covered to prevent pollination from others					
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★Cloning → <i>to produce genetically identical offspring</i> • Plant ◦ Producing genetically identical offspring by different methods to maintain desirable qualities ◦ <u>Tissue culturing</u>: pulverised tissue is regrown to produce new growth ◦ <u>Grafting</u>: cuttings of plants are replanted to produce more plants • Animal ◦ Producing genetically identical offspring, by using an egg of one individual and adding it into a somatic cytoplasm. It is stimulated to divide to produce an identical organism to the original mammal.					
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	★Impact of reproductive technologies - Though there is increased variation due to these technologies/methods, as the species with desirable traits are then “cloned” again there isa reduction of variation within the species - There is no evolution, no variation within the gene pool - Limits the adaptability of species to changing selection pressures → limiting the continuity of species.																														
5.2 Cell replication IQ: How important is it for genetic material to be replicated exactly?																															
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Interphase	- Chromosomes can not be seen individually, they are called chromatin material - They are thin, long and coiled together - DNA replication takes place
Prophase	- Individual Chromosomes start to become visible - They start to condense and thicken - The nuclear membrane starts to break down - Spindle fibres start to form
Metaphase	- The chromosomes start to line up in the middle of the cell (Equator) - The spindle fibres attach to each of the centromeres
Anaphase	- The spindle fibres start to pull the sister chromatids to opposite poles of the cell - They are now termed daughter chromosomes
Telophase	- The daughter chromatids are now on opposite sides of the cell - The nuclear membrane and the nucleolus start to reappear
Cytokinesis	- The division of the cytoplasm, separating each of the daughter nuclei so that now is in its own cell - There are now two identical diploid daughter cells

★ **Meiosis** → DNA replication = identical DNA produced = cell replication = Inherited, passing genome, recombination of gene, genetic variation, continuity of species

- Contributes to genetic variation
- Occurs in the testes and ovaries (respectively)
 - Sperm and egg cells
- Humans have 46 chromosomes altogether
 - 23 comes from the sperm and 23 comes from the egg
- Result: Diploid parent cells turn into 4 haploids (n) daughter cells



Crossing Over	The inner chromatids of each homologous pair exchange genetic information
Independent Assortment	Homologous pairs will line up independently of every other pair - There is an equal chance each homologous pair will end up on either pole of the cell - The first pair that lines up does not dictate how the rest of the pairs will line up.
Random Segregation	Within each homologous pair there are two alleles, alleles are separated from each other - The two alleles will never end up in the same gamete - Each gamete has half the amount of chromosomes, but there are various ways they can be split.
Random Fertilisation	It is a random fusion of the sperm and the egg, adding more variation

○ **Process (occurs in two stages)**

Meiosis 1	Meiosis 2
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Interphase I • DNA replication occurs • Sister Chromatids are made	Prophase II • Chromosomes condense
Prophase I • The chromosomes condense and thickens • Chromosomes pair up in homologous pairs ◦ They are the same size and have the same genes in the same/similar location • Crossing over occurs between homologous pair	Metaphase II • Line up along the centre
Metaphase I • Homologous pairs line up • Independent segregation and Randoms assortment occurs	Anaphase II • Sister chromatids are split by the spindle microtubules towards the opposite sides
Anaphase I • Homologous pairs are separated, pulled to opposite poles, by the spindle fibres	Telophase II • The two cells are now being divided in half leaving us with 4 new cells
Telophase I • Chromosomes arrive at opposite ends of the cell • Two diploid daughter cells are made	Cytokinesis II • Splits in 4 new cells with their own nuclear membranes

★ **Watson and Crick**

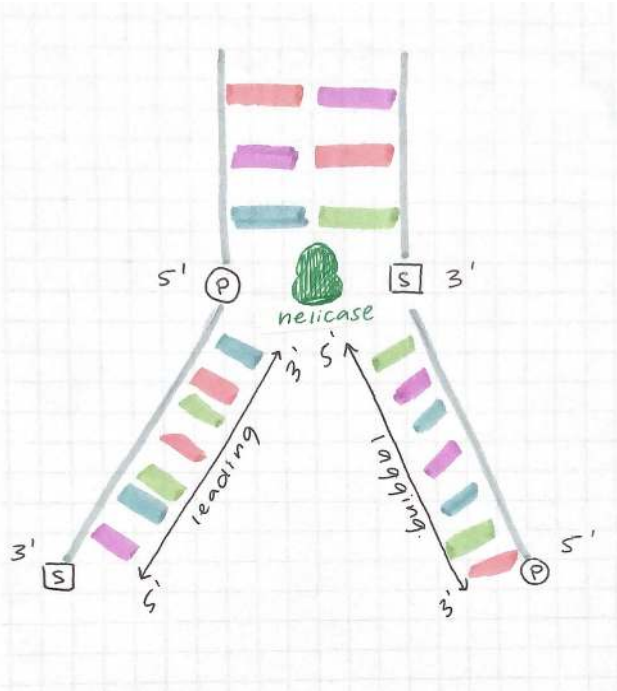
- Despite Rosalind Franklin being the first to crystallize and view the helical structure of DNA using X-ray technology, this image was given to Watson and Crick without Franklin knowing. This was the basis for the model that Watson and Crick developed, as they decided that the structure of DNA followed a double helix structure. After Franklin died from cancer, Watson and Crick received the Nobel Prize for the structure of DNA.

★ **DNA (Deoxyribonucleic Acid) structure**

- DNA is chemical code that is used for the production of proteins that express genetic information
- Made of nitrogenous base pairs and a sugar-phosphate backbone (nucleotides)
- **Nucleotides and Base pairs**
 - The basic unit that makes up DNA is the **nucleotide**, which is made up of a **phosphate molecule, covalently bonded with a deoxyribose sugar, which is bonded covalently to a nitrogenous base**.
 - Nitrogenous bases have a complementary base pair (A-T and G-C) and these pairs only work due to the hydrogen bonds that connect them together.
 - The adenine and the thymine base can **only make two hydrogen bonds**, while the **cytosine and the guanine can form three hydrogen bonds**
- **Double Helix Structures**
 - DNA is essentially a twisted ladder the base pairs making up the rungs and the sugar-phosphate backbone making the poles on either side.
 - DNA has 2 complementary strands either going $3' \rightarrow 5'$ or $5' \rightarrow 3'$
 - 3' ends in sugar
 - 5' ends in phosphate
 - This makes the DNA antiparallel

★ **DNA Replication**

- **The original strand of DNA:**
 - DNA has 2 complementary strands either going $3' \rightarrow 5'$ (leading) or $5' \rightarrow 3'$ (lagging)



- 3' ends in sugar
- 5' ends in phosphate
- Essentially both strands are antiparallel, both are trying to replicate from 5' → 3'
 - *Leading* stand will travel towards the continuing unzipping DNA
 - *Lagging* will travel away from the unravelling DNA

Process

1. Initiation

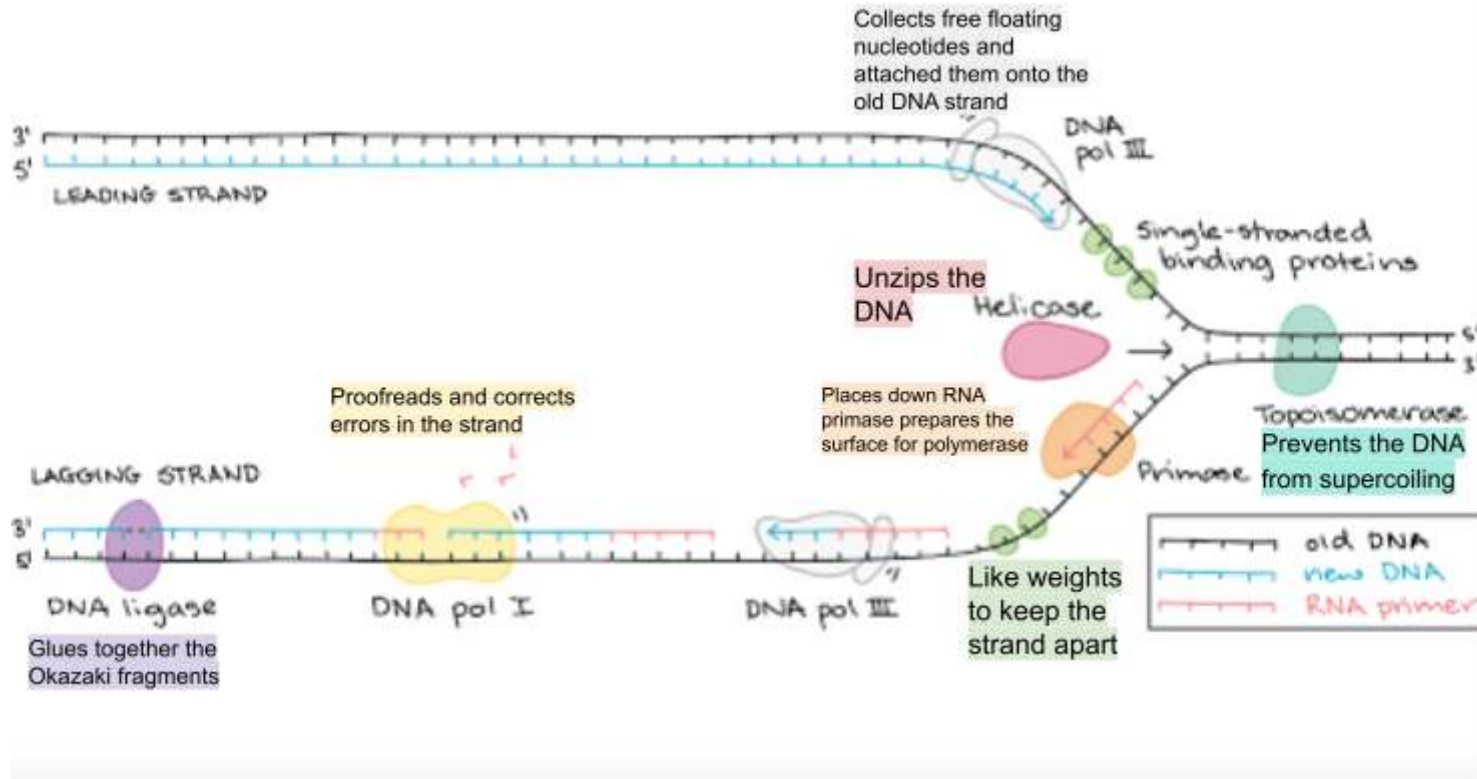
- Helicase uncoils/unzips the complementary strands of DNA.
- SSB proteins bind to the DNA strands to keep them separated
- Topoisomerase keeps the DNA from supercoiling (lays on top of the DNA as it unzips)

2. Elongation

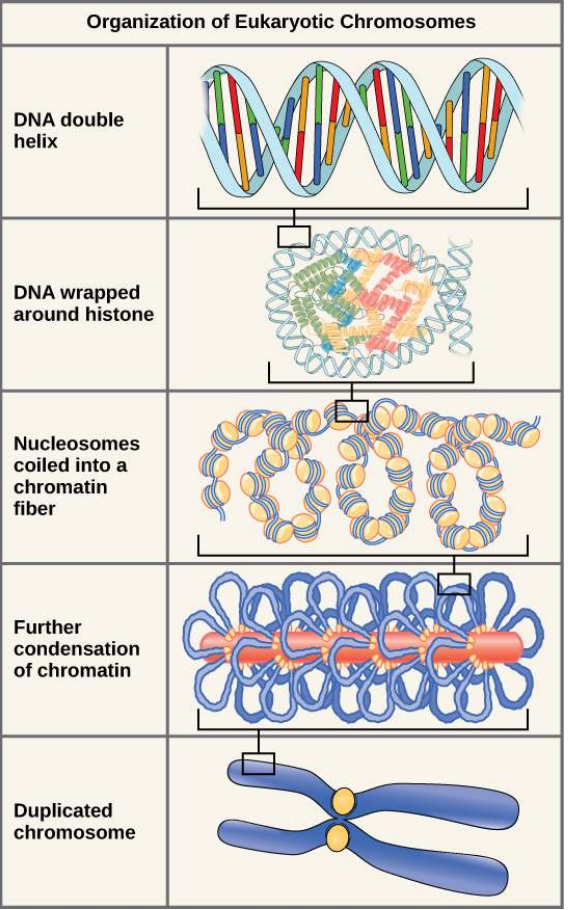
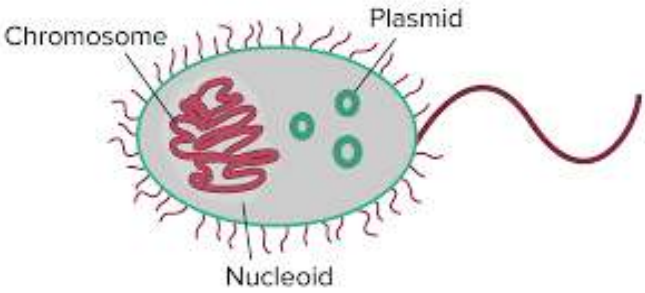
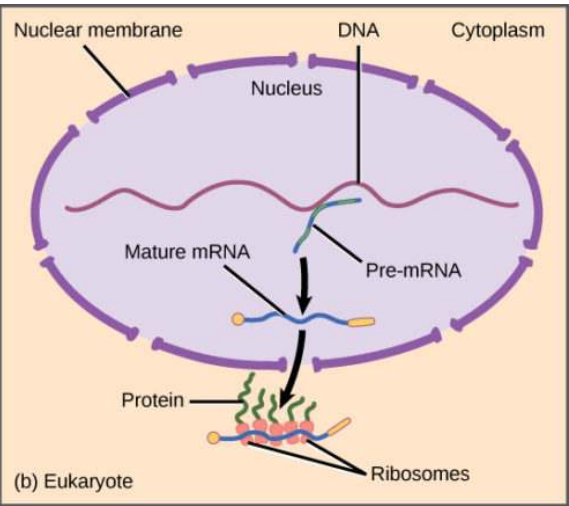
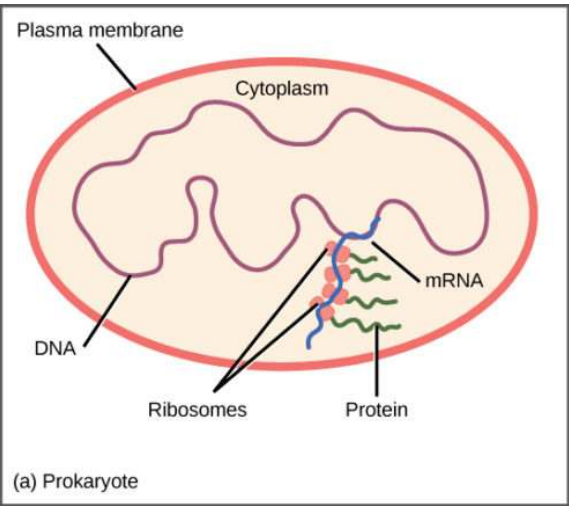
- Primase comes in and makes RNA primers on both strands
- Leading strand
 - DNA polymerase III adds free-floating nucleotide bases in the 5' → 3'
 - DNA polymerase I also proofread the strand to ensure that there aren't any errors
 - Ligase seals the gaps between the DNA (e.g bases that may have been removed while proofreading)
- Lagging strand
 - DNA polymerase III adds the bases to the RNA, though there are gaps as it has to keep going back leaving Okazaki fragments.
 - It will also get proofread and correct any errors
 - Ligase seals the gaps between the DNA
- All steps are repeated until the full strand is formed
 - The strands recoil into a double-helix shape

3. Termination

- Semi-conservative replication
 - There are little to no errors as each strand contains one original strand of DNA.

	 The diagram illustrates the process of DNA replication at a replication fork. The original DNA strands are labeled as 'old DNA' (black lines). New DNA strands are synthesized in two directions: the 'LEADING STRAND' (top) is synthesized continuously towards the fork, and the 'LAGGING STRAND' (bottom) is synthesized discontinuously away from the fork as Okazaki fragments. Key enzymes and their functions are labeled: Helicase (pink) 'Unzips the DNA'; Topoisomerase (green) 'Prevents the DNA from supercoiling'; Primase (orange) 'Places down RNA primer prepares the surface for polymerase'; DNA pol III (blue) 'Collects free floating nucleotides and attached them onto the old DNA strand' and 'Proofreads and corrects errors in the strand'; DNA pol I (yellow) 'Proofreads and corrects errors in the strand'; DNA ligase (purple) 'Glues together the Okazaki fragments'; and Single-stranded binding proteins (green dots) 'Like weights to keep the strand apart'. A legend indicates: black line = old DNA, blue line = new DNA, red line = RNA primer.
5.2.2 assess the effect of the cell replication processes on the continuity of species	Mitosis is the asexual replication of cells, in which one haploid cell, divides into two identical haploid daughter cells. This process is essential for growth and repair. Mitosis allows a species to grow into its mature/adult form. Without it, organisms will not be able to live and pass on their genes. It is very important to the survival of the species, as it allows them to grow to maturity and repair their damaged cells. For example, mitosis occurs starts when the egg of an organism has been fertilised and is starting to form into a baby; allowing it to grow, as somatic cells are starting to form and replicate. Meiosis is the process in which two diploid parents divide into four haploid daughter cells. Each of these cells is different from the other as they all contain a variation of the genes that they receive from the parent cells. This is essential for genetic variation within a species. Allows the organism to pass on its genes to its offspring. Genetic variation will allow the offspring to evolve and adapt to its environment. It is very important as without there will not be variation within the offspring, allowing it to have a greater chance of survival. For example, the parents of a child carry different genes/traits (alleles) these are crossed over, segregated and assorted in several different ways, in order for the child to receive a different set of genes that will allow it to survive in its environment.
5.3 DNA and Protein Synthesis IQ: How important is it for genetic material to be replicated exactly?	

5.3.1 construct appropriate representations to model and compare the forms in which DNA exists in eukaryotes and prokaryotes



5.3.2 model the process of polypeptide synthesis, including: (ACSBLO79)

- transcription and translation
- assessing the importance of mRNA and tRNA in transcription and translation

Term	Definition
Protein (polypeptide) Synthesis	The production of proteins from DNA. Involves two processes, transcription and translation.
Transcription	Process of producing single-stranded mRNA from DNA
Translation	Process in which the codons on mRNA are translated into a sequence of amino acids resulting in polypeptide
Exons	Coding regions of genes

(ACSBL079)
3. analysing the function and importance of polypeptide synthesis (ACSBL080)
4. assessing how genes and environment affect phenotypic expression (ACSBL081)

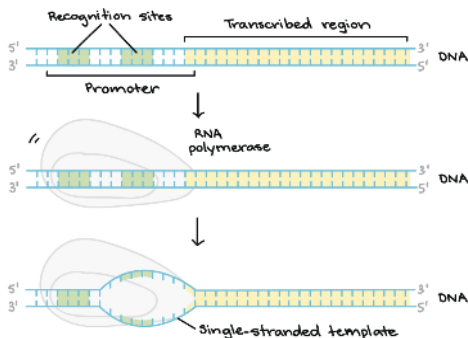
Introns	Non-coding regions of genes (these get spliced out)
Non-coding (sense strand)	The strand of DNA that serves as the template during transcription; its sequence is the same as the anticodons of tRNA
Coding (antisense strand)	Not used as a template, but its base sequence corresponds to the base sequence of the mRNA; contains codons
Anticodons	Three unpaired bases at one end of tRNA that attach the tRNA to its complementary bases on the mRNA strand
Codons	A triplet or sequence of three bases on DNA or mRNA that codes for a single amino acid
Promoter Region	A region of DNA that starts transcription of a gene

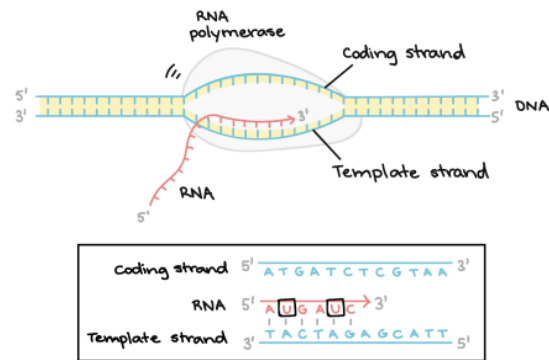
★ **Transcription** → DNA to mRNA

- Occurs within the nucleus, needed to create mRNA that will be able to leave the nucleus and then be translated for protein synthesis.
- RNA: a single-stranded sequence that has been made from the non-coding region of DNA
 - Uses ribose sugar
 - Uracil (U) instead of Thymine (T)
 - There are 3 types of RNA

Form	Function
mRNA (messenger)	Carries a complementary copy of the nucleotide sequence of DNA that specifies the amino acid sequence for a particular polypeptide.
tRNA (transfer)	Together with proteins, rRNA forms a ribosome which is where they are joined to form a polypeptide chain of amino acids (used in Translation)
rRNA (ribosomal)	Transfers amino acids from the cytoplasm to the ribosomes, where they are joined to form a polypeptide chain based on the sequence of nucleotides

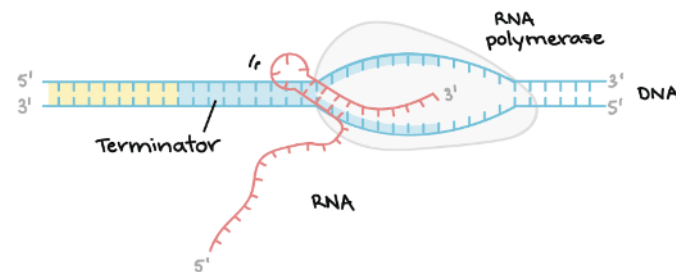
- RNA polymerase: an enzyme which uses a single strand of DNA to create a complementary RNA strand
 - Specifically in the 5' to 3' direction

Pre-mRNA	
	Initiation • RNA polymerase binds to a sequence of DNA called the promoter (tells it where to start, found near the beginning of a gene). Each gene has its own promoter. • It will then start to separate the DNA strands, creating a single-stranded template (non-coding) needed for the following steps.



Elongation

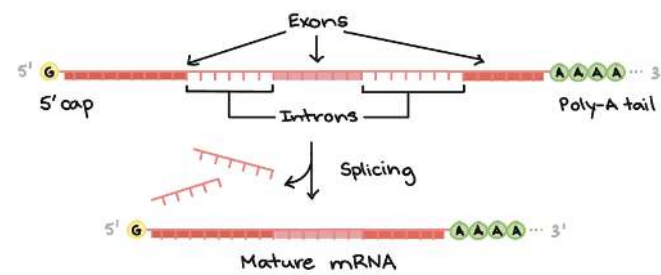
- The one strand of 3' to 5' acts as a template strand (non-coding) for RNA polymerase.
 - Reads each base at a time, going 5' to 3'
- Uses free-floating nucleotides to build the complementary bases.
- This RNA strand is the same as the top coding strand (non-template)



Termination

- Similar to promoters there are terminators at the end of the DNA, signalling that the process is complete
- There are several termination mechanisms that allow the RNA to be released
 - Hairpin: the RNA loops and creates complementary base pairs with itself, allowing it to let go.

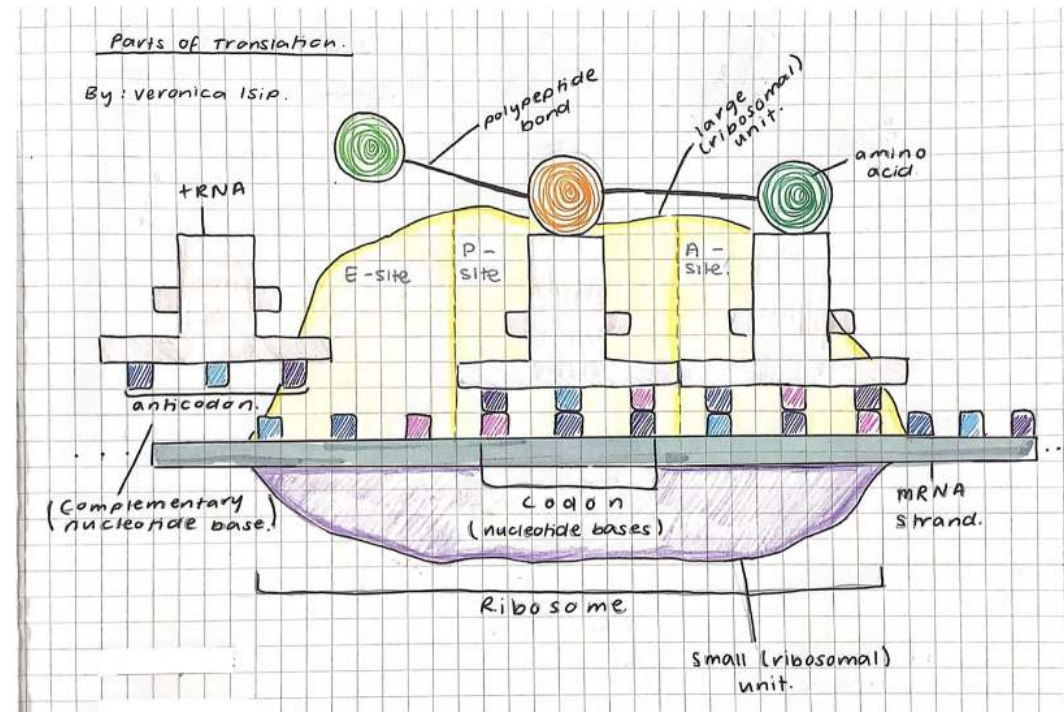
Mature mRNA (extra step in eukaryotes)



- pre-mRNA is modified with the 5' cap (at the beginning, Guanine) and 3' poly-A tail (at the end, a bunch of Adenines)
- Splicing out of introns occurs, leaving behind the exons.
 - Introns: segments of the gene that does not code for anything, "nonsense"
 - Exons: segments of the gene that code for a particular protein.
- This increased the stability of mRNA, creating the correct sequence.
 - If introns were not removed it could create a non-functional protein.
- The strand is now able to leave the nucleus through a pore, to be used in translation.

★ **Translation** → mRNA to protein

- The second step of protein synthesis
 - This is where the actual building of proteins starts
- Occurs within ribosomes (outside of the nucleus)
- The main components of translation is shown below ↴



Process	
Initiation	• The mRNA attaches to the small subunit of the ribosome. • The start codon AUG calls for the tRNA(1) w/ the amino acid Methionine and the anticodon of UAC (initiates the translation process) • The next tRNA will then enter through the A site of the ribosome attaching to the next set of codons. • A polypeptide bond is then formed between the two amino acids
Elongation	• tRNA (1) will leave its amino acid behind leaving via the E site of the ribosome. • The current tRNA is now holding two amino acids, it will then move along into the P site. • According to the next set of codons it will then call for the next tRNA (enters through the A site) ■ A bond is now formed between the one that is being held in the P site. • The old tRNA will drop the amino acid and exit via the E site • This process is repeated until the end of the mRNA strand
Termination	• The ribosome will then reach a stop codon on the mRNA strand, meaning translation of this specific protein is complete • These codons don't call of any amino acids instead it calls for a 'release factor' (similar to a tRNA but no amino acid only the anticodons) ■ The RF helps to release the newly made polypeptide chain and separates the mRNA from the ribosomal units ■ The newly formed polypeptide chain will then fold into a specific shape to then be sent to specific parts of the organism to fulfil function.

★ **Function and purpose of polypeptide synthesis**

- The main purpose for polypeptide synthesis is to create functional protein structures, that are then able to be used by an organism to fulfil other functions
 - E.g enzymes are a type of protein that is used to break down food

5.3.3 investigate the structure and function of proteins in living things

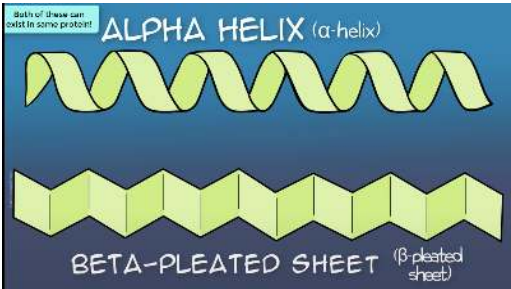
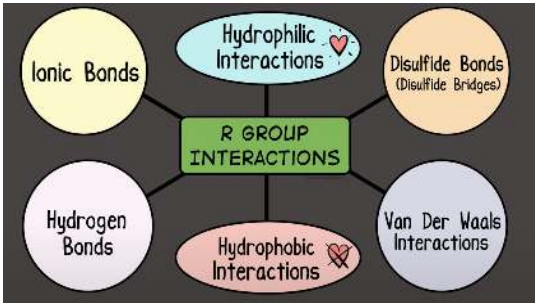
★ **Protein folding:** the way a certain polypeptide chain is folded affects the shape and functions of the final proteins being made

- There are 4 structures to protein folding
 - **Primary:** the sequence of amino acids in the polypeptide that build up the protein. They are held together with peptide bonds.
 - **Secondary:** the chain shape (how it is folded) two of the common ways are → Alpha Helix and Beta-pleated sheet.
 - **Tertiary:** Folding that occurs in the 3D shape of a functional protein, due to R groups. R groups can vary among different amino acids. They can define the amino acid and can make amino acids behave a certain way
 - **Quaternary structure:** how multiple polypeptide chains are folded together.

★ **Types of proteins and their functions**

- Proteins are a major component of every cell and have many roles in bodily functions

Type	Function + Example
Structural	Maintains the shape of the cells and creates connective tissues • Dietary intake of proteins is important as it is what allows us to grow, repair and maintain tissue ○ E.g Actin (found in muscle cells): muscle contractions and cell movements
Storage & Transport	Storage proteins bind to a certain substance to hold them in place and transport proteins bind to substances and carry them around the body • E.g haemoglobin: carries/deliver oxygen around the body
Signalling	-
Sensory	-
Motor	-
Enzymes	Biological catalysts • They act on molecules to either break it down into something more simple or they build it up into something more complex. Creating efficiency and an easier way to survive ○ E.g. Ligase (used in DNA replication): joins fragments together of newly synthesised DNA to form a new seamless strand.
Defensive	Protects cells and the body from foreign substances • E.g Antibodies: they have a specific binding site to a specific antigen on a pathogen to fight infection ○ E.g. Immunoglobulin E (initiates an allergic reaction when exposed to an allergen)



5.4 Genetic Variation

IQ: Can population genetic patterns be predicted with any accuracy?

5.4.1 conduct practical investigations to predict variations in the genotype offspring by modelling meiosis, including the crossing over of homologous chromosomes, fertilisation

★ **What creates genetic variation within a population (recap)**

A **mutation** is a change in the DNA of an individual. A single nucleotide polymorphism refers to a change in a single nucleotide in a section of DNA and is responsible for alternative alleles. The mutation is the raw material for evolutionary change. Mutations introduce new alleles into populations.

Sexual reproduction is the production of genetically unique gametes in meiosis resulting from **an independent assortment** of chromosomes and **genetic recombination** in crossing over. The **random union of gametes** in fertilisation further mixes the genetic material, creating genetic variation.

and mutations	Source	Effect																				
	Crossing over (prophase)	Homologous chromosomes exchanges genetic information creating <u>new combinations of alleles in the chromosome</u>																				
	Independent Assortment (metaphase I) & Random Segregation (anaphase I)	Homologous chromosomes arrange themselves and separate independently, leading to <u>random allele combinations</u> in the 4 non-identical gametes																				
	Random Fertilisation	Non-identical male and female gametes fuse to create an offspring with a <u>unique combination of alleles</u> to form unique genotypes.																				
	Mutations	Produce <u>new alleles</u> to increase the possible variations.																				
	Meiosis + Fertilisation + Mutations = recombination of genes ↓ Genetic variation ↓ Favourable adaptations ↓ Natural selection ↓ Reproductive maturity of fittest organisms ↓ Inheritance of favourable variations ↓ Genetic isolation ↓ Speciation (new species) ↓ Diversity • Genetic diversity → <i>variations in gene combinations</i> • Species diversity → <i>variations of species in an area</i> • Biodiversity → <i>variations of living things in an area</i> ↓ Evolution																					
5.4.2 model the formation of new combinations of genotypes produced during meiosis, including but not limited to a. interpreting examples of autosomal, sex-linkage, co-dominance, incomplete dominance and multiple alleles b. constructing and interpreting information and data from pedigrees and Punnett squares	<table><tr><th>Key terms</th><th>Key definitions</th></tr><tr><td>Autosomal inheritance</td><td>Inheritance on germline cells where the genes inherited are on the 1st -2nnd pairs of chromosomes and code for phenotypes</td></tr><tr><td>Sex-linked</td><td>Inheritance on germline cells where the genes inherited are on the 23rd pair of chromosomes. These genes code for gender and secondary sexual characteristics</td></tr><tr><td>Co-dominance</td><td>Both Alleles are dominant, so they show without blending</td></tr><tr><td>Simple dominance</td><td>One allele is completely dominant over the other</td></tr><tr><td>Incomplete dominance</td><td>Both alleles are dominant, so they have blended in the phenotype</td></tr><tr><td>Dominant</td><td>Will always be expressed over the recessive</td></tr><tr><td>Recessive</td><td>This trait is masked due to the dominant and can only be expressed if both alleles expressed are recessive.</td></tr><tr><td>Pedigree</td><td>Graphical representation of inheritance patterns of a trait in related individuals over generations</td></tr><tr><td>Multiple alleles</td><td></td></tr></table>	Key terms	Key definitions	Autosomal inheritance	Inheritance on germline cells where the genes inherited are on the 1st -2nnd pairs of chromosomes and code for phenotypes	Sex-linked	Inheritance on germline cells where the genes inherited are on the 23rd pair of chromosomes. These genes code for gender and secondary sexual characteristics	Co-dominance	Both Alleles are dominant, so they show without blending	Simple dominance	One allele is completely dominant over the other	Incomplete dominance	Both alleles are dominant, so they have blended in the phenotype	Dominant	Will always be expressed over the recessive	Recessive	This trait is masked due to the dominant and can only be expressed if both alleles expressed are recessive.	Pedigree	Graphical representation of inheritance patterns of a trait in related individuals over generations	Multiple alleles		
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Multiple alleles																						

Punnet Squares

A diagram/ method used in calculating and figuring out the phenotypic and genotypic expressions of an offspring.

→ 5.4.2a

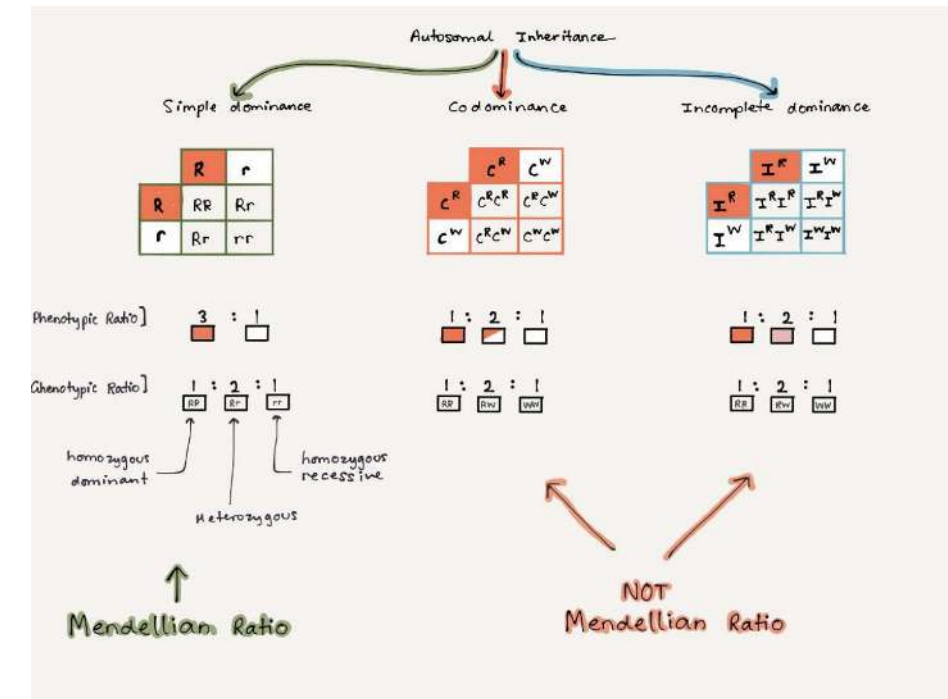
★ Chromosomes

- **Autosome** → contains genes responsible for body traits (skin, hair eye colour)
 - 1-22nd homologous pairs in humans
- **Sex chromosomes** → contains genes for gender and secondary sexual characteristics
 - 23rd pair in humans
 - XX (females)
 - XY (male)

★ Autosomal Inheritance

- In a specimen, there is a specific gene for each trait, and 2 variations of this genes are known as alleles
 - There is both a **dominant and recessive**
 - Dominant alleles are always shown over the recessive ones.
 - E.g T = dominant tall → t = recessive short
 - Dominant alleles will be expressed using a capital letter and lowercase for recessive.
- There are three types of inheritance (explained in diagram →)
 - **Simple dominance**
 -
 - **Codominance**
 -
 - **Incomplete Dominance**
 -
- **Multiple Alleles**
 - Though most genotypes have 2 alleles, some traits can be coded for more than 2 alleles
 - Multiple alleles result in a number of different phenotypes depending on the interaction of the alleles.
 - **Blood groups**
 - There are 3 alleles that code for blood types A, B and O
 - O is recessive
 - A and B are both dominant over O but equally dominant over each other co-dominant.
 - Therefore, there are 4 blood groups for the genotypes created:
 - $I^A \rightarrow A$
 - $I^B \rightarrow B$
 - $i \rightarrow O$

Genotype	Alleles	Phenotype
$I^A I^A$	I^A	Blood type A
$I^A i$	I^A and i	
$I^B I^B$	I^B	Blood type B
$I^B i$	I^B and i	
$i i$	i	Blood type O
$I^A I^B$	I^A and I^B	Blood type AB



★ Sex-linkage

- Due to a mutation, an autosomal gene can be transferred to the X chromosome of the 23rd (sex) pair.
 - It becomes an X-linked gene
 - Undergoes sex-linked inheritance
 - Never carried on the Y chromosomes
 - Only carried on the X chromosomes
- Females with one recessive allele and one dominant allele, for a gene that is unique to the X chromosome, will always display the dominant phenotype.
 - However, a male with a recessive allele for a gene unique to the X chromosome will always exhibit that recessive trait because there is no other corresponding allele on the Y chromosome.

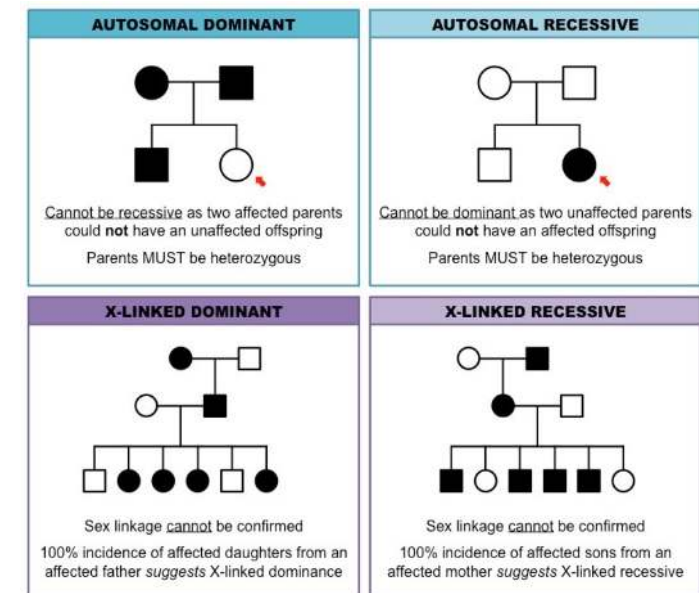
X^E = dominant f. allele (most dominant)	X^e = recessive f. allele	Y = m. allele (most recessive)
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Genotype	Alleles	Gender	Phenotype
$X^E X^E$	X^E	Female	Normal
$X^e X^e$	X^e	Female	Diseased
$X^E X^e$	X^E X^e	Female	Carrier
$X^E Y$	X^E Y	Male	Normal
$X^e Y$	X^e Y	Male	Diseased

→ 5.4.2b

★ Pedigrees

- **Autosomal Dominant Inheritance**
 - Trait should not skip generations (unless penetrance)
 - An affected person married to a “normal” person should have approx 50% of the offspring being affected (also indicates that the affected individual is heterozygous)
 - Distribution of the trait should be close to equal among the sexes
- **Autosomal Recessive Inheritance**
 - Trait often skips a generation
 - Distribution of the trait should be close to equal among the sexes
 - Traits are often found in pedigrees with consanguineous marriages
 - If both of the parents are affected, all of the children should be affected
 - Most affected individuals have “normal” parents
 - When a “normal” person is married to an affected individual, all of the children are normal (indicating the normal parent is homozygous dominant)
 - If a “normal” person is married to an affected individual and one or more of the children is affected, then approximately half of the children should be affected
- **X-linked recessive Inheritance**
 - Most of the individuals affected are males
 - For a female child to be affected, the father must be affected and the mother must be affected or a carrier
 - All of the sons of an affected mother must be affected
 - Approx half of the sons of a carrier female should be affected



5.4.3 collect, record and present data to represent frequencies of characteristics in a population, in order to identify trends, patterns, relationships and limitations in data, for example

- a. examining frequency data
- b. analysing single nucleotide polymorphism (SNP)

5.5 Inheritance Patterns

IQ: Can population genetic patterns be predicted with any accuracy?

5.5.1 investigate the use of technologies to determine inheritance patterns in a population using, for example:

- DNA sequencing and profiling

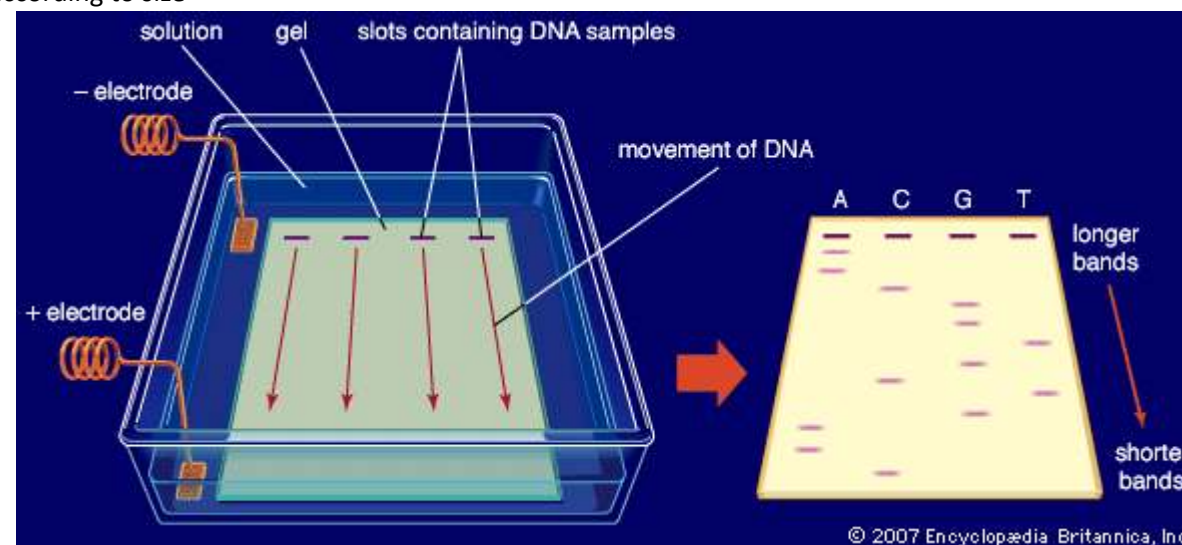
Term	Definition
Genome	-

★ DNA sequencing

- Process to determine the sequence of the nucleotide bases (ATCG) of a particular genome
- **Sanger method**
 - This method can determine sections of DNA that are up to 900 base pairs in length, and it is also called the chain termination method
 - This method was also used for the Human Genome Project

★ DNA profiling

- **Gel Electrophoresis**
 - Samples of fragmented DNA are placed in the wells of an agarose gel
 - The gel is placed in a buffering solution and an electrical current is passed across the gel
 - DNA, being negatively charged, moves to the positive electrode
 - Smaller fragments move faster through the gel (compared to larger ones)
 - The fragments are then separated according to size



★ Use of the technologies

- **DNA sequencing**
 - To find the cause of genetic diseases
 - Evolutionary relationship
- **DNA profiling**
 - **Disease testing**
 - To work out whether a person is a sufferer or a carrier of a particular disease.
 - **Forensics**
 - The DNA in crime suspects is compared with DNA from a crime scene (if it is a 100% match, the suspect is guilty)
 - **Paternity testing**
 - 50% of a child's DNA comes from the mother and the father
 - If one parent is known, the other parent is the one which completes the child's DNA sequence

5.5.2 investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns and relationships, for example: - the use of population genetics data in conservation management - population genetics studies used to determine the inheritance of a disease or disorder - population genetics relating to human evolution	
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Mod 6: Genetic Change (yr 12)

6.1 Mutations						
IQ: How does mutation introduce new alleles into a population?						
Syllabus Outcome	Summary of content					
6.1.1 explain how a range of mutagens operate, including but not limited to - electromagnetic radiation sources - chemicals - naturally occurring mutagens	<table><tr><th>Term</th><th>Definitions</th></tr><tr><td>Mutagen</td><td>Any environment factor that alters the genetic sequence of DNA</td></tr></table>		Term	Definitions	Mutagen	Any environment factor that alters the genetic sequence of DNA
	Term	Definitions				
	Mutagen	Any environment factor that alters the genetic sequence of DNA				
	** Upon our daily exposure to mutagens DNA repair systems are usually able to detect and fix minor alterations to DNA, but: when the repair system is compromised, mutagens may cause the development of diseases (such as cancer; mutagens that result in cancers are called carcinogenic.) **					
	★<u>Types of Mutagens</u>					
	Physical	Chemical	Biological			
Refers to heat and ionising radiation	Chemicals will cause mutations with prolonged exposure	Naturally occurring environmental factors capable of altering DNA				
Radiation = the transfer of energy; though	(if there is frequent or extensive exposure of cells) Examples:	Viruses: replicate by inserting their DNA				

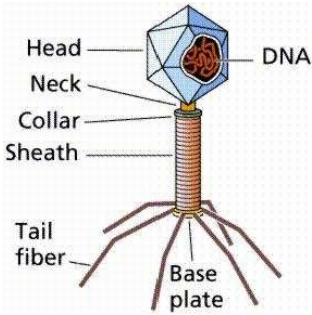
not all are harmful

Ionising radiation = any radiation that breaks down the chemical bonds between molecules

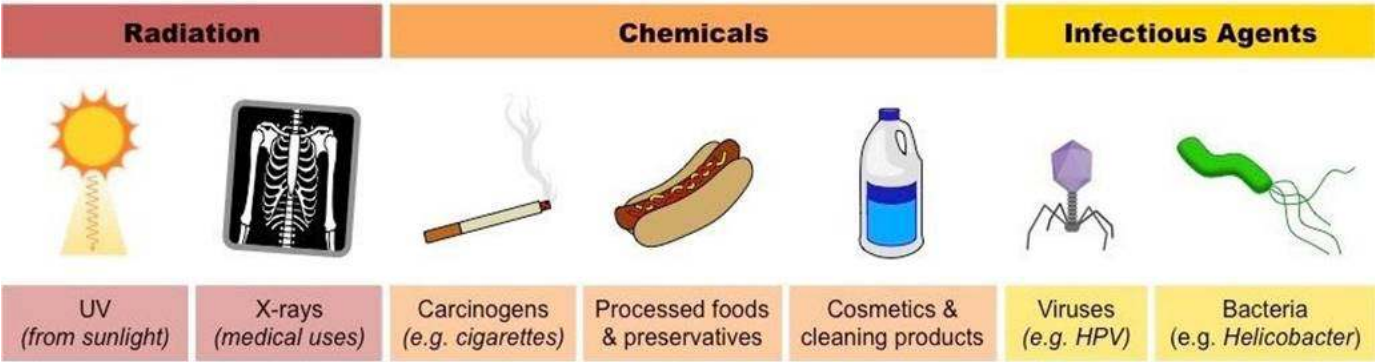
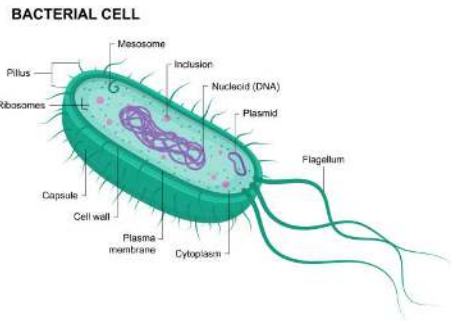
The **electromagnetic spectrum** made of 7 different types of radiation no all are harmful, however ultraviolet (UV), X-ray and Gamma ray radiation are ionising. They can break the hydrogen bonds between nitrogenous bases and the bonding of the Sugar phosphate backbone.

- Nicotine (in cigarettes)
- Pesticides
- Alcohol

into host cells, creating a disruption in regular cell function that may lead to lasting changes



- **Bacteria:** can induce inflammation, reducing the efficiency of DNA repair systems and increasing the potential for/rate of mutation



6.1.2 compare the causes, processes and effects of different types of mutation, including but not limited to:

- point mutation
- chromosomal mutation

Term	Definition
Mutation	A permanent, change within the base sequence of DNA • This can either be helpful, harmful, or neutral effect • Can be cause by mutagens or errors in biological processes
Point mutation	Mutations that only alter a single nucleotide of a nucleic acid sequence (at most a small grouping of nitrogenous bases)
Chromosomal Mutation	Affects large sections of chromosome, typically multiple copies of the same genes on that chromosome.

★ Point Mutation

- There are 3 main types of point mutation

Substitution (S)	When one nucleotide is switched out/replaced by another (for example, A is replaced by a T, G or C base)	
Insertion	When nucleotides are inserted (Added) into a sequence	Both insertion and deletion mutations are classified as frameshift
Deletion	When nucleotides are deleted (removed) from a sequence	

Possible effects of a point mutation

- Missense (S):** affect a singular codon causing the introduction of a different amino acid into the resulting polypeptide chain
- Nonsense (S):** prematurely introduces a stop codon that shortens the polypeptide chain and produced a dysfunctional protein as a result
- Silent (S):** occur within non-coding regions of DNA or as a result of nucleotide substitution where the base changed has no effect on the type of amino acid inserted into the sequence and so has no effect on the protein produced as a result
- Neutral:** result in the same amino acid being added to the thus unchanged polypeptide chain to result in no change in protein structure or function.

Frameshift mutation

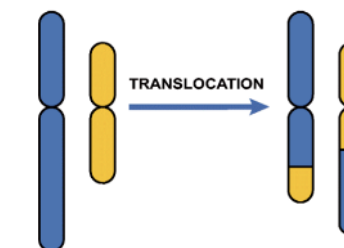
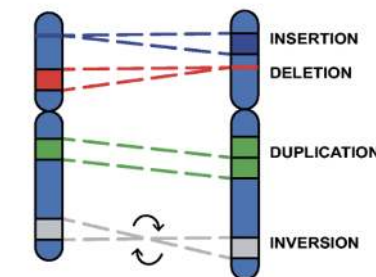
Insertion or deletion of a nucleotide base into or out of a nucleic acid sequence that thereafter shifts the entire sequence up or down causing codons to incorrectly realign

- More likely to cause an alteration to the phenotype as a frameshift effect than just one codon so has reduced likelihood being silent in nature.

★ Chromosomal Mutation

- Structural chromosomal mutation** → occurs when a portion of the chromosome is changed upon errors during the process of crossing over

Deletion	A section of a chromosome is deleted (Removed)
Insertion Duplication	A section of a chromosome is inserted (added) or duplicated (doubled)
Inversion	A section of a chromosome is inverted (turned upside down)
Translocation	A section of a chromosome is relocated (moved) to a different non-homologous chromosome



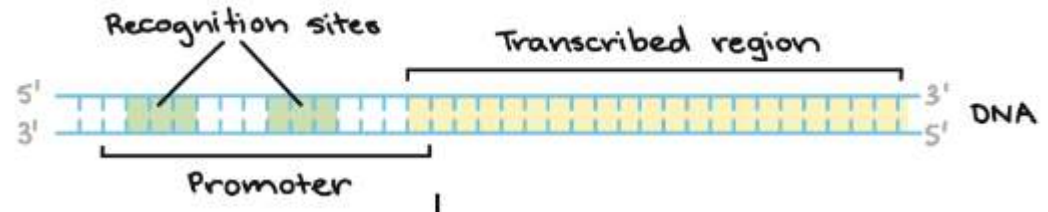
- Quantitative chromosomal mutation** → occur when sister chromatids separate incorrectly during anaphase to result in an incorrect number of chromosomes in 2 or more of the resulting gametes

	<table><tr><td>Aneuploidy (an-u-ploid-y)</td><td>The number of chromosomes in offspring differs from the parent by on or multiple copies of an entire chromosomes being added or removed from the genome of the offspring E.g. down syndrome results in the presence of 47 chromosome instead of 46 </td></tr><tr><td>Polyploidy (polly-ploid-y)</td><td>An organism contains more than two whole sets of homologous chromosomes</td></tr><tr><td>Nondisjunction</td><td>The failure of one or more pairs of homologous chromosome or sister chromatids to separate normally during nuclear division, usually resulting in an abnormal distribution of chromosomes between the resulting daughter (germline) cells</td></tr></table>	Aneuploidy (an-u-ploid-y)	The number of chromosomes in offspring differs from the parent by on or multiple copies of an entire chromosomes being added or removed from the genome of the offspring E.g. down syndrome results in the presence of 47 chromosome instead of 46	Polyploidy (polly-ploid-y)	An organism contains more than two whole sets of homologous chromosomes	Nondisjunction	The failure of one or more pairs of homologous chromosome or sister chromatids to separate normally during nuclear division, usually resulting in an abnormal distribution of chromosomes between the resulting daughter (germline) cells																					
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6.1.3 distinguish between somatic mutations and germ-line mutations and their effect on an organism	<table><tr><td></td><td>Germline mutation</td><td>Somatic mutation</td></tr><tr><td>Define</td><td>Mutation occurring in gamete/germline cells</td><td>mutation occurring in somatic cells - any cell that isn't germline</td></tr><tr><td>Locus</td><td>Germline (gamete) cells</td><td>somatic (body) cells</td></tr><tr><td>When does it occur?</td><td>occurs either during interphase via errors in the DNA replication process or by non-disjunction (where sister chromatids separate incorrectly during anaphase of meiosis II)</td><td>occurs solely during interphase as a result of errors in DNA replication</td></tr><tr><td>Effect?</td><td>always harmful</td><td>either helpful, harmful or neutral</td></tr><tr><td>Always appears in phenotype?</td><td>phenotypic ramifications are unavoidable: always present in the phenotype of the resulting offspring</td><td>phenotypic ramifications are avoidable: not always present in the phenotype (able to be hidden if the mutation is neutral, silent or located on a non-coding region of DNA)</td></tr><tr><td>Inheritable</td><td>yes</td><td>no</td></tr><tr><td>take-effect</td><td>during embryonic development (at birth)</td><td>after embryonic development/formation of the zygote</td></tr><tr><td>Example</td><td>E.g Huntingtons disease: the parent has an abnormal repetition of the code CAG for the protein huntingtin, when they have children the repetition increases in an amount within the offspring, causing them to most likely develop the disease.</td><td>Skin cancer: The oncogenes (regulates cell division), if mutations in these occur the cells will not know how to stop replicating. Forming a mass of unspecialised cells i.e. a tumour</td></tr></table> Germ-line mutation (in sperm only) Entire organism carries the mutation. Half of the gametes carry the mutation. Gametes Embryo Somatic mutation Organism Patch of affected area Gametes of the organism None of the gametes carry the mutation.		Germline mutation	Somatic mutation	Define	Mutation occurring in gamete/germline cells	mutation occurring in somatic cells - any cell that isn't germline	Locus	Germline (gamete) cells	somatic (body) cells	When does it occur?	occurs either during interphase via errors in the DNA replication process or by non-disjunction (where sister chromatids separate incorrectly during anaphase of meiosis II)	occurs solely during interphase as a result of errors in DNA replication	Effect?	always harmful	either helpful, harmful or neutral	Always appears in phenotype?	phenotypic ramifications are unavoidable: always present in the phenotype of the resulting offspring	phenotypic ramifications are avoidable: not always present in the phenotype (able to be hidden if the mutation is neutral, silent or located on a non-coding region of DNA)	Inheritable	yes	no	take-effect	during embryonic development (at birth)	after embryonic development/formation of the zygote	Example	E.g Huntingtons disease: the parent has an abnormal repetition of the code CAG for the protein huntingtin, when they have children the repetition increases in an amount within the offspring, causing them to most likely develop the disease.	Skin cancer: The oncogenes (regulates cell division), if mutations in these occur the cells will not know how to stop replicating. Forming a mass of unspecialised cells i.e. a tumour
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6.1.4 assess the significance of 'coding' and 'non-coding' DNA segments in the process of mutation

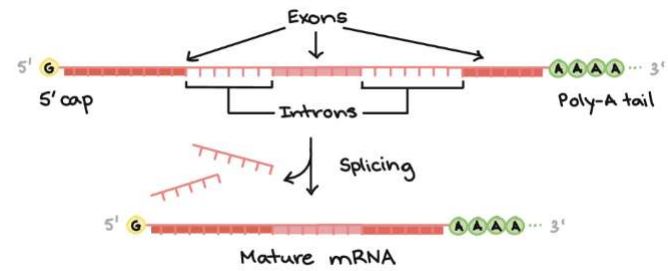
★ **Eukaryotic genes**

- A human genome is comprised of a mere 2% coding DNA (segments that code for proteins), while the remaining 98% is classified as non-coding (segments that do NOT code for proteins)
- Structure
 - Start and stop codons = nitrogenous base triplets at the beginning and the end of the coding regions that initiate + terminate the production of polypeptide chains accordingly
 - Promoter region = section of the gene where transcription begins; located before the start codon.
 - Exons = coding DNA regions directly responsible for the production of proteins
 - Introns = non-coding DNA regions that remain within the nucleus after the process of replication



★ **Mutations in Coding regions and their significance**

- Changes in the coding regions will affect the production of proteins through the nucleic acid sequence of mRNA
- This can alter the structure of the protein being produced and/or the function
 - Presenting as:
 - An altered trait, see through the phenotype
 - The development of a genetic disease
- If mutations occur to products involved in DNA repair there is generally an increase in the rate of mutation during DNA replication, prior to cell division
- Prokaryotes tend to have more coding regions and so its maintenance is essential to the organism's survival: when sections of DNA become inactive due to fluctuation environmental change there is an increase rate of mutation



★ **Mutations in non-coding regions and their significance**

- There is yet to be a biological purpose to be found, non-coding DNA is useful for DNA testing and forensic analysis as it is highly variable across individuals
- Some portions of non-coding DNA serve vital functional purpose:
 - Introns, non-coding RNA genes, promoter and terminator regions: if mutations occur in THESE regions, the overall function of the cell or organism could be compromised

STING: acronym to remember the types and functions of **NON-coding DNA**

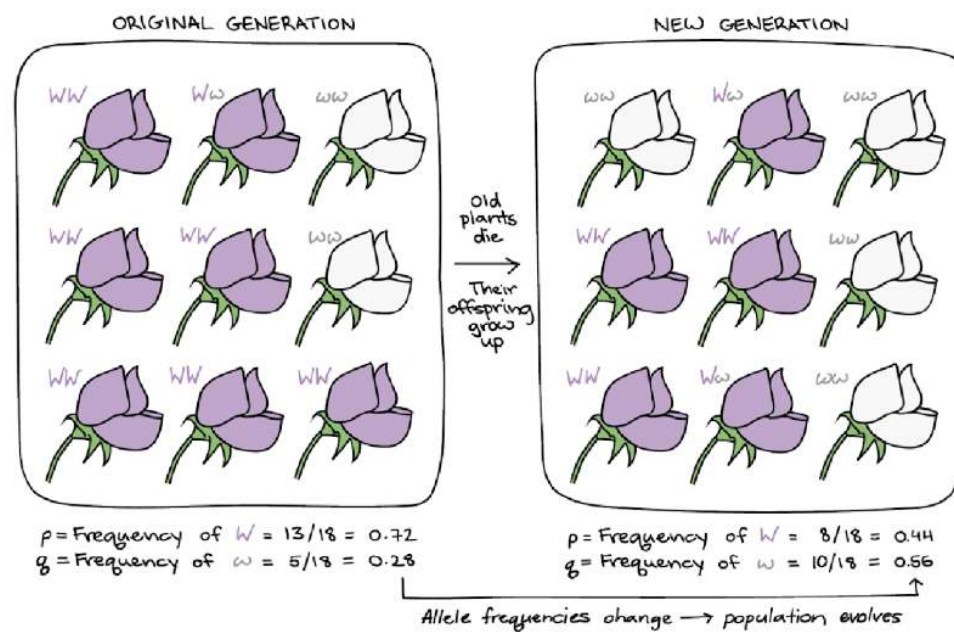
	non-coding DNA	description
S	satellite DNA	• tandemly repeating sequences of DNA • structural component of heterochromatin + centromeres • commonly utilised in DNA profiling
T	telomeres	• regions of repetitive DNA at the end of a chromosome • protect against chromosomal deterioration during replication
I	<u>introns</u>	• non-coding DNA segments/sequences • removed by RNA splicing during transcription to create a functional mRNA strand
N	non-coding RNA genes	• code for RNA molecules that are NOT translated into protein • structural genes, for example, that code for tRNA and rRNA
G	gene regulatory sequences	• sequences that are involved in the process of transcription (stage 1 of protein synthesis) • includes promoters, enhancers and silencers

Intron mutations:

- mutations can lead to introns failing to be removed (not spliced out), meaning that it will be read as part of the coding sequence during translation and will contribute to the eventual protein being produced
 - Introns that fail to be removed, are known as 'pseudo-exons' - and can cause the development of genetic disease

	<u>Non-coding RNA gene mutations</u> • Can produce tRNA and rRNA which is essential for the production of chains and proteins ◦ Mutations in these non-coding sections (non-coding RNA genes) are not well tolerated and can in some cases cause death. <u>Promoter/terminator regions</u> • Mutations within the promoter to terminator section of the non-coding DNA usually result in the over or under-expression of a gene leading to either an excess or shortage of the gene product.															
6.1.5 investigate the causes of genetic variation relating to the processes of fertilisation, meiosis and mutation	★ <u>Causes of genetic variation: mutation, meiosis and fertilisation</u> <table><tr><td colspan="2">Variation via MUTATION</td></tr><tr><td>chromosomal</td><td rowspan="2">These can either affect single nucleotides, groupings or whole chromosomes, they are the source of slightly different versions of the same gene, called alleles (this can affect populations if these mutations are passed down) • Alleles = variation of genes = the raw material for natural selection and evolution </td></tr><tr><td>point</td></tr></table> <table><tr><td colspan="2">Causes of variation in meiosis</td></tr><tr><td colspan="2">#1 DNA replication mutations: DNA is replicated (during interphase) before the process of meiosis occurs - whilst replicating DNA, random mutations may be introduced as a result of replication errors</td></tr><tr><td>#2 Crossing Over = homologous chromosomes (when in close proximity) exchange/trade corresponding genetic material to result in the production of ‘recombinant chromosomes’ • Blocks of genes are exchanged within the homologous pair • Occurs during prophase I • Causes variation from one daughter cell to another • Does NOT occur every time chromosomes line up mid-cell • Chromosomes further apart in characteristic are more likely to cross over; closer = less likely. </td><td>#3 Independent Assortment: alleles of different genes assort independently of one another during gamete formation - in other words, the allele a gamete receives for one gene does not influence the allele received for another gene</td></tr><tr><td colspan="2">#4 Random Segregation: homologous chromosomes align in pairs along the cell centre during metaphase I, randomly and independently of neighbouring chromosomes so that when pairs then move apart, during anaphase I, there is an equal chance that the resulting daughter cell will receive either chromosome</td></tr><tr><td colspan="2">Genetic material belonging to the parent is randomly, but equally, divided between the four (4) Gametes produced during meiosis One chromosome from each pair is placed into each of the resulting haploid sex/germline cells: so each individual gamete possesses a combination of genes that is entirely different from the other three (3) gametes.</td></tr></table> ★ <u>Fertilisation</u>: fusion of haploid male and female gametes to produce diploid offspring • a randomised half of one organism's genetic material fuses with that of an entirely separate other to create a diploid (2n) cell with the full number of chromosomes that forms their very own genetically unique genome • variation is increased amongst a population via this method [sexual reproduction] as ◦ it requires the contribution of 2 gametes: each belonging to 2 genetically varied parent organisms ◦ it is a random process, allowing for any number of potential combinations to occur • sexual reproduction can NOT produce new variations of a gene **the creation of new alleles is ONLY possible through mutation, but meiosis and fertilisation both serve to shuffle existing alleles so that each offspring (by chance) possesses a unique combination of alleles - for which the number of possibilities are endless.**	Variation via MUTATION		chromosomal	These can either affect single nucleotides, groupings or whole chromosomes, they are the source of slightly different versions of the same gene, called alleles (this can affect populations if these mutations are passed down) • Alleles = variation of genes = the raw material for natural selection and evolution	point	Causes of variation in meiosis		#1 DNA replication mutations: DNA is replicated (during interphase) before the process of meiosis occurs - whilst replicating DNA, random mutations may be introduced as a result of replication errors		#2 Crossing Over = homologous chromosomes (when in close proximity) exchange/trade corresponding genetic material to result in the production of ‘recombinant chromosomes’ • Blocks of genes are exchanged within the homologous pair • Occurs during prophase I • Causes variation from one daughter cell to another • Does NOT occur every time chromosomes line up mid-cell • Chromosomes further apart in characteristic are more likely to cross over; 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6.1.6 evaluate the effect of the mutation, gene flow and genetic drift on the gene	<table><tr><td><u>Term</u></td><td><u>Definition</u></td></tr></table>	<u>Term</u>	<u>Definition</u>													
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pool of populations	Gene each of two or more alternative forms of a gene that arise by mutation and are found at the same place on a chromosome (eye colour - brown and blue) Allele Genetic information which is transferred from a parent to offspring and is held to determine characteristics of the offspring Dominant one variant of a gene that will always be expressed Recessive variant of the gene that will not be expressed over the dominant, unless two of these alleles are inherited together Phenotype set of observable characteristics of an organism - how the gene has been expressed Genotype the genetic information and the expression of a variant of a certain type of trait Population Allele Frequency Refers to how frequently a particular allele appears in a population. Gene flow Also called migration, is the movement of genes from one population to another population - If different versions of a gene are carried to a population where those gene versions previously did not exist, gene flow can be very important source of genetic variation Gene flow is the exchange of genes between two populations. Population genetics → The study of genetic variation within a population and the changes in the frequency of genes and alleles within a population and among populations over time. ★ Gene Pool → <i>The collection of alleles within a population - combination of all the genes that are present in a given population.</i> • Cannot be identified by mere observation of a population as by sight you see only the phenotype of an individual, not the genotype • If little to no gene exchange occurs between gene pools they are considered separate (such as populations of the same animal living in different areas of the world or different species that live in the same area but do not interbreed) ○ LARGER → gene pools tend to have more genes and therefore more genetic variety between individuals of the population ■ A diverse population is better prepared to cope with sudden environmental change (members of the population are likely to have genes that protect them from adverse change and so allow their survival and eventually reproduction, by which favourable traits are passed to the next generation) ○ SMALLER → gene pools tend to have less genes and thus less genetic variety between individuals of the population ■ A population with lesser genes is poorly prepared to cope with sudden environmental change and are more susceptible to the risk of endangerment or extinction. ★ Allele Frequency • Calculated by converting the number of each allele into a percentage
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When the allele frequencies change, we know the population has evolved (microevolution)

- Factors that cause allele frequencies to change
 - Mutations** → occur during meiosis, affect gamete cells and are the primary source of inheritable genetic variation as they lead to the formation of new alleles
 - Gene flow** → the movement of alleles from one population to another changes how frequency existing alleles occur by introducing previously unseen (new) genes/alleles to the gene pool of one population from that of another
 - Genetic Drift** → when allele frequencies in a population change over generations: changes that take place because of random fluctuations of gene frequencies as a result a result of reduced/small population size

★ Mutations and its affect on the gene pool

- Likely to introduce or remove alleles from a population this eliminating a number of traits from the existing gene pool and resulting in an increase or decrease to the genetic variation present in the gene pool of the population
 - It is more likely that mutations have a negative impact than a positive one.

Possible outcomes of a mutation introducing a new allele:

- + New allele result sin a beneficial phenotype → allele is selected for → allele increases in frequency → more varied gene pool → greater likelihood of species survival
- New allee results in a detrimental phenotype → alleles is selected against → allele decreases in frequency or disappears → less varied gene pool → reduced likelihood of species survival

★ Gene Flow

Immigration

when new individuals enter the population (gain)

VS

Emmigration

when individuals leave the population (loss)

+ Increased allele frequency → more varies gene pool → greater likelihood of species survival

- Decreased allele frequency → less varied gene pool → reduced likelihood of species survival

★ Genetic Drift → a change in the frequency of a populations alleles due to random chance, making a particular allele more or less common in the population as a result

- Bottleneck effect → an abrupt reduction in number of individuals in population that causes drastic loss of diversity within the gene pool
 - Random organisms are chosen to survive at the occurrence of a natural disaster or wipeout event
 - E.g. habitat destroyed due to a fire
- Founder effect → a small number of individuals separate from larger population, establishing a new population with a loss genetic diversity
 - Random sample survives and gives rise to a new population
 - By chance, the new population will have different allele frequencies compared to the old population and lower genetic variation amidst the gene pool
 - Similarly, the original population will experience a decline in the genetic variation (a loss of allele frequency) due to the isolation.migration of a portion of its population that has formed the new, removed population.

6.2 Biotechnology

IQ: How do genetic techniques affect Earth's biodiversity?

6.2.1 investigate the uses and applications of biotechnology (past, present and future), including: (ACSBLO87)

- analysing the social implications and ethical uses of biotechnology, including plant and animal examples
- researching future directions of the use of biotechnology
- evaluating the potential benefits for society of research using genetic technologies
- evaluating the changes to the Earth's biodiversity due to genetic techniques

Biotechnology	The use of biological organisms or enzymes in the synthesis, breakdown or transformation of materials
Gene technology	A modern branch of biotechnology that involves the direct modification or removal of a gene, or the transfer of a gene from one species to another - When altered via the use of gene technology, an organism (plant or animal) is referred to as a GMO: genetically modified organism
Biodiversity	The variety of plant and animal life within an ecosystem

 USES of biotech

- **Agricultural/farming:** manipulation of animal breeding and plants to produce quality, high-yield crops and animal-products (cheese, milk, meat)
- **Medicine & pharmaceuticals:** diagnosis, treatment +/- prevention of disease (w/specific relation to the drugs utilised to do so)
- **Industry:** processing of raw materials and the manufacture of goods (e.g. wine, beef and coffee industries)

 Social Implications (4):

- **Environmental harm:** what impact will the manipulation of biological material have on the environment
- **Bioterrorism:** if the wrong handsome be used against humankind? Could a potentially deadly disease be made?
- **Lab/production safety:** are all the necessary precautions takes when handling genetically modified organism
- **Ethical issues:** are involved organisms (animals) treated fairly? Is humankind playing God? Conflict if animal genes are placed into vegetables?

 Ethical Implications (4) → generally overlap w/ social issues (people ethics are largely influenced by personal circumstances/factors: religion, past, experiences, understanding of scientific processes)

- Ethics = the study of why certain decision are understood to be morally wrong, and the reasoned views behind making these judgements
- Bioethics = using an ethical approach to make decisions about biological issues.

Ethical FRAMEWORKS → are developed by organisations to help ensure ethical considerations are made

- Rights + responsibilities: the rights of one imply the responsibilities (duties) of another to ensure said rights
- Consequentialism: weighing the benefits and risks/harmful results of our actions regarding biotechnology
- Autonomy: debate of the individual's right to choose for themselves, or does one decision count for all?
- Virtue ethics: (virtue = something the community accepts are “good” or “right” - honesty, for instance, or kindness and patience) virtue ethics emphasise decisions that aligns with these commonly accepted characteristics/virtues

 Future directions of biotech

- Medicine: improved detection and treatment of disease
- Agriculture: sustainable food production methods designed to meet demands of an increasing population

Societal Benefits

- + Improved medicine
 - improved/early diagnosis of disease and predisposition to disease = early intervention and prevention = better community health
 - New and improved vaccines
 - Current examples: anticancer drugs are delivered to specific cancer cells (w/out damaging the surrounding healthy cells.)
- + Improve agriculture
 - Quality, high-yielding food = improved nutrition
 - Genetically modified crops possess favourable traits that make them disease resistant and pest tolerant, more efficient and better quality
 - Current example: GM crops have genes that increase vitamin levels to help battle malnutrition

 effect of biotechnologies on Earths Biodiversity

- Biodiversity is generally indicative of an environmental health
- Biologists and agricultural breeders realise that to ensure the long-term survival of species in order to meet the demands of the growing human population, diversity amongst living organisms must be conserved and maintained
- Modern biotechnologies in the hands of humans, holds the potential to alter the paths of evolution of artificially combining the qualities of organisms that were once entirely separate species

- Increasing or conserving biodiversity in the short-term
- Reducing it in the long, as characteristics are reproduced and GM organisms are artificially bred with help of reproductive technology

CHANGES to biodiversity:

Short-term	Long-term
● temporary increase of biodiversity upon the introduction of previously foreign genes to the gene pool of one species from that of another	● genetically modified species develops a resistance to certain environmental factors and survives ● favourable traits are passed through successive generations via the reproduction of the GM organisms ● other variations of the species die out creating a monoculture (single-species culture): lone survival of the resistant plant ● reduced variation within the gene pool of the species

6.3 Genetic Technologies

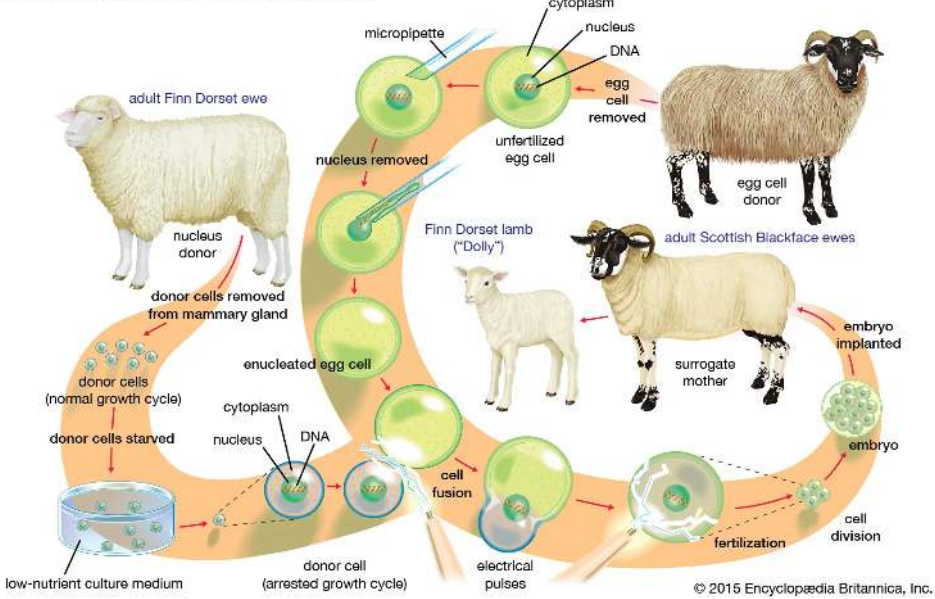
IQ: does artificial manipulation of DNA have the potential to change populations forever?

6.3.1 - investigate the uses and advantages of current genetic technologies that induce genetic change	Gene technologies		Any technology capable of manipulating the genetic sequence of a living organism’s DNA	
	(Current) methods		USES	ADVANTAGES
	Reproductive technologies <i>Any process that aids organisms in reproduction, genetic or artificial transfer or gametes</i> Purpose: Promotes sexual reproduction to: - Increase population size - Promote diversity within the gene pool - Pass on + increase allele frequency of favourable traits	Artificial insemination	- livestock industry (animal production) - fertility treatment (humans)	+ efficient + able to synchronise pregnancies and bypass issues with fertility
		IVF (in vitro fertilisation)	fertility treatment (humans)	+ the freezing of embryos + genetic screening
		Artificial pollination	- crop pollination - genetic experimentation	+ controlled inheritance of favourable traits
	Cloning techniques <i>The production of a , or many replicas of a particular animal, plant or part thereof</i> Purpose: Typically clones will possess desirable traits of importance to humans	Whole organism cloning	livestock industry (production of genetically identical offspring)	+ ensures definite trait inheritance (favourable)
		Therapeutic cloning	medicine (stem cell technologies)	+ stem cells capable of differentiating into any somatic cell
		Gene cloning	medicinal and industrial (large-scale production of important molecules)	+ production of biologically relevant proteins in high demand (insulin, enzymes for industrial purposes)

	Recombinant DNA techniques <i>The process by which genetic engineering is made possible to transplant genes from one organism to another</i>	Transgenesis	● agriculture (development of pest-resistant crops) ● environmental biotechnology (bioremediation)	+ creation of multi-functioning organisms + transfer of favourable traits + use of pesticides reduced + exploitation of biological phenomena
	Purpose: The receiving organism is known as ‘transgenic’ {a synonym for genetically modified}	Gene sequencing	● medicine (development of personalised treatments) ● genetic research (improving understanding of genomes, assisting the identification of newly discovered genes)	+ identification of genetic disorders and associated risk factors + understanding of evolutionary relationships + forensic biology (kinship tests)
		Gene therapy	● medicine (reprogramming of dysfunctional tissues/cells)	+ treatment of disease like cystic fibrosis
6.3.2 - compare the processes and outcomes of reproductive technologies, including but not limited to: a) artificial insemination b) artificial pollination	★Artificial Insemination: <i>the collection, storage and transport of semen from male animals with desirable characteristics so that more females can be impregnated and larger numbers of offspring with the desirable characteristics produced.</i>			
	Pros		Cons	
	● Desired characteristic selected and bred through multiple generations to increase the frequency of favourable alleles ● Can prevent the spread of certain diseases and sterility as a result of genital disease ● Increases rate and likelihood of conception ● Can overcome barriers of male sterility ● Higher productivity - one prizes stud can fertilise: ○ Many more females than under natural conditions ○ Females located distances apart ○ Female animals are able to produce offspring long after the death of the stud ● Variety can be developed by introducing semen of distantly related species ● Can increase population size of endangered species		● Constant use of the same male or female may limit genetic variability within the resultant population ○ decreasing diversity, in turn, increasing the susceptibility of the population to endangerment/extinction in the face of environmental change ○ the use of a small number of stud males in a breeding program means that the genetic variation is reduced compared with the situation that would exist if random mating occurred ○ because stud animals are chosen for their superiority in a limited number of inherited traits, several consequences from the widespread use of a few stud animals may result including: ■ specific alleles of a few selected genes will become predominant in the herd and alternative alleles of the genes concerned will be lost. ■ potentially valuable alleles may be unwillingly lost from the genetic composition of the herd because other inherited features are ignored	
	★Artificial Pollination → <i>pollen is collected from the anther of a male plant and brushed directly onto the ripe stigma of a female, covered, then, to prevent cross pollination and fostered until grown</i>			
	Pros		Cons	
● The development of desired traits through the production of hybrid plants ○ E.g increases growth rate, increased uniformity of phenotype ● Potential to increase the variety of plants ● There will be a consistency in offspring ● Solution to the declining bee population ● Useful in conditions where natural carriers of pollen grains such as insects are fewer in number.		● Influence genetic diversity of the plant species ● There is chance of a possible loss of variation within the plant species ○ Leading to monoculture → there is only 1 type of crop being grown. ● If the plant become particularly susceptible to a disease, there is an increase to the endangerment - extinction of the plant species ○ Like animals they will lose certain genetics that allow to fight away these diseases ● results are unexpected ● Time-consuming		

	● Infertility of cross-species hybrids				
6.3.3 - investigate and assess the effectiveness of cloning, including but not limited to: a) whole organism cloning b) gene cloning	<table><tr><th>Term</th><th>Definition</th></tr><tr><td>Cloning</td><td>Biotechnology/ a method used to create an identical replica of a chosen plant or animal, that presents desirable characteristics/purposes that render them important for human needs.</td></tr></table> ★ Whole Organism Cloning (2 methods) ● Artificial Embryo Twinning ○ relatively low-tech to make clones -mimics the natural process that creates identical twins ○ Carried out in a Petri dish instead of inside the mother. ○ A very early embryo is separated into two individuals cells, which are allowed to divide and develop for a short time in the Petri dish ○ The embryos are placed into a surrogate mother, where they finish developing. ○ Since the embryos all came from the same egg they are all genetically identical ● Somatic Cell Nuclear Transfer (SCNT) ○ Produces the same result as artificial embryo twinning ○ Somatic cell: in any cell in the body other than sperm and egg (reproductive cells) ○ Nuclear : holds the DNA, divided into packages called chromosomes, and it contains all the information needed to form an organism. ○ Transfer: isolate a somatic cell from an adult female, removed the nucleus and all of its DNA from an egg cell. Then transferred into the nucleus from the somatic to the egg cell. ■ Chemicals are added and they start to act like a freshly fertilised egg. Which is them implanted into a surrogate mother. ○ In SCNT the egg cell's single set of chromosomes is removed. It is replaced by the nucleus from a somatic cell, which already contains two complete sets of chromosomes. ■ And so the resulting embryo has both sets of chromosomes from the somatic cell. 🔴 Method to cloning organisms 1. an empty egg cell (it's haploid nucleus destroyed) is obtained from a female organism [enucleation] 2. the somatic cell of a different organism of the same species is selected: containing the genome of the organism which scientists wishes to replicate/clone 3. the (diploid) nucleus of said cell is removed and fused with the empty egg cell via injection 4. the resulting egg cell contains a diploid number of chromosomes and is genetically identical to the somatic cell donor 5. an electric shock is used to stimulate cell division of the egg cell such that it develops into an embryo 6. the embryo is implanted into a surrogate mother organism, whereby the surrogate mother will give birth to the offspring that is genetically identical to the organism that contributed the DNA-containing somatic cell	Term	Definition	Cloning	Biotechnology/ a method used to create an identical replica of a chosen plant or animal, that presents desirable characteristics/purposes that render them important for human needs.
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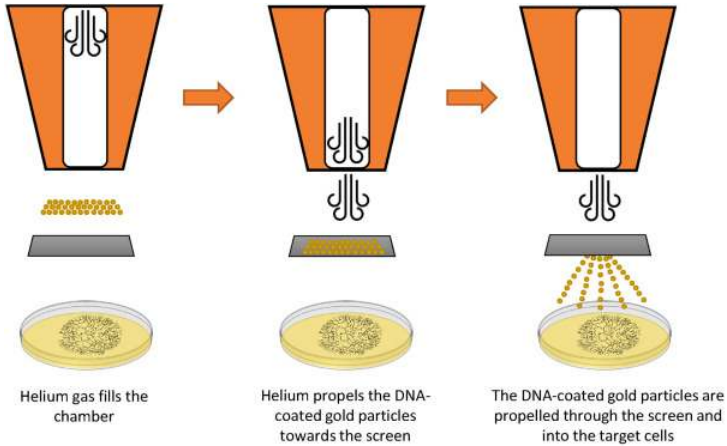
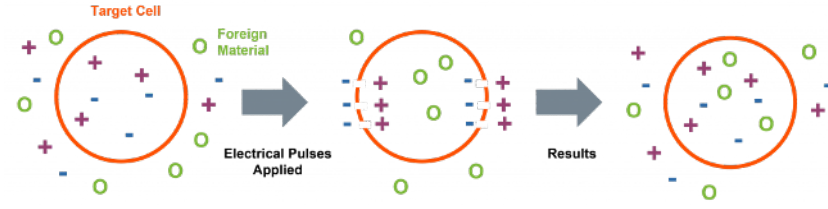
Dolly: The Cloning of a Sheep, 1996



Pros	Cons
- Allow the relocation of organisms that have favourable characteristics at a large scale. Could provide higher yield of products and higher quality of products obtained - E.g merino sheep can be cloned as they produce high quality wool, prized cow that make better quality milk, prized pigs that can produce better quality meat - May have benefits in preserving endangered species	- Produces offspring that are genetically identical to the organisms that donate the somatic cell with DNA. lowers the genetic diversity and variation. If performed on a mass scale, it could lead to mass decline in species population and also major loss to farmers that have many clones animals - Surrogate mothers suffer increased health risks - Any mutation that may occur in the DNA contained in the original somatic cell is transferred into the cloned offspring. - Ethical issues such as religious issues concerning that humans should not interrupt with evolution by playing god - Case studies involving clones that die earlier than expected - Expensive - High pregnancy losses and high morbidity and mortality

★**Gene Cloning**

- Gene cloning is the copying of a particular gene.
 - The gene is cut from one organism using a restriction enzyme and inserted into a bacterial plasmid.
 - The plasmid is inserted back into the bacteria.
 - As the bacteria reproduces it clones the gene that has been inserted
 - Once multiple copies of the gene have been made, the gen can be inserted into an egg cell of another species, and after fertilisation and becomes a part of the newly formed organisms DNA.
- An efficient way to produce hormones and enzymes found in living organisms in vitro as they are being used for promoting certain mechanisms in industrial processes as well.
 - E.g insulin which is a hormone used to control blood sugar levels and is VITAL for aiding patients with diabetes → the industrial production of insulin has made it more accessible for patients
- **There are 4 way to administer a cloned gene**

	Micro injection	DNA is directly injected into the nucleus of a single cell - usually done under an optical microscope to introduce DNA into eggs cells when creating transgenic species	
	Biolistics	mechanically delivering DNA on microscopic particles into target tissues and cells by 'firing' them from a gene 'gun'.	 Helium gas fills the chamber Helium propels the DNA-coated gold particles towards the screen The DNA-coated gold particles are propelled through the screen and into the target cells
	Electroporation	increasing the membrane permeability by applying an electrical current	 Target Cell Foreign Material Electrical Pulses Applied Results
	Transduction by vector	DNA is carried into cells by a viral vector such as an adenovirus, liposome or bacterial plasmid. These vectors may be injected directly into the bloodstream or may be delivered by aerosol delivery	

6.3.3 - describe techniques and applications used in recombinant DNA

Term	Definition
------	------------

technology, for example:
a) the development of transgenic organisms in agricultural and medical applications

Recombinant DNA Technology

Plasmids

The joining together of DNA molecules from two different species

A circular self-replicating form of DNA found in Bacteria

(Restriction) Enzymes

Used as a catalyst to cut a specific DNA sequence. This process is often called digestion
- E.g. like scissors

Transgenic species

One whose genome has been deliberately altered by the transfer of a gene or genes from another species or breed

★ Recombinant DNA Technology

- DNA: molecules formed by laboratory methods of genetic recombination (e.g. molecular cloning) to bring together genetic material from multiple sources, creating sequences that would not otherwise be found in biological organisms
- Taking a piece of one DNA, and combining it with another strand of DNA
- Benefit → creating synthetic drugs, including insulin

🔴 Process

1. Target gene is isolated from the human gene/DNA, whilst the plasmid of a bacterial cell is removed
2. Enzymes are applied
 1. Cuts from enzyme creates sticky ends, the sticky ends of the human gene fit onto the sticky ends of the plasmid
3. Sticky ends come together forming a complete plasmid
 1. Plasmid product is called recombinant DNA
4. Recombinant DNA plasmid is inserted into a bacterial cell
 1. The cell will copy itself many time over and therefore the recombinant DNA is also copied

★ Transgenic Species (transgenesis)

Example → Milk in cows

The scientists have successfully introduced **human genes** into 300 **dairy cows** to produce **milk** with the same properties as **human breast milk**. **Human milk** contains high quantities of key nutrients that can help to boost the immune system of babies and reduce the risk of infections.

The human gene (for milk) is microinjected into the eggs of the cow. This will then be developed within the host cow, leading to the creation of a transgenic cow. This then means that the cow will express the milk that contains the human gene.

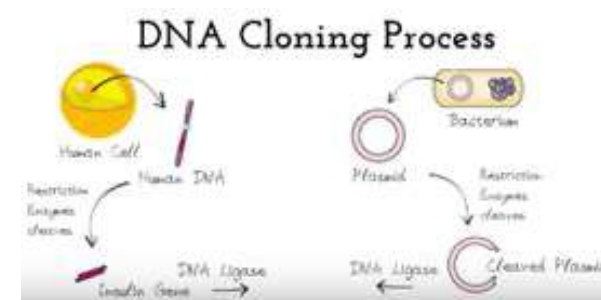
Process

Step 1:

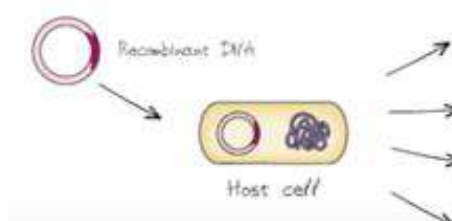
- 2n nuclei from adult transgenic cow cells are injected into the eggs of the egg donor
- These eggs will then be developed within the host cow
- Eventually becoming cloned transgenic cows
 - They will/should also express the same milk as the original transgenic cow

Step 2:

- Human DNA and plasmid are isolated
- Restriction enzymes cleave out the gene that is needed e.g. insulin gene and will also cleave the plasmid leaving sticky ends that the genes will recognise.
- DNA ligase will join the gene and plasmid together to form the new recombinant gene



DNA Cloning Process (cont.)

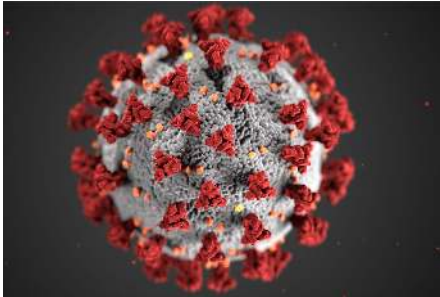
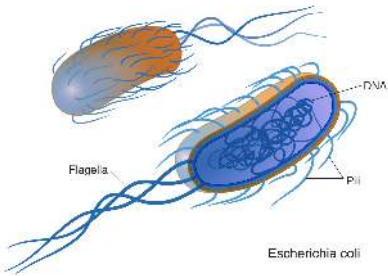


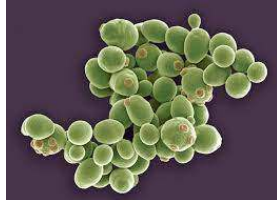
	● Medical uses of Transgenic Organisms ○ Many ethical issues concerning the use of transgenic organisms in medical applications ○ Transgenic organisms also provide us with opportunities to safely test and develop new treatments ○ E.g. transgenic sheep in Australia have been given a gene for the blood-clotting factor that is lacking in people who suffer from haemophilia. The factor can be extracted from the sheep’s milk and used for human treatment ○ Transgenic mice are often used to study a range of diseases including cancer, obesity, heart disease, diabetes, Alzheimer;s disease and Parkinson’s disease. ○ E.g. demand for human organ transplants grows, there is a need to develop non-human organs for transplantation, as the waiting lists for viable organs. Organs from transgenic swine that have human complementary surface markers on their cells would inhibit the activation of organ rejection. This would enable people on long waiting lists to access viable organs.								
6.3.4 - evaluate the benefits of using genetic technologies in agricultural, medical and industrial applications	<table><tr><th>Application</th><th>Benefits</th></tr><tr><td>Agricultural</td><td> ● Higher yield of crops obtained ● Crops are disease resistant and can survive in harsh conditions ● Increases variety within species by different techniques such as mutation breeding, cross breeding, somatic hybridization etc. ● Some varieties of crop have been produced that could directly fix atmospheric nitrogen decreasing the plants dependency on external fertilisers </td></tr><tr><td>Medical</td><td> ● Genetic technologies have enabled the production of vaccines which are currently being used to prevent various infectious diseases. ● Microorganisms and plant-based substances are now being manipulated to produce a large number of useful drugs, enzymes and hormones at low costs. The most prominent example is the production of vast amounts of Insulin, a hormone that plays a role in balancing sugar level in blood. ● Genetic technologies are now being applied to produce human growth hormones, follicle stimulating hormones (for treating infertility), human albumin, monoclonal antibodies, antihemophilic factors etc. ● Gene therapy, a process in which defective genes are replaced to treat genetic disorders, is currently one of the major sectors of research in medicine. Clinical research using somatic gene therapy has been conducted with several diseases, including X-linked SCID, chronic lymphocytic leukaemia (CLL), and Parkinson's disease. In 2012, Alipogene tiparvovec became the first gene therapy treatment to be approved for clinical use. In 2015 a virus was used to insert a healthy gene into the skin cells of a boy suffering from a rare skin disease, epidermolysis bullosa, in order to grow, and then graft healthy skin onto 80 percent of the boy's body which was affected by the illness. If successful, a wide range of diseases can be cured with gene therapy. </td></tr><tr><td>Industrial</td><td> ● Some enzymes and proteins are required to carry out fermentations on a large scale in industries. Genes for such enzymes and proteins can be introduced in an organism using genetic technologies and can be multiplied in larger numbers in a bioreactor to produce the moderate amount of that substance for industrial fermentation. ● These techniques are used to produce medicines such as insulin, human growth hormone, and vaccines, supplements such as tryptophan, aid in the production of food (chymosin in cheese making) and fuels. ● Other applications with genetically engineered bacteria could involve making them perform tasks outside their natural cycle, such as making biofuels, cleaning up oil spills, carbon and other toxic waste and detecting arsenic in drinking water. ● Certain genetically modified microbes can also be used in biomining and bioremediation, due to their ability to extract heavy metals from their environment and incorporate them into compounds that are more easily recoverable. </td></tr></table>	Application	Benefits	Agricultural	● Higher yield of crops obtained ● Crops are disease resistant and can survive in harsh conditions ● Increases variety within species by different techniques such as mutation breeding, cross breeding, somatic hybridization etc. ● Some varieties of crop have been produced that could directly fix atmospheric nitrogen decreasing the plants dependency on external fertilisers	Medical	● Genetic technologies have enabled the production of vaccines which are currently being used to prevent various infectious diseases. ● Microorganisms and plant-based substances are now being manipulated to produce a large number of useful drugs, enzymes and hormones at low costs. 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6.3.5 - evaluate the effect on biodiversity of using biotechnology in agriculture	● Herbicide resistant crops produced using recombinant DNA technology promotes the use of herbicides in plant fields because these modified plants are not affected by them. ● These herbicides may not cause harm to the modified plants but can kill native plants that feed animals and sicken amphibians directly, causing a decrease in biodiversity. ● Due to the use of different breeding methods, wild type agricultural crops may cross with the genetically modified ones. ● The engineered organisms often dominate, resulting in only a modified species over several generations, reducing the diversity that is available. In other words, due to the extensive number of modified traits, the possibility of losing the genuine, wild type plants may decrease. ● There is also concern with the nutritional value of genetically modified crops and also whether the crops are harmless to intake in dietary form. The cause of it would be because some traits in plants are taken from microorganisms. An example would be Bt brinjals and tomatoes whose insect/disease resistance comes from a gene from the bacteria Bacillus thuringiensis. Thus whether a population of crop having a bacterial gene in them is suitable for human consumption or not has raised a couple of eyebrow								
6.3.6 - interpret a range of secondary sources to assess the influence of social, economic and cultural	● Harm to the Environment: - Whether a GMO (genetically modified organism) may or may not cause harm to the environment and its adaptability with the changing climatic conditions still cannot be predicted. In some cases, the effect of the existence of a GMO on other organisms raises questions too. For example, A strain of corn has been created with a gene that encodes a natural pesticide. On the positive side, the transgenic corn is not eaten by insects, so there is more corn for people to eat. The corn also doesn’t need to be sprayed with chemical pesticides, which can harm people and other living things. On								

contexts on a range of biotechnologies	the negative side, the transgenic corn has been shown to cross-pollinate nearby milkweed plants. Offspring of the cross-pollinated milkweed plants are now known to be toxic to monarch butterfly caterpillars that depend on them for food. Scientists are concerned that this may threaten the monarch species as well as other species that normally eat monarchs. • Bioterrorism: - Governments are worried that terrorists will use biotechnology to create new Superbugs, infectious viruses, or toxins, for which we have no cure. • Laboratory/production safety: - It's hard to protect oneself if you don't know what you're working with. Some new technologies, usually non biologicals such as nanoparticles, make commercial production lines before they have been sufficiently tested for safety. There is also concern about technician safety in laboratories, even under secured conditions, when working with organisms of unknown virulence. • Protecting Human Subjects in Clinical Trials: - At times, human trials don't work the way they are planned which may cause the subjects on which the products / technologies are used are harmed and in severe cases, these tests and trials turn out to be lethal. - This issue has generated considerable debate since 1999, when 18-year-old Jesse Gelsinger died while participating in a gene therapy trial at the University of Pennsylvania. • Affordability: - A wide number of clinical biotech products and treatments cannot be afforded by the mass public.
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Mod 7: Infectious Disease

7.1 Causes of infectious disease		
IQ: How are diseases transmitted?		
Syllabus Outcomes	Summary of information	
7.1.1 describe a variety of infectious diseases caused by pathogens, including microorganisms, microorganisms and non-cellular pathogens, and collect primary and secondary-sourced data and information relating to disease transmission, including: A. classifying different pathogens that cause disease in plants and animals (ACSBL117) B. investigating the transmission of a disease during an epidemic C. design and conduct a practical investigation relating to the microbial testing of water or food	<u>Term</u>	<u>Definition</u>
	Disease	Any process or condition that adversely affects the normal functioning of a living thing or parts of a living thing
	Infectious Disease	Caused by another organism or an infective agent known as a pathogen
	Pathogen	Organism capable of causing disease
	Virulence	The measure of the pathogenicity of an organism; virulence is related to a number of variables, including the number of the particular pathogen needed to cause the disease
	Microbes	Microorganisms
	Parasitic	Living in or on another organism and obtaining nutrients from that organism
	Epidemic	An outbreak or unusually high occurrence of a disease in a population or region
	Pandemic	An outbreak of a disease over a while county or on a global scale
	★ <u>Classifying different pathogens that cause disease in plants and animals</u> ● Pathogens can be categorised into 6 (7) types and sub categorised into either - microscopic or macroscopic	
Types of Pathogens	Description	Examples

samples D. investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission	Prions Non-cellular infectious proteins, which are abnormally folded versions of a protein needed within an organism. 	● Key disease mechanism ○ Prions can lead to serious problems depending on the structure, function and location of the normal protein ○ Prions usually affect brain and neural tissue, which leads to neurodegeneration ● Prion proteins are most abundant within the brain ○ Prions in the blood interact with the normal version of the protein and pass on the deformation. ○ The prions build up in the brain - leading to a brain disease ○ In this case it can lead to rare neurodegenerative disorders. ● Can be found in both animals and humans	● Animal prion diseases ○ Chronic wasting disease (deer) ○ Scrapie (sheep) ○ Transmissible mink encephalopathy ● Human Prion Diseases ○ Kuru ○ Creutzfeldt-Jakob disease (CJD) - hereditary
	Virus Non-cellular entities of a single type of nucleic acid (DNA or RNA) encased in a protein coat (capsid) 	● Key disease mechanism ○ Needs a host cell to reproduce ○ The virus attaches to a host cell, which it then penetrates ○ It hijacks the host's enzymes and nutrients to make its own viral proteins and nucleic acid ○ These viral materials assemble into new viruses which are released from the host cell and infect other cells ● Size ranges from 20-40 nanometers ● On its own it can't do a lot of harm, though once it enters a living organism and starts to replicate itself it can start to cause disease/illness ● They are able to mutate and evolve	● Rotavirus ○ Causes diarrhoea and gastroenteritis, mainly in children between six months and two years old ○ <u>Direct transmission</u> → person to person contact, particularly with someone who hasn't washed their hands ● Rhinovirus ○ Cold ○ <u>Direct</u> → transmitted through direct contact (shaking hands - used their hand to wipe their dripping nose) ● Strains of the Ebola virus ○ EVD is a serious often fatal; previously called Ebola haemorrhagic fever ○ <u>Direct</u> → contact with blood, body fluids and tissues of infected animals or humans; transmission through sexual contact may occur up to seven weeks after clinical recovery
	Bacteria Prokaryotic, unicellular organisms, which don't have membrane bound organelles and are each made of a cell 	● Key disease mechanism ○ Pathogenic bacteria can cause disease, usually by releasing toxin or damaging host tissues ○ Can cause disease by: ■ Chemical damage (releasing toxins) ■ Physical damage (invading cells) ● Not all bacteria cause disease and many are beneficial ○ Only about 1% of known bacteria are disease-causing microbes ● 1000 nanometers in size ● Bacteria are classified according to their shape ○ Spherical (cocci) ○ Rod (bacilli) ○ Spiral (spirilla) ○ Comma (vibrios) ○ Corkscrew (spirochaetes)	● Vibrio Cholerae ○ Causes cholera ○ <u>Indirect</u> → contaminated water ● Chlamydia trachomatis ○ Chlamydia ○ <u>Direct</u> → via sexual contact ● Yersinia pestis ○ Bubonic plague ○ <u>Vector</u> → transmitted through fleas
	Protozoan	● Key disease mechanism	● Plasmodium falciparum: a sporozoan protozoa

eukaryotic, unicellular microorganisms without a cell wall 	○ None ● Many protozoans are free living and do not cause disease	○ Causes malaria ○ Vector: transmitted from person to person by Anopheles species of female mosquito carrying the protozoa - kills more than 400 000 people a year worldwide
Fungi eukaryotic, non-photosynthetic organism with a cell wall 	● Key disease mechanism ○ The damage caused by fungi mostly arises from the enzymes they produce ■ They can secrete digestive enzymes and chemicals into their surrounding in order to break down organic matter ■ They can absorb the simple nutrients ● Includes: ○ Yeasts ○ Moulds ○ Crop-destroying rusts and smuts ● Like bacteria, many fungi are beneficial and play a key role as decomposers in ecosystems and in the fermentation process.	● Candida albicans ○ Vaginal thrush and mucous membrane infection ● Cryptococcus gatti ○ Meningitis ● Trichophyton ○ Athletes foot
Macroparasites Multicellular pathogens which can be seen with the naked eye 	● Key disease mechanism ○ They tend to cause disease by competing for nutrients from the host ○ They can also produce toxin and damage host tissue ● Includes: ○ Roundworms ○ Flatworms ○ Flukes ○ Ticks ○ Lice and fleas	● Tick paralysis in dogs ○ Certain species of female tick infests the skin of the dog and feeds on its blood, released toxins through its saliva ● Body lice ○ Head lice feed on human blood and cause infection called pediculosis ● Termtods (liver fluke) ○ Causes inflammatory disease of the bile ducts; sheep can cattle are most commonly infected with liver flukes but humans can also be infected

★**Modes of transmission - infectious disease**

↳ refers to the movement or the transmission of pathogens from a reservoir to a susceptible host.

- Once the pathogen has exited the reservoir, it needs a mode of transmission to the host through a portal of entry.
- Can either be by direct or indirect contact or through vector

Direct contact When there is contact between people (through touch, saliva, mucus, contaminated blood or any other bodily fluids	Indirect contact Involves contact with a surface or substance that is contaminated with pathogens.	Vector transmission An organism that transmits a disease causing pathogen e.g. mosquitoes
● Direct droplet transmission ○ Occurs when an individual coughs or sneezes and small droplets of mucus, which may contain pathogens, are ejected as a fine spray ● Person to person contact ○ Infected individual touches or exchanges body fluids with another individuals.	● Airborne Transmission ○ The transmission of droplets containing pathogens over distances greater than one metre. ● Contaminated objects ○ Surface or object containing the pathogen is transferred to the mouth, or nose or eyes before washing the hands	● A vector is an organism that is not the pathogen but can spread the disease from one host to another (carrier) (fleas, ticks etc, they carry the disease (they aren't affected) and spread the pathogen between hosts)

- E.g kissing, sexual contact, biting, oral secretions or body lesions
- **Food and drinking water**
 - Several pathogens are transmitted as a result of improper treatment, handling or storage of food and water.

★**Investigating the transmission of a disease during an epidemic**

- **Epidemic**
 - Occur when the number of people affected by a particular disease is higher than usual
 - Rapid spread of (usually) an infectious disease
 - Spread in a short period of time
 - Endemic level, level of a particular disease is steady and its occurrence is not increasing (or decreasing) within population.
 - Can also include non-infectious disease e.g. obesity, drug addiction
 - An epidemic that spreads to an entire nation or other countries are classified as a pandemic

Microbial testing of water and food supplies

- Completed as prac

CASE STUDY: CHOLERA

- Caused by vibrio bacteria, direct life cycle (no vectors)
- Recent outbreak in Nigeria (September 2018)
- Transmission
 - Ingested through contaminated food or water □ bacteria infects bowel of human □ excreted in feces which can further spread disease
 - Developing countries at higher risk due to lack of infrastructure, insufficient education, poor sanitary conditions, and lack of drinking water
- Major symptoms
 - Cramps and diarrhea
 - Vomiting
 - Dehydration
- Treatment
 - Fluids and salts replaced (oral rehydration solution)
 - Antibiotics (in very early stages)
- Prevention
 - Vaccines and education
 - Avoidance of consuming contaminated water/food (purification)
- Control
 - Effective sewerage systems and proper infrastructure
 - Chlorination of water

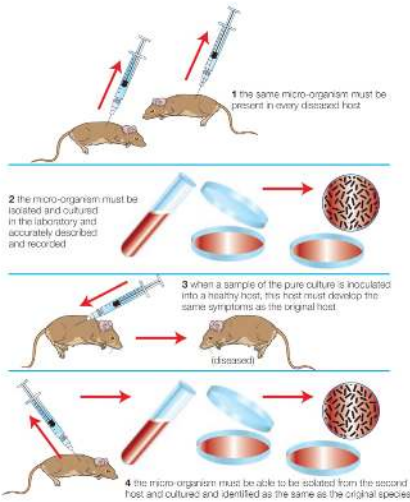
7.1.2 investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including:

1. Koch’s postulates
2. Pasteur’s experiments on microbial contamination

Term	Definitions
Generate Spontaneously	(spontaneous generation) an obsolete idea that certain life forms could arise from dust or dead tissue
Contamination	In microbiology, the unintended or accidental introduction of infectious material or pathogens, commonly occurring through poor hygiene

★**Koch’s postulates**

- **Contribution**
 - Koch’s work gave a system of producing pure culture of disease microbes and a procedure for the identification of the pathogen that was causing disease.
 - Discovered that diseases could be caused from water and set up methods for ensuring safe water supplies.
 - His work has led to the modern practice of epidemiology where the prevention and control of disease over a population is carried out.
 - Koch introduced/developed the use of agar plates into the field of bacteriology - has allowed a more closer and detailed understanding of the growth and types of bacteria. He established a series of steps known as Koch's postulates which are still followed today to link a particular microorganism to a particular disease.
- **Germ theory of disease**
 - He had carried out a study of the anthrax bacillus, in doing so by examining the blood type of sheep he identified rod-shaped bacteria. In collecting this bacteria and cultivating it in healthy sheep he found that it would kill them. This experiment added weight to the germ theory of disease as he found that microorganisms that grew out of the body would be able to kill another
 - Was able to identify the specific microorganism that causes a particular disease postulates
- **There are 4 postulates** that are followed to determine the particular microorganism that causes a specific disease
 1. The same microorganism must be present in every diseased host.
 2. The micro-organism must be isolated and cultured in the laboratory and accurately described and recorded.
 3. When a sample of the pure culture is inoculated into a healthy host, this host must develop the same symptoms as the original host.



4. The micro-organism must be able to be isolated from the second host and cultured and identified as the same as the original species.

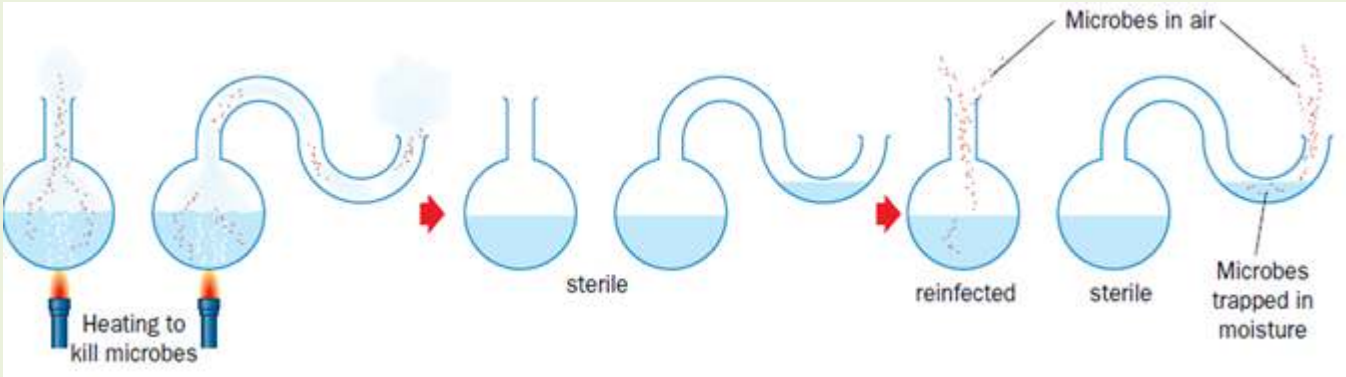
★ **Pastuer's experiments on microbial contamination**

- **Contribution**
 - Before the discovery of lenses and microscopes it was believed that all living matter, was/could generate spontaneously - from non-living matter
 - Through both pastures and Koch's experiments they had provided evidence that many diseases are caused by microorganisms.
 - Pastuer discovered that microbes such as bacteria and mould can cause contamination and disease.
- **Famous Swan-necked flask experiment**
 - Through this experiment he discovered that microbes were responsible for the souring of alcohol and developed the process called pasteurisation
 - Led to the development of vaccinations for fowl cholera, anthrax and rabies
- **The Experiment**

Aim:
to model pasteur's experiment to identify role of microbes in contamination

Hypothesis:
The flask that has the S-shaped glass will not show signs of microbial growth. The flask with straight glass tubing that is open to the air will show signs of microbial growth.

- Method:**
1. Add the filtered nutrient broth to each of the flasks until they are approximately one-third full.
 2. Fit the stoppers, one with the straight tubing and the other with the S-shaped tubing, to the flasks.
 3. Heat each flask so that it boils gently for 15 minutes. Ensure that after boiling there is a small amount of water trapped in the S-bend.
 4. Leave both flasks in a warm position out of direct sunlight for several weeks.
 5. Every 2 or 3 days, observe the contents of each flask. Look for cloudiness, scum, bubbles and fungal colonies. Record any changes



Results:
The liquid in the container opened directly to airborne microorganisms went cloudy, developed scum, bubbles and produced unpleasant odours; that is, became contaminated. The container with the S-shaped tubing closed to the air remained unchanged; that is, not contaminated.

Conclusion:
Microbes cannot grow spontaneously. Contamination relies on the entry of microbes to the medium .

7.1.3 assess the causes and effects of diseases on agricultural production, including but not limited to:

1. plant diseases
2. animal diseases

Term	Definition
Agriculture	Agriculture is a form of primary industry that involves the cultivation of crops and pastures and the rearing of animals to provide meat, milk, fibres and other products for humans.
Biosecurity	all measures to prevent the introduction of pathogens (bio-exclusion) and reduce the spread of pathogens (bio-containment)

- **Importance of Agriculture in Australia**
- Due to Australia's unique position of being isolated from the rest of the world, our agriculture is relatively free from infectious diseases that affect our animals and plants compared to other countries.

	- This makes out agricultural product highly sought after around the world - Stringent Biosecurity measures are in place to reduce the likelihood of disease transmission from pathogens. - The introduction of new plant diseases could potentially devastate the horticulture industry, as well as forestry and agriculture. ★Plant diseases (agricultural) - If there is a new virus, fungus or other pathogen that enters,, it can potentially reduce the variation, meaning that crops have a less chance of survival because there is most likely not a variant in the crop to survive and reproduce. This can mean that whole crops will be wiped out, destroying the agricultural industry. Studied Plant Disease Panama Disease - Bananas Cause Fusarium Oxysporum Effects on the Banana This pathogen causes fungus to grow within the roots of the banana. It chokes the banana water supply (cutting it off) and killing the plant. The pathogen originates and starts within the solid of the plant and makes its way up into the plant Effects on agricultural production - concern for the farmers The panama disease is a major threat (makes a significant impact) on the banana agricultural industry, with a high risk of causing extinction to the cavendish bananas. Bananas are all produced through the process of asexual reproduction, they are all clones of the same type of banana known as the cavendish. This means that all bananas essentially have the same exact DNA and yielding clones reduces the amount of variation within its population, reducing biodiversity and its inefficiency to evolve resistance through natural selection against the Panama disease. This eventually leads to the entire crop farm of the cavendish bananas to be eradicated and all die, reducing the overall yield of bananas. For farmers this is a significant concern as they need to grow these surplus amounts in order to export them around the globe. ★Animal Diseases Studied Animal Disease Sheep Lice Cause Insect → Bovicola Ovis When environmental conditions are favourable female lice lay two eggs every three days. The eggs are white, microscopic and are attached to the wool fibre. Immature lice hatch from the eggs after about 10 days at which they are much smaller than the adults. Effects on the Sheep Lice feeds on the dead skin, secretions and bacteria normally found at the surface of the skin. They do not suck on the blood. They do not eat the wool fibres but they do cause the thickening of the skin. The presence of the lice irritates the sheep, causing them to bite, scratch and rub the trees and fences. This damages the wool, causing breakage and coting of the fibres. Sheep infested with lice have a ragged appearance - often with tags of wool hanging from the fleece. Effects on agricultural production - concern for farmers Sheep lice is seen to make a significant impact on the agriculture industry. Australia is one of the largest wool exporting countries, producing \$658.4 million in 2021 for the exportation of the product. The lice forces farmers (in NSW) to lose over \$100 million per year in lost production and treatment costs. The farmers are unable to export and sell sheep wool due to its diminished qualities as sheep start biting and scratching, irritated by the lice. Infected sheep produce 10% less wool, as it becomes more felted or matte - making it 10% less valuable. Infected sheep are stressed through the discomfort of irritated skin, causes rubbing, affecting the feed intake (leading to malnourishment) , and producing cockle damaged hides; causing them to be more susceptible to flystrike.
7.1.4 compare the adaptations of different pathogens that facilitate their entry into and transmission	★Adaptations of different pathogens that facilitate their entry into hosts - For a pathogen to successfully affect the host, the following must occur: - Entry into host (by passing first line of defence, e.g. nose hairs) - Adhesion to host cell (some also need to enter cells e.g. virus)

between hosts

3. Resist/not stimulate secondary host mechanisms (white blood cells)

4. Survive to reproduce, colonise other tissues, and multiply (takes control of cell replication processes)

5. Damage to host in order to leave the host and be transmitted to a new host

• Adaptations of pathogens causes **virulence**

• Pathogens have developed an array of strategies or adaptation to enable them to gain entry into and persist in their host

- It is though that the evolutionary strategies (through natural selection) of pathogens are just slightly ahead of the development of host resistance strategies

• Pathogens possess many adaptations to penetrate the defence barriers - enhances their transmission

Mode of transmission	Pathogenic adaptations	Example
Airborne	Mainly viruses due to small size Suspended in air for long periods (small size) Resistant to dehydration Causes sneezing/coughing: increases transmission Aerotolerant (survive range of oxygen concentrations)	Influenza
Waterborne	Mostly bacteria Modified outer surfaces (flagella) enable movement (different to fimbria and pili) Halotolerant (survive in saline environments – produce synthesis chemicals balancing osmotic effect) Temperature resistant (can't be boiled off or destroyed by fever)	Giardia Campylobacter
Venereal	Capable of uterine invasion	Chlamydia HIV/AIDS Gonorrhoea
Vector-borne	Life cycle is synchronised with host Surface pathogens allow attachment of pathogen to vector tissue Form reservoirs in digestive tract or salivary glands	Rickettsia felis Malaria Zika
Faeco-oral	Stable in hostile environments (e.g. stomach acid, low-oxygen intestines) Antimicrobial resistance genes Induce vomiting and diarrhoea, increasing transmission	E. coli Salmonella
Soil-borne	Form endospores to lie dormant, activate in favourable conditions Stable in variety of conditions	Clostridium tetani Pathogenic fungi
Vertical	Capable of movement across placenta (unknown transmission method)	Parvovirus B19 Rubella

Pathogen	Adaption
Bacteria	Adhesins (cell-surface components – fimbria, pili) facilitate adhesion to other cells Long pili enable DNA exchange between bacteria Protective capsule prevents macrophage (white blood cell) invasion Resistance to washing effects of secretions e.g. mucus, urine Produce enzymes (collagenase, lecithinase) to dissolve cell walls
Viruses	Surface proteins connect to white blood cell receptor sites, immobilising immune system Enter cell nucleus to replace host genome with viral DNA (e.g. polio) Invades using receptor-mediated endocytosis, binding with cell surface to be engulfed
Prions	Misfolded proteins produced by host body, change neighbouring proteins to prions White blood cells produce tumour necrosis factor, helping prions invade dendrites
Protozoans	No common features (individual examples) Toxoplasma Gondii causes toxoplasmosis, transmitted through cat faeces and contaminated food Possess microtubules, form structures enabling invasion of host cells
Fungi	Cell wall contains mannoproteins, shields from immune system detection Thermotolerance (production of heat shock proteins) resists heat i.e. fever Capsule protects from being engulfed by macrophage Produce hydrolytic enzymes to damage cell walls
Microparasites	No common features (individual examples) Hookworm secrete proteins reducing immune system response, teeth anchor worm to gut lining Ticks use vasoconstriction inhibitors to prevent blood clotting (same as leeches)

7.2 Responses to Pathogens

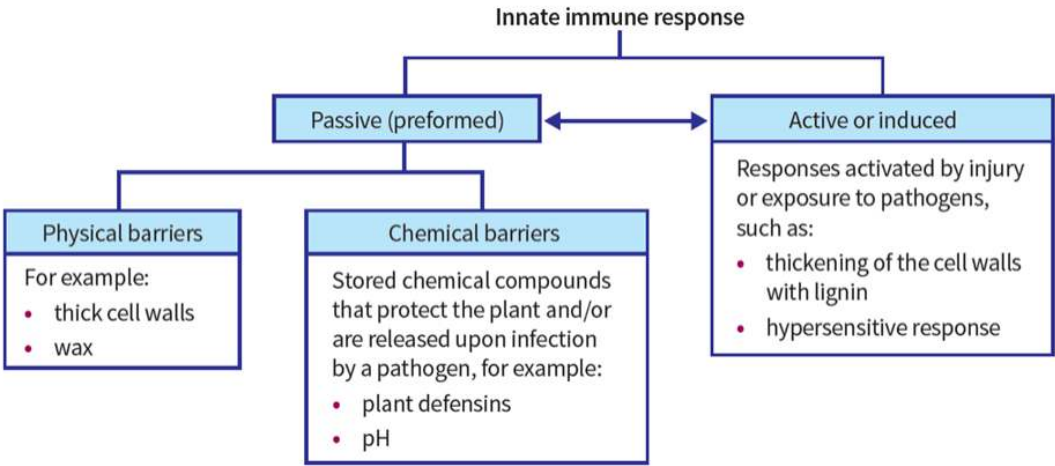
IQ: How does a plant or animal respond to infection?

7.2.1 investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-sourced investigation, for example: a. fungal pathogens b. viral pathogens	★The response of a named Australian plant - Plants can only rely on their innate immune responses. - They do not have an adaptive response system <u>General defences and responses of plants to pathogens</u> Passive defence performed defence mechanisms are the physical and chemical defences that protect against the entry of pathogens Physical - Properties of the plant surface - Waxy leaf cuticles, and thickened, tough cell walls, - Bark: protects pathogens from invading food source, sap, in the phloem beneath the bark tree - Vertically hanging leaves prevent the formation of moisture in the leaf which provides some protection from water moulds
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Chemical	- Antimicrobial chemicals - E.g the young fruit of mangoes and avocados have an antifungal and antimicrobial compounds, make the unripe fruit less susceptible to disease - Naturally occurring chemical inhibitors (pathogen development) - E.g. marigold secrete chemicals into the soil around their roots which are toxic to nematodes and provides protection for other plants growing in the same root zone - Phenolic compounds → compounds containing phenolic acid; compounds are used in treatment of cancers and are thought to have antioxidant and anti-inflammatory properties - The chemical discourages herbivores and inhibit some plant microorganisms + pathogens - Peptides against fungi, bacteria, insects and insect transmitted viruses these are called defensins - Defensins → antimicrobial or antifungal plant peptides
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- When the plants protective barriers ARE breached, the invading pathogen encounters the active line of defence in the plant

Active defence	
Rapid active response	- Plant recognitions → change in permeability of cell wall - Allows movement of ions & activates certain genes - Release hydrogen peroxide in an oxidative burst → kill microbes directly - Reinforcement of cell wall w/ aggregates of cytoplasm → cell wall apposition - Apoptosis: cluster of dead plant cells to accumulate around and isolate pathogens, followed by secretion of antimicrobial compounds.
Delayed active response	- Lysozyme-like chemicals released - Salicylic acid act as signals agent of subsequent infections and play a role in the plant's 'memory' of a particular pathogen → system acquired resistance



★Fungal and Viral Pathogens

Pathogens a risk to the Australian Eucalypt - Phytophthora Cinnamomi (P. Cinnamomi)

What is it?

- It is a soil borne water mould that produces a disease called root rot or die back.
- It is one of the worlds worst most invasive species, affecting native forests, agricultures etc
- In australia the pathogen has caused extensive damage to native ecosystems e.g. the Eucalyptus marginata (jarrah) forests WA

Plant response

- Mass production of defence proteins e.g Phytoalexins synthesises cells and accumulate rapidly at the site of infection
- Activation defence pathways that involve the production of defence hormones (e.g. salicylic acid)
- Hypersensitive response, cell death, at or near the infected site
- Lignin synthesis to reinforce physical barriers such as cell walls

7.2.2 analyse responses to the presence of pathogens by assessing the physical and chemical changes that occur in the host animals cells and tissues

★**Animal/Human Response to pathogens**

As a pathogen approaches and tries to enter the body, it **encounters each level of protection known as the lines of defence.** The first line of defence consists of physical and chemical barriers to pathogens, to prevent them from entering the body in the first place formed by a range of body tissues. Physical barriers just prevent movement/contact whereas chemical barriers act to interfere with microbes in some way (breaking them down or inactivating them).

★**Innate Immunity → First line of defence**

- *Non-specific:* reacts the same way to ALL entering pathogens
- *Innate:* Always present, and are not produced from prior exposure, no memory

🔴**Physical Barriers**

Skin	A physical Barrier • The skin is the largest organ of the human body and forms an outer barrier that covers the body and prevents penetration by microbes. It is fairly dry, and has a layer of dead cells which helps to prevent the growth of pathogens. Biological Barrier • The skin also contains its own population of harmless bacteria that help to stop invading microbes from multiplying. This is done by competing for food and shelter with the pathogenic microorganism therefore reducing the available requirements resulting in a reduction of number of pathogenic microbes
Mucous Membranes	• The respiratory, digestive, reproductive and urinary tracts are covered with membranes that produce a thick layer mucus which traps the entering pathogens. • The pathogens are held in the mucus until they are removed by processes of coughing and or sneezing. • The mucus can contain an antibody that prevents bacteria and viruses from attaching to the surfaces • The mucus also provides a moist, nutritious layer in which the harmless microbes, live and produce substances that inhibit the growth and entry of pathogens,
Cilia	• Tiny hairs that line the respiratory surfaces of the trachea and bronchial tubes. • Cilia are constantly beating in an upwards direction to move the mucus containing the trapped pathogens toward the throat, where they are removed by coughing, sneezing or swallowing

🔴**Chemical Barriers**

Sebum and sweat	• Waterproof lubricates the skin → hinders bacterial growth • Lysozyme is secreted in perspiration and lyses or breaks down bacterial cell walls
Saliva	• Salivas has a flushing action against microbes as well as chemical activity against them due to antimicrobial molecules contained within it.
Stomach Acid	• Stomach wall secretes hydrochloric acid → highly acidic environment in the stomach ◦ Discourages the growth and survival of microbes
Tears	• The tear film contains the following chemical substances, which have antimicrobial properties: ◦ Lysozyme, lactoferrin, lipcallin ◦ AMP's ◦ IgA ◦ Mucins
Urine	• Antimicrobial peptides secreted by the cells lining the urinary tract to prevent binding of bacteria to epithelial cells and lyse - bacterial cells • Flushing activity that takes place during urination assists in keeping pathogens away from the bladder • As a urine is acidic, this washing of the walls of the ureter and bladder helps kill and hinder the growth of microbes

★**Innate → Second line of defence**

- Also Non-specific and innate
- Activates once pathogens gain entry into the body

Main/key parts of this line of defence

	Inflammatory response	1. Occurs at the site of infection/penetration 2. Wounded cells release chemicals that signal mast cells to release histamine 3. The chemicals will cause the blood vessels to dilate and leak fluid, causing an inflammatory response (redness and hotness) 4. Phagocytes from the capillaries makes its way (rushes) to the penetrates site a. They detect bacteria, foreign substances 5. Phagocytes perform phagocytosis a. They start to engulf the bacteria, breaking it down, reducing the swelling of the penetrated site
	Phagocytosis	1. Bacteria is engulfed - endocytosis 2. Lysosomes provide enzymes to ingest/breakdown the bacteria 3. Exocytosis - removes the broken down bacteria 4. The antigens from the bacteria displayed on the surface of the macrophage a. Indicated to the T and B cells
	The lymph system	• A drainage system • 3 main duties ○ Has white blood vessels called lymphocytes which are used to fight infection ○ Acts as a one-way drainage system - transporting fluid from body tissues into the body circulation ○ Gets rid of waste produced by other cells • Responsible for filtering the lymph fluids and removing pathogens, dead cells and other debris. ○ Occurs in the lymph nodes, which also contain phagocytes that destroy any foreign material that is brought into the lymph node by the tissue fluid
	Cell death (Apoptosis)	• When the body is struggling to fight off an infected area sometimes it will decided to kill off the cells within the vicinity to seal the infection off • This is then followed by a layer of macrophages as well • The pathogens eventually die and are then consumed by the macrophages
	Supportive parts	
	Complement Proteins	• Specifically causes death to pathogenic cells • The cascade is composed of many plasma proteins, synthesised in the liver. The functions of proteins include: ○ Triggering the recruitment of inflammatory cells ○ Selecting pathogens from destruction by other cells by opsonizing (process where foreign particles are marked for phagocytosis), or coating, the surface of the pathogen ○ Form holes in the plasma membrane of the pathogen, resulting in cytolysis of the pathogen cell, causing the death of the pathogen ○ Remove the “corpse” of neutralised antigen-antibody complexes
	Natural killer cells	• Class of non-specific lymphocytes that can target and destroy infected body cells or tumour cells ○ Infected cells release chemicals called interferons, which function to promote the activation of natural killer cells ○ Natural killer cells induce apoptosis (cell death) in the infected cell
	Fever	• Macrophages release chemicals which reset the body’s thermostat in the hypothalamus; the high temperature can kill some pathogens, increase the activity of some lymphocytes and phagocytes and speed up the death of the virus infected cells

7.3 Immunity		
IQ:How does the human immune system respond to exposure to a pathogen?		

7.3.1 investigate and model the innate and adaptive immune systems in the human body

The Immune System: Three Lines of Defense

NON-SPECIFIC DEFENCES (INNATE IMMUNITY)		SPECIFIC DEFENCES (ADAPTIVE IMMUNITY)
First line of defense	Second line of defense	Third line of defense
• Skin • Mucous membranes • Secretions of skin and mucous membranes	• Phagocytic leukocytes • Antimicrobial proteins • Inflammatory response • Fever	• Lymphocytes • Antibodies • Memory cells

★ **Innate (non-specific)**

- This system is non-specific. Therefore its main role is to prevent pathogens from spreading and entering the body.
- Provides a front line of defence by engaging pathogens directly and acting immediately.
- Unaltered in its ability to fight the same or different pathogens time and time again.
- Innate immune system is both the 1st and 2nd line of defence

🎯 **1st line of defence**

- **Includes Physical and Chemical Barriers**
 - **Skin** = forms a tougher outer barrier that covers the body and prevents penetration by microbes
 - **Mucous membrane** = covered with membranes that produce a thick layer of mucus which traps the entering of pathogens
 - **Chemical barrier in the skin** = oil and sweat glands produce antibacterial and antifungal substances that inhibit the growth of pathogens
 - **Cilia** = tiny hairs in the respiratory surfaces that constantly beat in an upwards direction to move the mucus containing the trapped pathogens towards the throat
 - **Urine** = sterile and slightly acidic that flushes and cleans the ureters
- The first line of defence will do its best in trying to prevent a pathogen from entering into the body.

2nd line of defence

This line of defence is initiated when the pathogen has entered into the body.

- ★ **Inflammatory response**
 - When the first line of defence is penetrated (e.g metal stick), bacteria is able to enter the body. This causes the wounded cells to release chemicals that signal mast cells to release histamines (hormones). Thus, triggering vasodilation (dilation of the capillaries) and increasing capillary permeability (allows the flow of fluids in and out of the vessel), causing an inflammatory response (signs of redness and hotness).
- ★ **Phagocytosis**
 - Phagocytes (type of WBC) from the capillaries rush to the penetrated site, after detecting the bacteria/foreign substance/ pathogen. It then performs the process phagocytosis - engulfing the bacteria (endocytosis), breaking it down and removing it (exocytosis). Thus, swelling of the penetrated site has been reduced and area free of bacterial pathogens.
- ★ **Lymph system**
 - The lymph system acts as a drainage system, cleaning and washing away foreign particles, dead cells or bacterial debris, they move with the tissue fluid. To stop the backflow the vessels have one-way valves and muscular walls that contract to force the lymph forwards The substances flow with the tissue fluid, making its way into the lymph nodes, where everything is destroyed. Therefore, it prevents these harmful substances from being fed back into the body.
- ★ **Cell death (apoptosis)**
 - When the body is struggling to fight off an infection, in order to rid off the bacteria the cells surrounding the area will die - forming a capsule or cyst. This seals off the infection and pathogens from the rest of the body, which will eventually die and are then consumed by the macrophages.

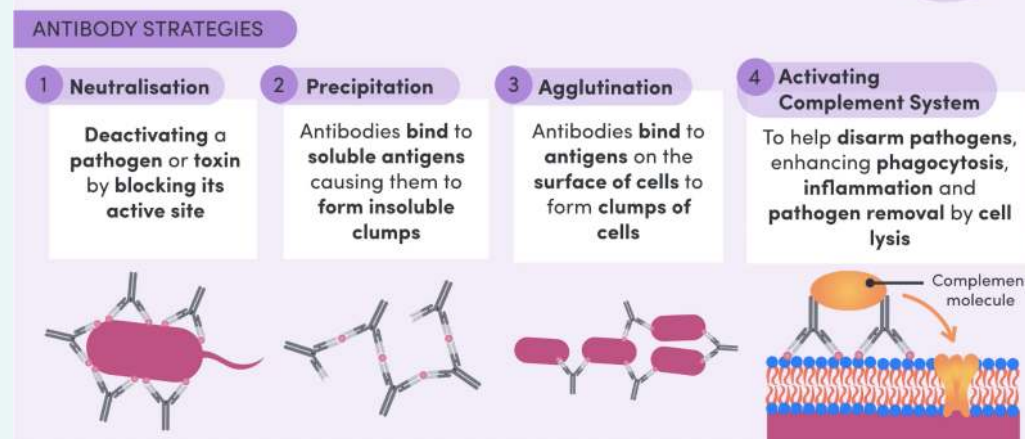
★ **Adaptive Immunity (Specific)**

- Specific
 - Able to recognise and respond to specific antigens
- Adaptive
 - Has memory
 - Second exposure to the same antigen (produces larger and more rapid response than occurred at the primary exposure)
- If the innate immune system fails to clear the infection the adaptive system takes over
 - This is initiated by the macrophages from the initial response from the 1st line of defence

🔴 3rd line of defence

★ Humoral Response (B-cells)

- B-cells are lymphocytes produced in the bone marrow
 - After they mature they are released into the bloodstream, spleen, tonsils and lymph nodes
- On the surface of B cell there are different antibody that only responds to specific antigens
- Once they become triggered by the antigen it begins to proliferate (clone, replicate) into millions of cells
 - They only live for a few days, so if they don't come across an antigen they will die
- After cloning each, clone forms plasma cells that produce antibodies specific to the particular antigen



- The antibodies that are made move to the site of infection and combine with the antigen to form that antigen-antibody complex
 - Deactivate the antigen and the pathogen
- B memory cells have a specific antigen receptor located on their surface
 - Provides immunological memory

• TYPES OF B-CELLS

Type of B Cell	Role in Acquired or Adaptive Immunity (3 rd Line of Defence)
B Cell plasma	defends the body against bacteria and viruses that are outside the cell. By producing antibodies that are specific to the antigen, which deactivated the antigen/pathogen
B Memory	is designed to live a lot longer and remain in the body, up to many years. They allow for a more faster and stronger response if the pathogen were to re-enter

★ Cell mediated immunity (T-cells)

- Cytokines are secreted by the helper T cells to activate (initiate process of cell replication through mitosis) more of the same helper T cells and macrophages
- A specific cytokine (interleukin) produced by the helper T cells, activate the production of cloning the B cells that are specific to the antigen
- Interleukin also activated the production of proliferation of clones of the cytotoxic t cells that have the particular antigen receptor on their surface
- The suppressor T - cells are then responsible for suppressing the activity of the B-cells and cytotoxic T cells once the immune response had defeated the infection
- Similar to the B-cells they have specific receptor protein that can recognise a specific antigen
- When the matching T helper cell comes across the antigen in the body that matches their receptor protein, they are activate and produce many clones of cytotoxic (killer) t cells specific to the

- antigen
 - T helper cells are triggered by an antigen presenting dendritic cell
 - They then move to the site of the infection and release chemical that destroy the infected cells
- TYPES OF T-CELLS

Type of T Cell	Role in Acquired or Adaptive Immunity (3 rd Line of Defence)
T Helper Cell	release chemical that activate the cloning of cytotoxic T cells and B cells and increase macrophage activity after the antigen has been recognised
T Killer Cell (Cytotoxic T cell)	move to the site of infection and release chemicals that destroy infected cells
T Suppressor Cell	suppress the immune response when the infection has been defeated
T Memory cell	remain in the body to respond to future infections by the same antigen

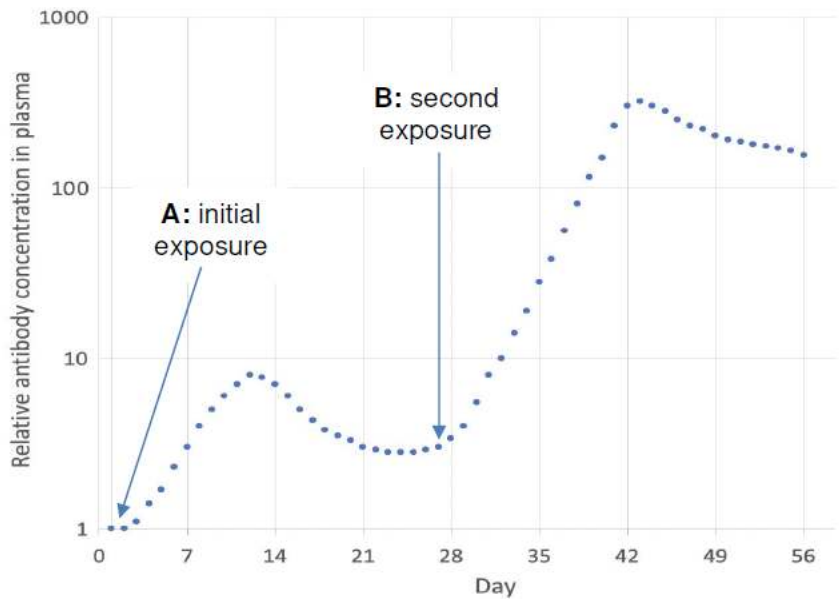
★Similarities and Differences (systems)

	Innate	Adaptive
Similarities	Both attack entering pathogens	
Differences	- Non-specific: reacts the same way to all entering pathogens - Includes physical, chemical barriers, phagocytosis, inflammatory response, apoptosis, lymph system - Doesn't produce memory cells as a response after attacking pathogen - Unaltered in its ability to fight against pathogens	- Specific: is only reacts when a specific pathogen is found entering the body - Includes humoral and cell mediated responses - Produces memory cells as a response after attacking pathogen (immunology) - Altered in its ability to fight against pathogens

7.3.2 explain how the immune system responds after primary exposure to a pathogen, including innate and acquired immunity

Question 32. (8 marks)

- (a) The following graph indicates the relative concentration of an antibody following the initial (A) and second (B) exposure to a pathogen. Discuss what is happening at the level of the immune system to explain the differences in antibody response (4 marks)



Acquired

When the body is initially exposed to the pathogen, the body does not have the B cells to create antibodies. At this initial exposure point the body is still trying to recognise and create antibodies to fight off the pathogen, this can be seen in the graph where on day 0 there is no relative antibody concentration in plasma, meaning that there are no B-cells present at this initial exposure point to create antibodies. At the body's second exposure to the pathogen T-cells will start to proliferate and clone in order to create memory B-cells. These memory cells allow the body to recognise the entering pathogen, the B-cells now have a faster and stronger response - symptoms do not occur/ or are minimal.

7.4 Prevention, Treatment and Control
IQ:How can the spread of infectious diseases be controlled?

7.4.1 investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease

★ **Local, regional and global (COVID-19)**

Local	Regional	Global
Covering the mouth and nose when coughing, washing hands with soap and water, use of PPE (e.g. masks, gloves), social distancing. Cheap however requires education, campaigns and widespread cooperation	Limiting person to person transmission by closing borders and entry screening. Requires extensive resources	Travel bans and restrictions, screening travellers existing areas with the disease.
DURING COVID		
Public health policies for tracing close contacts for early detection (signing into places, QR codes), cleaning contaminated facilities and closing down schools/affected workplaces. Can be costly to businesses/families who may be affected by shutdowns.	Widespread use of antiviral medication, surveillance of community quarantine, restrictions on "hotspots", common infection control guidelines - Border between states closed - The vulnerable were provided with antiviral medication	Surveillance, international networking, liaising with governments of other countries to determine travel guidelines redistributing antiviral medications to areas of demand, emergency preparedness for new strains. Countries closing their international borders.

7.4.2 investigate procedures that can be employed to prevent the spread of disease, including but not

Term	Definition
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limited to:

a. hygiene practices

b. quarantine

c. vaccination, including passive and active immunity

d. public health campaigns

e. use of pesticides

f. genetic engineering

Hygiene	
Quarantine	a public health measure based on restricting the movement of people, plants and/or animals through ports of entry to a country, to prevent or limit the spread of pests and contagious disease
Immunity	state in which the individual has become resistant to being infected by a pathogen and develops the corresponding disease
Vaccination	A procedure that uses vaccines to prevent diseases in populations. - Key purpose of immunisation programs
Public Health campaigns	Procedures aimed at persuading the public to engage in or to stop specific behaviours and to communicate information about symptoms, treatments and public health support
Pesticides	Pesticides are chemicals that are used to kill the pests of plants and animals, pathogens, and vectors that transmit pathogens from one organism to another
Genetic engineering	Process that involves altering the genetic composition of an organism in some way

★Hygiene practices

Personal

● Hands should always be washed with soap and water before preparing and eating food and after going to the bathroom

● The body and hair should be washed regularly and teeth cleaned, to prevent the build-up of pathogens (spec. bacteria)

E.g build up of plaque (bacteria) can cause bleeding and swelling of the gums, loss of teeth

● Always cough or sneeze into elbow, tissue to prevent airborne droplets form spreading to others

Community

● Sewage and garbage disposal reduce the risk of pathogen numbers increasing and spreading throughout the community

● Sterilisation and disinfection of equipment in hospitals, doctor surgeries dentists and hairdressers/barbers reduces the risk of the spread of pathogens from one person to another

● City planning reduces overcrowding and therefore reduces the risk of the transmission of disease throughout the population- this is important in controlling disease

Food and water

● Food and water cleanliness are designed to control the outbreak of disease by reducing the pool of pathogens in food and the environment

● Cooking food properly

● Eating using sanitise utensils

● Throwing out old food

★Quarantine

● **Purpose** → the role is to minimise the risk of exotic pests and disease entering Australia, in order to protect native flora and fauna, our agricultural industries, our environment and health

Types of Quarantine	
Animal	- Involves all animals that come into Australia spending time at quarantine stations to ensure that they are free of disease before they are released - To bring pets into Australia, you have to leave them in quarantine for a number of weeks
Plant	- Examining all plants, parts of plants and plant products brought into Australia for pests/diseases - Many of these items are refused entry, though in some cases items are allowed after treatment by quarantine - Live plants must be kept in quarantine until any diseases that may be present have had time to develop
Human	- Captains of aircrafts and ships are required to notify AQIS if any passenger or crew are displaying any symptoms of prohibited diseases, such as arabies, yellow fever malaria SARS or avian influenza. - Also aircraft are sprayed with insecticide to kill any pests that have AUstralia with the aircraft

- Australian international airports have mosquito-trapping programs to detect any entering the country as they could be acting vectors

★ **Vaccination - including passive and active immunity**

- An effective vaccine protects and individual against specific infectious disease and its various complications
 - Some vaccines only require a single vaccination to provide lifelong protection, e.g Hepatitis B.
 - Other vaccines require a booster shot to maintain its protection e.g Polio, flu vaccine
- Reduces the need for antiviral and antibiotics
- **Herd Immunity**
 - Comes about when a high proportion of the population has been immunised and has acquired immunity.
 - The small number of people who have not been vaccinated will be protected as there are not enough susceptible people to allow transmission of disease.
- **Active immunity** → when the body uses its immune response and there is a creation of memory cells

Natural Active Immunity	Acquired (artificial) Active Immunity
• Antibodies are made • Antigens are recognised • When a person is infected with a pathogen, specific antibodies are produced to combat the pathogen. • These antibodies remain in the blood long after the infection has been fought, sometimes for the rest of the person's life. • If the person is infected with pathogen again, the antibodies that have been made will act fast and destroy it almost immediately • If the person gets infected, but fights off the infection without getting sick, the person is said to be immune to that disease	• A vaccination is given. ◦ A vaccine can be made out of an attenuated pathogen (meaning weakened), it doesn't have the full strength of the pathogen itself, but it still is enough to activate an immune response • Through the vaccination the person is prompted to produce antibodies without the disadvantage of becoming ill, which usually leads to immunity • The use of vaccinations has resulted in many diseases becoming extremely rare or eradicated.

- **Passive Immunity** → short-term immune response, antibodies are inherited through growth and development

Natural Passive immunity	Artificial Passive immunity
• Naturally acquired passive immunity occurs during pregnancy, when antibodies are passed from the maternal blood into the fetal bloodstream.	• Artificial passive immunisation is normally administered by injection and is used if there has been a recent outbreak of a particular disease or as an emergency treatment for toxicity, as in for tetanus. The antibodies can be produced in animals, called "serum therapy," although there is a high chance of anaphylactic shock because of immunity against animal serum itself. Thus, humanised antibodies produced in vitro by cell culture are used instead if available.

★ **Public health campaigns**

- Public Health Campaigns are advertised and/or provided to the public to convey a health message and generate a desire from the target audience.
 - Health promotion information focusing on behavioural risk/protective factors of transmitting a disease
 - I.e encouraging people to wear mask, sanitise, stay home to mitigate the spread COVID-19
 - Immunisation campaigns such as the New South Wales 'Save the date to vaccinate' campaign that includes television advertisements and brochures available from public health centres and www.health.nsw.gov.au in multiple languages
 - Disease prevention/education programs run through public organisations such as hospitals, health centres and schools.
- Campaigns reduce antibiotic resistance by providing prescription guidelines to practitioners and the public, in particular parents with young children
- To run a campaign is cheaper than having to research and treat a disease

	★<u>Use of pesticides</u> • There are several types of pesticides ◦ <i>insecticides</i>, which kill insects ◦ <i>fungicides</i>, which kill fungal pathogens ◦ <i>herbicides</i>, which kill weeds. • DDT ◦ widely used during World War II to kill lice that transmitted the pathogen that caused the disease Typhus ◦ DDT has also been widely used to kill the Anopheles mosquito, which carries the plasmodium that causes malaria. ■ However, the effectiveness of DDT was reduced as mosquitoes developed resistance to it due to natural selection. • However, due to the inevitable development of resistance by the pesticides, pesticides’ negative impact on human health and the environment, alternative solutions are currently being researched. ◦ Alternative approaches: ■ Include integrated crop protection and integrated pest management → understanding the life cycles of pathogens ■ Better farm sanitation, using resistant or tolerant plant varieties ■ Etc. ★<u>Genetic engineering</u> • Altering the genetic make-up of organism, it is possible to make them resistant to disease • Organisms with genes from other organism inserted into their own genetic material are called transgenic species • Genetic engineering is one option for creating organisms that are resistant to the effects of the pathogen without having to rely on costly vaccination programs → aka transgenic species ◦ Concern: reduction in biodiversity • engineering the genes of vectors such as mosquitoes can control the spread of diseases including malaria, dengue and Zika → CRISPR-Cas9				
7.4.3 investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example: a. antivirals b. antibiotics	★<u>Antibiotics</u> • Antibiotics are drugs (natural or synthetic) used to fight bacterial infections. They are one of the most frequently prescribed medications in medicine • Chemicals that are capable of destroying or inhibiting growth of bacteria that cause disease • They target bacteria without destroying the host and are <u>not at all effective against viruses.</u> • Types of antibiotics ◦ <u>Broad spectrum</u> ■ Acts on a wide range of bacteria ■ Use when the identity of the bacteria causing the infection is not known ◦ <u>Narrow spectrum</u> ■ Acts on only one or two types of bacteria • Bacteriostatic antibiotics stop bacteria from multiplying by interfering with DNA replication, protein production or other cellular metabolic processes, e.g. amoxicillin <table><tr><th>Positives</th><th>Negatives</th></tr><tr><td> • The discovery of penicillin rates as one of humanity's greatest discoveries. Estimates are that 200 million lives have been directly saved from antibiotics. • Due to different cell structures, e.g. cell walls, bacteria are more easily targeted than viruses. </td><td> • Antibiotics only work on bacterial infections. • Misuse and overuse have led to multiple examples of antibiotic resistance. • People can be allergic to antibiotics </td></tr></table> ★<u>Antivirals</u> • Controls viral infections • Inhibit their development inside infected cells • Slow down progress of the disease to allow the body’s natural defences to take over	Positives	Negatives	• The discovery of penicillin rates as one of humanity's greatest discoveries. Estimates are that 200 million lives have been directly saved from antibiotics. • Due to different cell structures, e.g. cell walls, bacteria are more easily targeted than viruses.	• Antibiotics only work on bacterial infections. • Misuse and overuse have led to multiple examples of antibiotic resistance. • People can be allergic to antibiotics
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	● How they work ○ The drugs keep the virus from escaping from one cell to infect the neighbouring cells ○ They are the most effective within a day or two of the onset of symptoms ○ 1. Preventing a virus from entering a cell by binding to a cell receptors that allow the virus to enter ○ 2. They can inhibit enzymes that aid in the replication of the virus ○ 3. They block transcription and translation of viral proteins ○ 4. They prevent the virus from leaving the ce <table><tr><th>Positives</th><th>Negatives</th></tr><tr><td> ● Designed to target specific viral pathogens ● They can significantly improve the health outcomes and life expectancy of infected individuals → i.e. HIV used to be a death sentence, antivirals allowed for improved life outcomes </td><td> ● Research is on small no. of diseases: HIV/AIDS, herpes, influenza ● Developing safe and effective antiviral drugs is challenging as viruses use the host cell's machinery to reproduce. ● Further complicated by the high mutation rates of virus → drug-resistance </td></tr></table>	Positives	Negatives	● Designed to target specific viral pathogens ● They can significantly improve the health outcomes and life expectancy of infected individuals → i.e. HIV used to be a death sentence, antivirals allowed for improved life outcomes	● Research is on small no. of diseases: HIV/AIDS, herpes, influenza ● Developing safe and effective antiviral drugs is challenging as viruses use the host cell's machinery to reproduce. ● Further complicated by the high mutation rates of virus → drug-resistance
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7.4.4 investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic	● Environmental management seeks to change the environment in order to prevent OR minimise vector propagation and human contact with the vector-pathogen by destroying, altering, removing or recycling non-essential containers that provide egg/larvae/pupal habits				
7.4.5 interpret data relating to the incidence and prevalence of infectious disease in populations, for example: a. mobility of individuals and the portion that are immune or immunised b. Malaria or Dengue Fever in South East Asia	★mobility of individuals and the portion that are immune or immunised ● Incidence → the number of NEW cases occurring during a specified time. → the infection rate, or the probability (risk) of contracting the disease. ○ Expressed as a percentage or as a number per 100 000 of the population ○ To calculate the incidence of a disease as a percentage, the following formula can be used: $\frac{\text{number of new cases during a specified time}}{\text{size of population at start of monitoring period}} \times 100$ ● Prevalence → the proportion of the population that have the disease at a particular point in time. ○ Measure of disease that allows us to determine a person’s likelihood of having a disease. Therefore, the number of prevalent case is the total number of cases if disease existing in a population. ○ The prevalence of a disease can be calculated using the following equation: $\frac{\text{all new and previous cases during a time period}}{\text{population during the time period}} \times 100$ ● Mobility ○ The mobility of a population is an important factor in the assessment of potential disease outbreaks. ○ Humans act as carriers for pathogens and may spread the disease to new locations when they move. ○ In managing/monitoring humans, countries track the social trends of their people e.g. the number of Australians who travel overseas by plane and boat each year, how often are people going between states each year. ● Morbidity → term for illness ● Mortality → another term for death ○ Mortality rate refers to the number of deaths due to a disease divided by the total population. ■ If there are 25 lung cancer deaths in one year in a population of 30,000, then the mortality rate for that population is 83 per 1000,000 ★Malaria or Dengue Fever in South East Asia Malaria				

	● Approx 40% of the world population, mostly those living in 3rd world countries, are at risk of malaria. Every year more than 500 million people become severely ill with the disease. ○ Post eradication epidemics surfaced in the mid 1970s and in 2015 there were 296 million cases of malaria worldwide, resulting in an estimated 731 000 deaths. ● More than 80% of the population is at risk of contracting malaria in Asia and the South-Pacific. ○ Malaria incidence is between 2.5 and 2.8 million cases and high-risk groups include infants, young children, pregnant women and mobile population groups	
	Pathogen	Caused by a parasite - transmits ● P. Falciparum ○ The deadliest malaria parasite ● P.vivax ○ Dominant malaria parasite
	Mode of transmission	Vector - Female Mosquito
	Adaptations of the pathogens to aid transmission	● The female mosquito (Anopheles) ● The Mosquito bites the individual and the pathogen is then transfused into the bloodstream of the individual
	Mobility	It is spread through a vector, therefore depending on the amount of vectors there are can affect how quickly the disease can spread. ● Though it doesn't spread through person to person contact
	Symptoms	● Fever ● Headache and chills ● Start within 10-15 days of getting bitten ● Infants, children under 5 years, pregnant women, travellers and people with HIV or AIDS are at higher risk <u>More severe symptoms</u> ● Extreme tiredness and fatigue ● Impaired consciousness ● Multiple convulsions ● Difficulty breathing ● Dark or bloody urine ● Jaundice (yellowing of eyes and skin) ● Abnormal bleeding
	Treatment	It is important for those experiencing signs of malaria to be tested immediately in order to be treated properly. <u>Most common medicines for malaria:</u> ● Artemisinin-based combination therapy medicines like artemether-lumefantrine are usually the most effective medicines. ● Chloroquine is recommended for treatment of infection with the <i>P. vivax</i> parasite only in places where it is still sensitive to this medicine. ● Primaquine should be added to the main treatment to prevent relapses of infection with the <i>P. vivax</i> and <i>P. ovale</i> parasites.
	Prevention	<u>Simple controls</u> ● Use mosquito nets when sleeping in places where malaria is present ● Use mosquito repellents (containing DEET, IR3535 or Icaridin) after dusk ● Use coils and vaporizers. ● Wear protective clothing. ● Use window screens. <u>Other controls</u> ● Vector Controls ● Chemoprophylaxis ● Preventative chemotherapies

	Vaccine															
	Dengue Fever - The global prevalence of dengue and severe dengue has grown dramatically over recent decades - Before 1970 only nine countries had experienced dengue fever epidemics but the disease is now endemic in more than 120 countries, with almost half of the global population at risk. ¾ of those most at risk live in Asia-Pacific region where vector populations are increasing - Dengue is found mainly in tropical and subtropical climates world-wide, mostly in urban and semi-urban areas. - Severe dengue can lead to death in young children in some Asian and Latin American countries <table><tr><td>Pathogen</td><td>Dengue- part of the Flaviviridae family</td></tr><tr><td>Mode of transmission</td><td>vector - Spread through a female mosquitos (Aedes aegypti) Airborne disease</td></tr><tr><td>Adaptations of the pathogens to aid transmission</td><td> - The pathogen is able to spread through a vector (mosquito) without affecting the vector - The pathogen is able to phagocytosed by macrophages when it enters into the host, changing the pH of the macrophage and forming a pore - The formation of the pore allows the virus to enter into the host's cytoplasm - The macrophage unintentionally duplicates the virus (budding) - The antibodies aren't able to find as well to the antigens of the virus </td></tr><tr><td>Mobility</td><td>Spreads through a vector - A single female mosquito can feed off 3-4 hosts in their lifetime --> disease can be easily spread </td></tr><tr><td>Symptoms</td><td> - Rashes - Painful joint - Muscle aches - Vomiting - Haemorrhagic fever - Dengue shock fever </td></tr><tr><td>Treatment</td><td>A vaccine must be effective against all four stereotypes simultaneously</td></tr><tr><td>Prevention</td><td rowspan="2"> - Know your surroundings - Cover up - Insect repellent - Dump standing water </td></tr><tr><td>Control</td></tr></table>	Pathogen	Dengue- part of the Flaviviridae family	Mode of transmission	vector Spread through a female mosquitos (Aedes aegypti) Airborne disease	Adaptations of the pathogens to aid transmission	- The pathogen is able to spread through a vector (mosquito) without affecting the vector - The pathogen is able to phagocytosed by macrophages when it enters into the host, changing the pH of the macrophage and forming a pore - The formation of the pore allows the virus to enter into the host's cytoplasm - The macrophage unintentionally duplicates the virus (budding) - The antibodies aren't able to find as well to the antigens of the virus	Mobility	Spreads through a vector A single female mosquito can feed off 3-4 hosts in their lifetime --> disease can be easily spread	Symptoms	- Rashes - Painful joint - Muscle aches - Vomiting - Haemorrhagic fever - Dengue shock fever	Treatment	A vaccine must be effective against all four stereotypes simultaneously	Prevention	- Know your surroundings - Cover up - Insect repellent - Dump standing water	Control
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7.4.6 evaluate historical, culturally diverse and current strategies to predict and control the spread of disease	★ Historical strategies to predict and control the spread of disease - There was very little understanding of the causes of disease and therefore minimal knowledge of how to prevent, treat or control them. - E.g. the Miasma theory → indirect and airborne transmission held that diseases like cholera were caused by the smell of the bad air, miasmata, and poisonous vapour with the suspended particle of decaying matter and a foul smell.															

	• The earliest account of mathematical modelling of the spread of disease was carried in 1766 by Daniel Bernoulli ◦ Created a model to predict increased life expectancy if a population was inoculated against smallpox. ★ Culturally diverse strategies to predict and control the spread of disease • Different cultures have different understandings about diseases and therefore different strategies to predict and control their spread. • This includes: ◦ Religious belief about health ◦ Health care ◦ Disease ◦ Pharmaceuticals ◦ And vaccinations • Cultural beliefs must be understood in order to predict and control the spread of disease ◦ E.g cultural practices and belief played a key role in the spread of the African ebola epidemic (2013 - 16). ■ African culture prides themselves on strong family and community relationships where hugs and handshakes are common ways of greeting. ■ In addition, individuals are cared for by their immediate families - not seeking professional medical treatment or advice. • They did so because they believed that they would die isolated from their loved ones. ■ Cultural practices also include: • The way one is buried (burials and mournings) • Rituals that involve members of the family having very close contact with the corpse of the deceased. ★ Current strategies to predict and control the spread of disease • Ensuring, monitoring and supply of clean water supply. • Appropriate handling and treatment of sewage waste. • Antibiotics & Anti-viral drugs. • Vaccines • Public Health Campaigns – including the importance of personal hygiene. • Global Health Maps • Quarantine • Isolation – In hospitals – Tuberculosis patients. 30 • Pesticides • Genetic Engineering
7.4.7 investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example: a. bush medicine b. smoke bush in Western Australia	★ Aboriginal protocols, medicines and biological materials • <u>Protocols in the development of medicines</u> ◦ Many indigenous cultures around the world have used traditional methods to treat infectious and non-infectious diseases. ◦ Aboriginal Australians are the custodians of a rich and detailed knowledge base of medicinal native Aus herbs, fruits and vegetables. ◦ Substances from plants such as tannous, mucilage, oil latex and alkaloids were used for medicine ◦ The plant material was usually crushed and used as poultice or infused with water to drink. ★ Recognition and protection of indigenous cultural and intellectual property • Aboriginal people feel as though that their traditional knowledge is being exploited and that their indigenous cultural and intellectual property right are not being recognised • These concerns include: ◦ Cultural knowledge of plants, animals and the environment are being used by scientists, medical researchers, nutritionists and pharmaceutical companies for commercial gain, WITHOUT THEIR INFORMED CONSENT.

- Is this fair?
 - It isn't fair that they don't get any financial benefits, as they are the people who had discovered the healing properties of the smoke bush, it was information that was passed down through the generations.
- The indigenous were the first to be on Australian land and therefore have vast knowledge about the plants (species) that can benefit humans in terms of health, which have been passed down through generations. It is therefore important to recognise that when taking and using their knowledge for wider purposes (developing medicine), that without their prior knowledge we wouldn't be able to develop and come to know the uses of Australian plants, in terms of their healing properties.

★**Bush medicine**

- Many plants contain antibacterial and anti-inflammatory compounds that are known to Western medicine

Plant	Use by indigenous peoples
Lemon scented gum	• Has citronella in it - insect repellent • Rub over the body to keep mosquitoes away
peach leaved quinine	• Good for sore throats e.g. as tea
Gumby Gumby	• Good for sore throats e.g. as tea

★**Smoke bush in Western Australia**

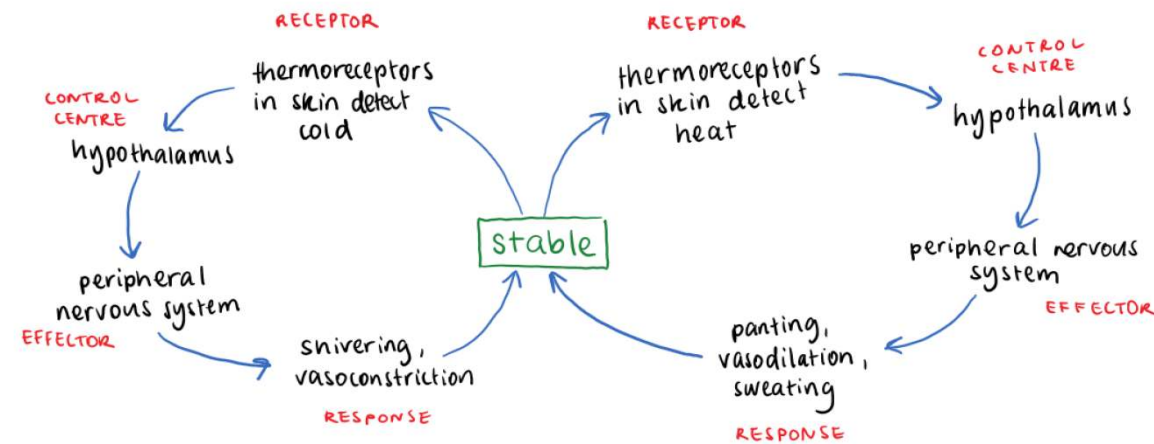
- Traditionally used by Aboriginal Australians for healing purposes
- 1960s- 1980s USA's national cancer institute was given licensing rights to collect specimens of smokebush for research into its effectiveness for treating cancer and HIV
- Research discovered that the plant contained Conocurovone, a property which has potential to destroy low concentrations of HIV
- Indigenous people, who originally discovered the healing properties receive little to no acknowledgement or financial return.

Mod 8: Non-infectious Disease and Disorders

8.1 Homeostasis	
IQ: How is an organism’s internal environment maintained in response to a changing external environment?	
Syllabus Outcomes	Summary of information
8.1.1 construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to: a. temperature b. glucose	★What is Homeostasis Homeostasis is defined as the maintenance of a stable, or constant internal environment, despite external and internal environmental changes. The nervous and endocrine systems work together to ensure many aspects of your internal environment remain within the narrow range required to maintain your health. One example is how your body keeps your internal temperature close to 37°C. Your body also tries to maintain a consistent blood volume, blood pressure and blood solute concentration by balancing fluid and ion levels. Another example of homeostasis is how your body excretes (removes) carbon dioxide and nitrogenous wastes (urea through the kidney to help ensure that the blood pH stays close to 7.4. Similarly, your body uses hormones to help regulate blood glucose levels. ★Negative Feedback Loops • Are diagrams that show how the body detects and counteracts changes in order to maintain homeostasis • There are multiple correct ways to draw a negative feedback loop ◦ In a ‘figure 8’ shape. ■ This shows the optimal condition in the centre, with one loop depicting detection and response to a change above the optimal, and the other loop depicting detection and response to a

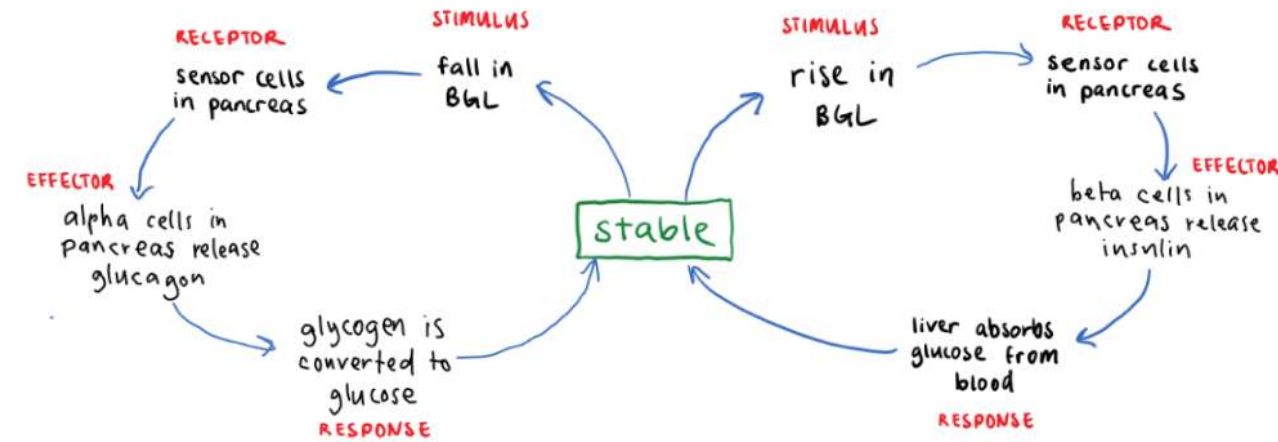
change below the optimal.

★ **Temperature**



★ **Glucose**

- Optimal BGL → Between 70 mg/dL - 100 mg/dL
- Organ receptor → the pancreas
- Cells that detect changes in BGL → Beta cells
- Hormone released by Beta cells → Insulin
- Specific changes that occur when insulin is released → Blood glucose levels fall, muscle and fat cells store excess glucose
- Hormones released by *alpha* cells → glucagon



8.1.2 investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including:

a. trends and patterns in behavioural, structural and physiological adaptations in endotherms that assist in maintaining homeostasis

Term	Definition
Adaption	Feature that will enable an organism to survive and reproduce in different environments - Always like these adaptations back to homeostasis
Endotherm	An endotherm that can regulate its internal temperature
Structural	Physical feature of an organism
Behavioural	The way an organism acts
Physiological	Internal processes that maintain homeostasis

- b. internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways
- c. mechanisms in plants that allow water balance to be maintained

★Trends and patterns (endotherms)

- Kangaroo
 - Structural → fur maintains temperature when its cold at night, large ears have more surface area exposed to the environment so capillaries can exchange heat efficiently with environment
 - Behavioural → move into shade, dig holes to reach cooler dirt and lay down, lick their wrists results evaporate cooling as there is a high density of capillaries to exchange heat
 - Physiological → Vasodilation in capillaries of the kangaroo, evaporative cooling from exchange due to high density of capillaries in wrist
- Penguin
 - Structure → high density of feathers to trap heat, no muscles in feet only tendons, bone, keratin, limits exposure of muscle to cold environment as the proteins would freeze
 - Behavioural → huddle together and rotate as the middle gets too hot to maintain internal temperature, they can also fluff their feathers up which can trap warmth, and can pull them down to keep cool.
 - Physiological → countercurrent exchange, blood flow to the feet but it's so cold that if it travelled to the heart it would cause damage, so the capillaries travelling up and down are placed side by side to exchange heat, which warms up cool blood, increases the temperature of blood.
- Whale
 - Structural: blubber thick fat-like material that insulate the whale from the cold water
 - Behavioural: migrate to cooler or warmer waters depending on the season
 - Physiological: salt regulation, salty environment it is important to get rid of extra salt, osmoregulation to maintain homeostasis of alt concentration and water levels by producing concentrated salty urine

★How the nervous system responds to stimuli

- The nerves allow the brain to receive and respond to stimuli from both the internal and external environments
 - E.g. light, sound, touch, pressure, heat and chemical concentrations
- Sensory receptors help to detect stimuli.
 - In humans we have sensory organs (ears, eyes, nose, skin and tongue)
- Sensory neurons (nerve cells) in the nerves transmit the information as electrical signals that travel to the control centre (usually the brain) where the info is processed.
- The CNS responds to the stimulus by activating more neurons in the nerves to transmit the information to an effector organ which produces a response to the stimulus.

★Coordination systems

- Body receptors and effectors are controlled via the CNA and they travel throughout the body via nerves in the peripheral nervous system (outside the brain and spinal cord) and hormones in the endocrine system
- Both nervous systems are made from a network of long strand neuron cells

Internal coordination systems for the maintenance of homeostasis	
Nervous systems (transmits signals between different parts of the body)	Endocrine system (produces hormones for metabolism growth etc.)
- Uses electrochemical impulses to relay messages from stimuli detected by a receptor and the appropriate reaction to be performed by the effectors - Reception of stimulus, transmission of messages, interpretation and generation of responses - Provides neural pathways which receptors can coordinate with effectors - CNS consists of the hypothalamus and the spinal cord.	- Hormones are chemical messenger molecules which are transported by the bloodstream - Endocrine glands are pituitary glands which release hormones in the direction of the hypothalamus. - Hypothalamus joins this system with the nervous system - Linking endocrine and nervous

★Homeostasis mechanisms in plants

- Controlled stomatal opening and closing - in early morning and late afternoon when temperature is cool and sunlight is less intense, can open stomata for CO2 necessary to carry out photosynthesis. Prevents dehydration
 - Thick and waxy cuticle insulates water from excessive sunlight as it reduces internal temperature, therefore decreasing the evaporation rate.
 - Shiny leaves help to reflect sunlight and therefore help to reduce the spike in internal temperature due to heat energy from sunlight, minimising rate of transpiration.

8.2.1 investigate the causes and effects of non-infectious diseases in humans, including but not limited to:

- a. genetic diseases
- b. diseases caused by environmental exposure
- c. nutritional diseases
- d. cancer

Term	Definition
Treatment	This is when you have a disease and medical care is given to a patient for an illness or injury
Management	The disease is a long-term illness (e.g. genetic disease) management is how to limit the symptoms or to help the person have an increased quality of life
Prevention	The action of stopping something from happening or arising

★Non-infectious diseases

- Known as a non-communicable disease
 - It cannot spread from person to person
 - Not caused by a pathogen
- Generally caused by lifestyle factors and are largely preventable
- There are several types of non-infectious diseases
 - Genetic
 - Environmental
 - Nutritional
 - Cancer

	Definition	Example	Cause	Effects
Genetic	Is inherited	Trisomy 21 (down syndrome)	Caused by trisomy 21 (3 chromosome 21's) caused by abnormal cell division	Heart defects, hearing/vision problems, learning problems and facial features
Environmental	Disease caused by factor in the environment	Hay fever	Allergic reaction to allergens in the environment	Cold-like signs and symptoms, such as a runny nose, itchy eyes
		Asbestosis	- Chronic lung condition that is caused by prolonged exposure to high concentration of asbestos fibres in the air - Fibres in building products that have been disturbed are small enough to penetrate the nucleus and cause mutations	Lungs are damaged, cancer forming in the lungs
Nutritional	Lack or excess of nutrient either micro or macro	Iron deficiency	Blood loss, lack of iron in diet, inability to absorb iron or pregnancy	Fatigue, pale skin, brittle nails and muscle pain
		Scurvy	- Insufficient intake of Vitamin C	Bleeding gums, loss of teeth,
		Obesity		
Cancer	Uncontrollable cell division	Melanoma	- Likely combination of environmental and genetic factors - Prolonged exposure to UV rays	Illness and death

★Genetic

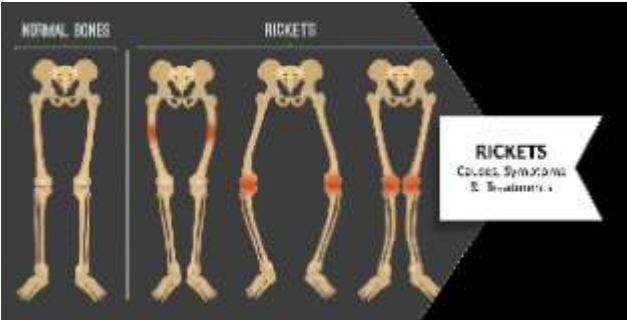
- These diseases are a result of an altered or incorrect expression of a gene or genes somewhere in the genome
- Mutations to the genetic material during gamete formation - errors during cell division such as unseparated chromosomes
 - This can cause disruption to normal gene expression and protein production, which results in the individual having a disease

★Cancer

- Refers to the a group of diseases that involve unregulated and abnormal cell growth leading to the growth of tumours
- Cancerous cells do not differentiate (turns our genes on and off to determine what the cell will be)
- **Malignant** → where upon uncontrolled growth spreads to more vital organs of the body
- There are many treatments that are used for different types of cancers such as:
 - Surgery - to remove tumour
 - Radiation - uses x-rays to damage DNA of cancer cells
 - Chemotherapy - drugs slows growth of cells (not as effective for melanoma as other types of cancers)
 - **Future directions (still in the process of research and testing)**
 - Targeted therapies - drugs that interrupt the pathways that cause uncontrolled cell division. Uses drugs to target specific genes and proteins that are involved in the growth and survival of cancer cells
 - Immunotherapy- uses the body's own immune system to fight the cancer
 - Personalised cancer vaccinations - Unlike pathogenic diseases, cancerous growths contain unique genetic sequences and biomarkers depending between individuals. By comparing the DNA sequences of tumour vs healthy cells, vaccine companies use recombinant DNA technology

- 8.2.2 diseases, for example: –
- a. nutritional diseases
 - b. diseases caused by environmental exposure

- ★ **Nutritional Diseases**
- An individual with unbalanced diet can result in disease
 - **Insufficient intake** of nutrients e.g. anorexia nervosa or malnutrition
 - **Excessive intake** of nutrients e.g. obesity or type II diabetes
 - **Imbalanced diet** for particular vitamin or mineral e.g. scurvy (Vitamin C)

Disease	Features	Symptoms
Nutritional Examples include: Rickets	Cause: Lack of vitamin D in the diet • Overuse of sun protection • Born to a woman deficient in vitamin D • Sick/unable to go outdoors • Lack of vitamin D in the diet Treatment • Improving exposure to sunlight • Improve consumption of vitamin D in the diet • Surgery to correct bone deformities • Treatment for underlying disorders • Oral vitamin D supplements Prevention • Mother should ensure that they intake appropriate amounts of vitamin D during pregnancy • People who live in environments without much sun (e.g. extreme hemispheres) should focus on prioritising vitamin D supplements Management • Increase daily vitamin D ◦ Vitamin D fortified infant formula • Monitor progress with x-rays and blood test <i>*most cases are curable when caught early*</i>	 • Poor development of bones leading to ◦ Weakened bones ◦ Fractures ◦ Bone and muscle pain ◦ Bony deformities

- ★ **Environmental diseases**
- Caused by factors in the environment that can cause harm.
 - **Lifestyle disease** - cardiovascular disease and diseases caused by substance abuse such as alcoholism and smoking related diseases
 - **Exposure to chemicals** - in the home, environment and or workplace. Such as lead poisoning caused by exposure to high levels of lead in the atmosphere, soil and products used in everyday life
 - **Physical factors** - skin cancer caused by excessive exposure to ultraviolet radiation in the sunlight

Disease	Features	Symptoms
Emphysema	Cause: Cigarette smoking or long-term exposure to certain industrial pollutants or dusts. Treatment • Steroids: inhaled as an aerosol spray, helps to relieve symptoms of the disease associated with asthma and bronchitis ◦ But can cause side effects e.g. weakened bones, high blood pressure, diabetes and cataracts • Protein Therapy: patients with an alpha-1 antitrypsin	◦ Type of lung disease that causes breathlessness ◦ Caused by cigarette smoking ◦ Shortness of breath or a long-term productive cough (cough that produces mucus or phlegm) ◦ Fatigue ◦ Barrel-shaped chest ◦ Cyanosis (a blue tinge to the skin) due to lack of oxygen,

		deficiency may be given infusions of AAT to help slow the progression of lung damage. Prevention • Stop smoking • Vaccine: receive a flu shot annually and pneumonia shot every five to seven years to prevent infections. Management • By taking medications and adjusting one's lifestyle	
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8.3 Epidemiology

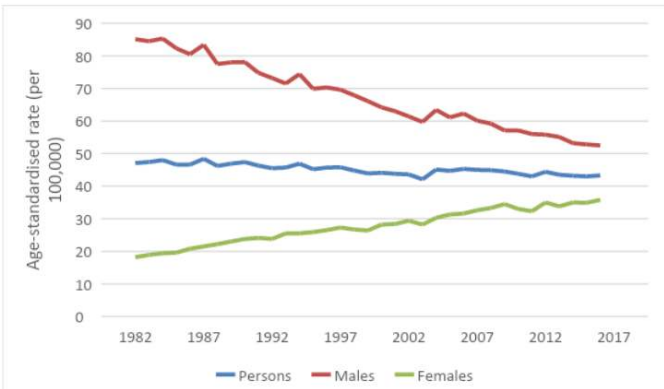
IQ: Why are epidemiological studies used?

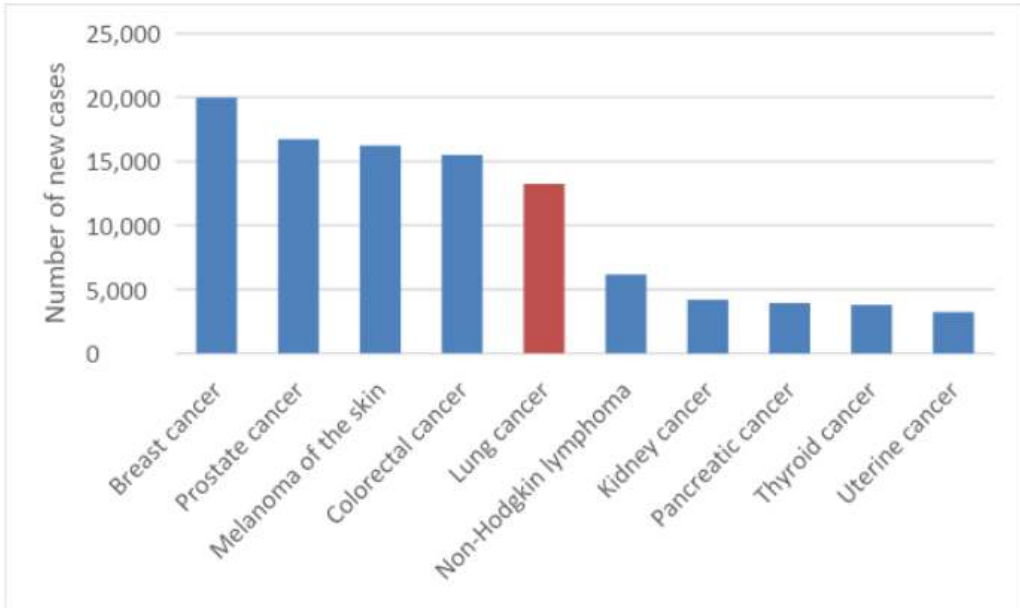
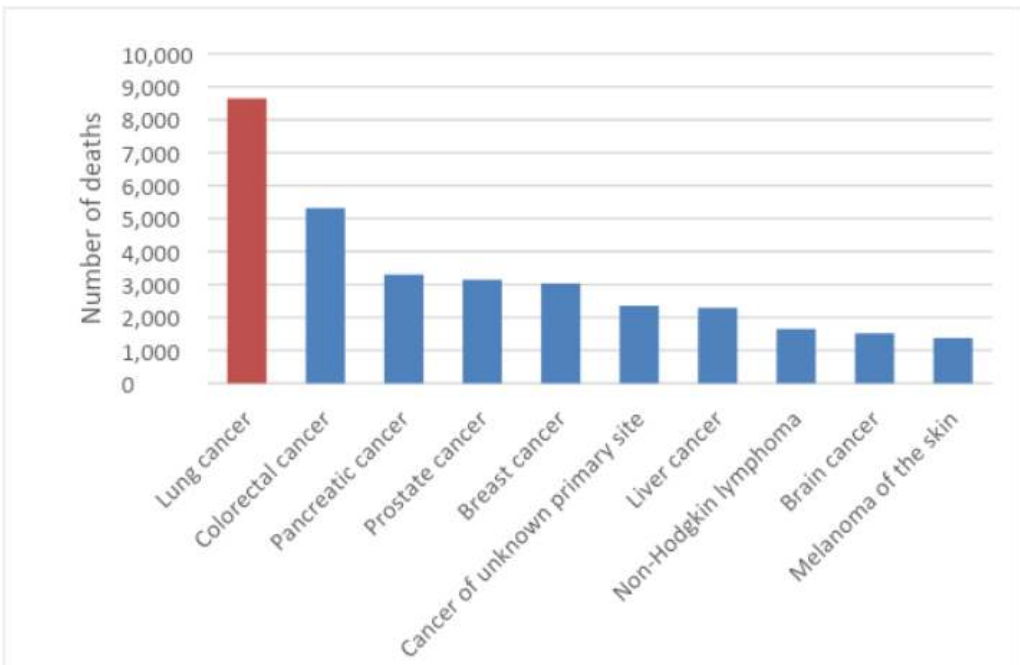
8.3.1 analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to:

- a. nutritional diseases
- b. diseases caused by environmental exposure

Term	Definition
Nutritional Disease	Any health impairments which are caused by the lack of fulfilling dietary requirements
Incidence	The timeframe in which a disease is present
Prevalence	The presence of a disease in a population and among other diseases
“Patterns of non-infectious diseases in populations”	The presence of disease within a population, relating to: - Inheritance - Lifestyle choices - Environmental condition or Disease caused by environmental exposure

- ★**Nutritional diseases**
- **Essential Nutrients**
 - Minerals- e.g. calcium and magnesium
 - Vitamins- e.g. vitamins A, C, D and K
 - Amino acids- e.g. tryptophan and histidine
 - Fatty acids- e.g. omega-3 fatty acids
 - **Many of the nutritional diseases result from a lack of, imbalance or sometime over consumption of certain nutrients.**
 - **Leads to different types of malnutrition**
 - **Scurvy**
 - A lack of vitamin C in the diet
 - Common among sailors of the past due to a lack of fresh fruit and veggies
 - Humans cannot synthesis the compound and must consume in their diet
 - Approx. 50mg per day
 - **Rickets**
 - Lack of vitamin D
 - Vitamin D is required to absorb calcium in the digestive system
 - Can be made by the body if exposed to small amounts of UV light
 - **Coronary Heart Disease**

	○ Build up of plaque inside coronary arteries of the heart ○ Fatty diet, high cholesterol and fatty substance can build up inside the arteries causing them to narrow ○ This places strain on the heart ○ Can cause blood clots and/or heart attack																																					
	★ Diseases caused by environmental exposure																																					
	Lung Cancer																																					
	Cause	- Smoking - Exposure to harmful particles (IE Asbestos) - Old age - HIV infection																																				
	Effect to humans (symptoms)	- Chronic coughing + coughing up blood - Pain in the chest, coughing, breathing etc. - Persistent fatigue and headaches																																				
Find a graph or other data showing the incidence for this condition (Australian and/or global data)	 <table><caption>Approximate Age-standardised incidence rates for Lung Cancer in Australia (per 100,000)</caption><thead><tr><th>Year</th><th>Persons</th><th>Males</th><th>Females</th></tr></thead><tbody><tr><td>1982</td><td>48</td><td>85</td><td>18</td></tr><tr><td>1987</td><td>47</td><td>80</td><td>22</td></tr><tr><td>1992</td><td>46</td><td>72</td><td>25</td></tr><tr><td>1997</td><td>45</td><td>65</td><td>28</td></tr><tr><td>2002</td><td>44</td><td>60</td><td>30</td></tr><tr><td>2007</td><td>45</td><td>58</td><td>33</td></tr><tr><td>2012</td><td>44</td><td>55</td><td>34</td></tr><tr><td>2017</td><td>45</td><td>55</td><td>35</td></tr></tbody></table>		Year	Persons	Males	Females	1982	48	85	18	1987	47	80	22	1992	46	72	25	1997	45	65	28	2002	44	60	30	2007	45	58	33	2012	44	55	34	2017	45	55	35
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Describe some of the trends in the incidence data that you collect (ie does it occur in particular populations relating to age or gender etc)	- Incidence rates for males are decreasing while rates for females are increasing - Male rates are higher than female rates - Overall rates have remained relatively the same with minor variation																																					

	Find a graph or other data showing the prevalence for this condition (Australian and/or global data)	 <table><caption>Number of new cancer cases</caption><thead><tr><th>Cancer Type</th><th>Number of new cases</th></tr></thead><tbody><tr><td>Breast cancer</td><td>20,000</td></tr><tr><td>Prostate cancer</td><td>17,000</td></tr><tr><td>Melanoma of the skin</td><td>16,500</td></tr><tr><td>Colorectal cancer</td><td>15,500</td></tr><tr><td>Lung cancer</td><td>13,500</td></tr><tr><td>Non-Hodgkin lymphoma</td><td>6,500</td></tr><tr><td>Kidney cancer</td><td>4,500</td></tr><tr><td>Pancreatic cancer</td><td>4,000</td></tr><tr><td>Thyroid cancer</td><td>3,800</td></tr><tr><td>Uterine cancer</td><td>3,500</td></tr></tbody></table>	Cancer Type	Number of new cases	Breast cancer	20,000	Prostate cancer	17,000	Melanoma of the skin	16,500	Colorectal cancer	15,500	Lung cancer	13,500	Non-Hodgkin lymphoma	6,500	Kidney cancer	4,500	Pancreatic cancer	4,000	Thyroid cancer	3,800	Uterine cancer	3,500
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Describe some of the trends in the prevalence data that you collect (ie does it occur in particular populations relating to age or gender etc)	- Lung cancer is the last “major” cancer - Lung cancer isn’t as prevalent as other types of cancer, though not by much																							
Find a graph or other data showing the mortality rates for this condition (Australian and/or global data)	 <table><caption>Number of deaths from cancer</caption><thead><tr><th>Cancer Type</th><th>Number of deaths</th></tr></thead><tbody><tr><td>Lung cancer</td><td>8,800</td></tr><tr><td>Colorectal cancer</td><td>5,500</td></tr><tr><td>Pancreatic cancer</td><td>3,500</td></tr><tr><td>Prostate cancer</td><td>3,200</td></tr><tr><td>Breast cancer</td><td>3,100</td></tr><tr><td>Cancer of unknown primary site</td><td>2,500</td></tr><tr><td>Liver cancer</td><td>2,400</td></tr><tr><td>Non-Hodgkin lymphoma</td><td>1,800</td></tr><tr><td>Brain cancer</td><td>1,600</td></tr><tr><td>Melanoma of the skin</td><td>1,500</td></tr></tbody></table>	Cancer Type	Number of deaths	Lung cancer	8,800	Colorectal cancer	5,500	Pancreatic cancer	3,500	Prostate cancer	3,200	Breast cancer	3,100	Cancer of unknown primary site	2,500	Liver cancer	2,400	Non-Hodgkin lymphoma	1,800	Brain cancer	1,600	Melanoma of the skin	1,500	
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Describe some of the trends in the mortality data that you collect (ie does it occur in particular	- Obviously, lung cancer has the highest mortality rate of any cancer - Lung cancer deaths are also the highest by a large margin																							

	populations relating to age or gender etc)	
	Explain why you think there have been changes in the incidence, prevalence or mortality.	Due to improving medical practices and accessibility to said services, plus awareness being spread about the causes of lung cancer and how to avoid getting it (IE Anti-smoking campaigns, removal of asbestos from infrastructure etc.)
8.3.2 investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above	Case study - Type 2 Diabetes	
	Description Type 2 diabetes is a condition whereby the body becomes resistant to the effects of insulin, and gradually becomes unable to produce insulin effectively. The result of this is that individuals are unable to regulate their blood glucose levels effectively. Type 2 diabetes may lead to other medical complications, such as vision problems, cardiovascular disease, and nerve and kidney damage. A number of genetic family related risk factors are known to contribute to the development of the disease, as well as lifestyle behaviours, such as diet, exercise and weight.	
	Prevention	● Maintaining a normal weight ● Eating a healthy and varied diet ● Exercising regularly
	Treatment and management	Lifestyle changes ● Eating well ○ Helps to manage levels of blood glucose. Diets should include high-fibre, low-fat foods and avoid refined sugars and carbohydrates ○ Low glycaemic index foods are also recommended as they help to maintain more stable blood sugar levels ● Exercising ○ Lowers blood pressure and helps insulin to work effectively, reducing the rate of heart disease ○ Lower blood sugar levels, as glucose is being used by cells to enact aerobic respiration Monitoring ● Texting blood glucose levels regularly allows patients to effectively manage their treatment ensuring that these levels stay within a healthy range Treatment 1. Medication called metformin (decreases amount of glucose produced in liver, and increases insulin sensitivity in many tissues). Overall, it decrease blood sugar levels. 2. As it progresses, pancreas continues to become less effective, patients must take insulin injections. 3. For obese patients: bariatric surgery helps in reducing its affects.
	Future directions for research	● Increased understanding of the factors contributing to the development of type 2 diabetes → helps design better prevention measures and create effective treatment and management strategies ● Understanding genetic risks and analysing genomes to identify disease causing genes → allows for preemptive and personalised medicines which prevent the disease. ● Type 2 diabetes is normally treated by insulin shots, hence research into the production of insulin to be more efficient, less expensive and faster in order to increase world-wide access. ● Researching genetic engineering, cloning, large-scale synthetic protein production for insulin shots. ● Whole organ pancreas transplantation. Research into creating artificial pancreas. ● Islet cell transplants instead of whole organ transplants
8.3.3 evaluate the method		

used in an example of an epidemiological study

Term	Definition
Epidemiology	The study and analysis of the distribution, patterns and determinants of health and disease conditions in defined populations.
Validity	How accurately a method measures what it is intended to measure
Reliability	Degree of consistency of a measure
Evaluate	Make a judgement based on criteria; determine the value of

 What makes an epidemiological study valid and reliable?

1. Conducted over a long period of time
2. Study a large sample size (*reliability*)
3. Range of relevant data of both affected and unaffected, including factors such as age sex, diet, occupation etc
4. Representatives of a broad range of society and lifestyles
5. Control groups not exposed to cause of disease but similar in other respects
6. Correlation vs causation
7. Lack of bias

 Discussion about validity must

- Identify what validity is
- They are close to the true value of the quantity being measured
- Identify factors that affect the validity of the experiment (variables to be controlled, appropriate equipment, the range of values etc)
- They can be substantiated in secondary sources

 Discussion about reliability must

- Identify what reliability is
- Identify that repetition minimises the effect of random errors/outliers and/or allows them to be removed or disregarded
- It investigates what you think it will investigate, i.e., the procedure actually tests the hypothesis and it has an wide range of values.
- Give examples of possible random errors that may have crept into the experiment you are considering
- Variables not in the hypothesis are controlled (kept the same)
- Appropriate measuring procedures are included
- Assess the overall reliability of the experiment

Example: Wakefield's study (Lancet)

Connected the MMR vaccine with autism in children.

1. Conducted over a long period of time
 - Conducted within medical treatment period - *less than a year* - not valid
2. Study a large sample size (*reliability*)
 - Only 12 children, no control all affected and all had vaccine = **not reliable**
 - **Reliability is improved via repetition, 12 for an epidemiological study is not enough (sample sizes are usually 10% of the population- so 10% of all people who received the MMR vaccine)**
3. Range of relevant data of both affected and unaffected, including factors such as age sex, diet, occupation etc
 - 11 boys, 1 girl , all individuals had same abdominal issues, no information about family, genetics, lifestyle
 - Majority of the group were male
4. Representatives of a broad range of society and lifestyles
 - No information about lifestyles provided, histology focussed
 - All were children between the ages of 3-10 years old (*not a broad range*)

	- Information provided by parents was used (behavioural change etc) - Behavioural changes were not described or observed over a long period therefore judgements about whether the children had autism cannot be made 5. Control groups not exposed to cause of disease but similar in other respects - No control group, no other variables - Every individual had the MMR vaccine and had abdominal issues 6. Correlation vs causation - Kids may have had autism prior to the vaccine, no lineage information given - Autism is significantly more common in boys than in girls - <i>the sample size was primarily boys</i> 7. Bias - Wakefield was designing a vaccine which lead him to denounce the current vaccine to promote his own - All of the peer reviewers of his article were aware of the vaccine Wakefield was designing - Notorious anti-vaccine activist (<i>even today</i>) = METHOD IS NOT VALID OR RELIABLE										
8.3.4 evaluate, using examples, the benefits of engaging in an epidemiological study	<table><tr><th>Term</th><th>Definition</th></tr><tr><td>Epidemiology</td><td>The study of health trends within and across populations.</td></tr><tr><td>Descriptive study</td><td>An epidemiological study that provides information on the incidence of a disease and some causative factors.</td></tr><tr><td>Analytical study</td><td>An epidemiological study that involves establishing a hypothesis based on the data obtained in the descriptive study. It seeks to either support or refute a hypothesis for further study.</td></tr><tr><td>Intervention study</td><td>Involves epidemiologists testing out various methods and strategies that can be used to prevent the causative agent found in the analytical study. This is generally done through public health campaigns.</td></tr></table> - o Benefits of an epidemiological study. - Benefits of using descriptive epidemiology studies: It allows us to identify and understand prevalence, priority health groups and variations in disease frequency. - Benefits of using analytic epidemiology: Allows us to identify the risk factors and potential causes of the disease. - Benefits of using intervention epidemiology: Determines the effectiveness of preventive and control measures that have been implemented to prevent and control the disease respectively.	Term	Definition	Epidemiology	The study of health trends within and across populations.	Descriptive study	An epidemiological study that provides information on the incidence of a disease and some causative factors.	Analytical study	An epidemiological study that involves establishing a hypothesis based on the data obtained in the descriptive study. It seeks to either support or refute a hypothesis for further study.	Intervention study	Involves epidemiologists testing out various methods and strategies that can be used to prevent the causative agent found in the analytical study. This is generally done through public health campaigns.
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8.4 Prevention IQ: How can non-infectious diseases be prevented?											
8.4.1 use secondary sources to evaluate the effectiveness of current disease-prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to: a. educational programs and campaigns b. genetic engineering											

8.5 Technologies and Disorders

IQ: How can technologies be used to assist people who experience disorders?

8.5.1 explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example:

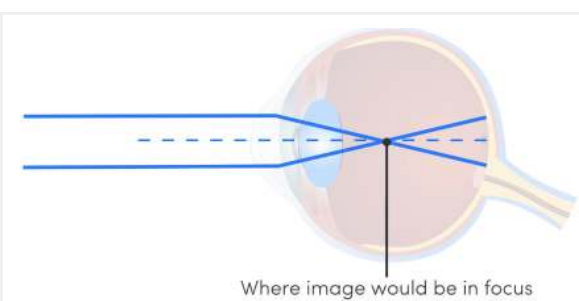
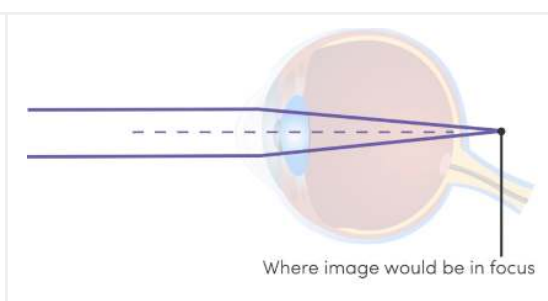
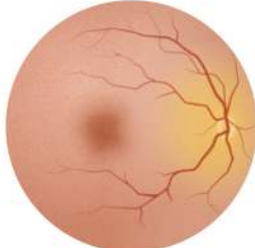
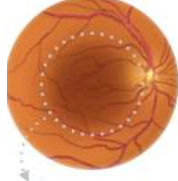
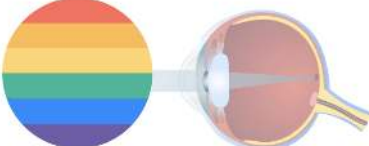
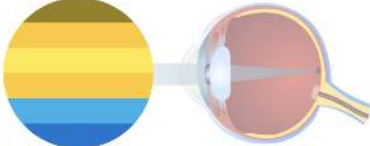
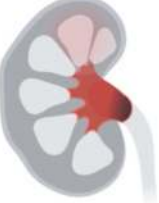
- A. hearing loss
- B. visual disorders
- C. loss of kidney function

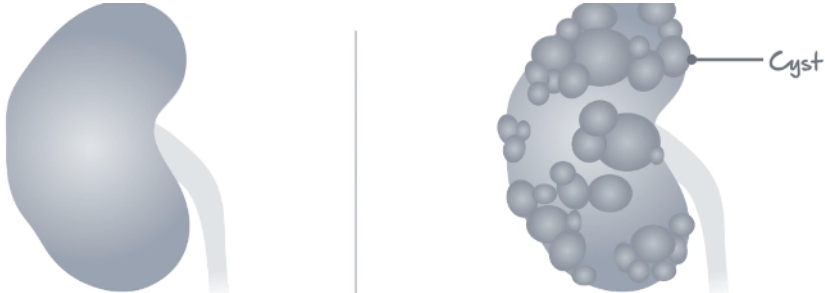
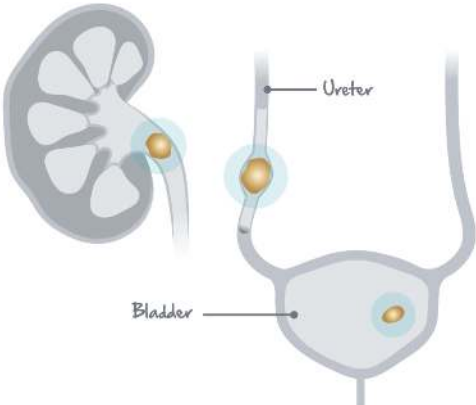
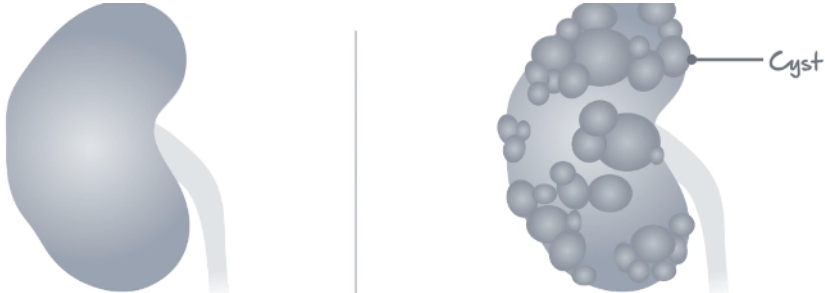
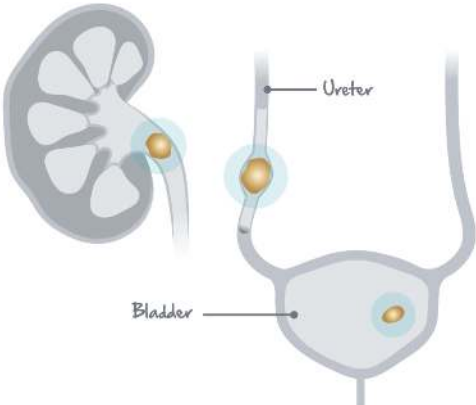
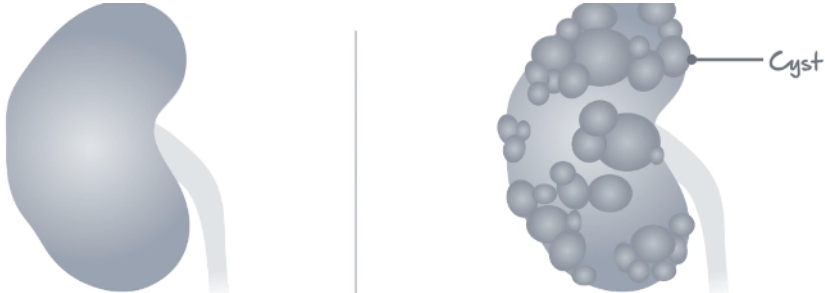
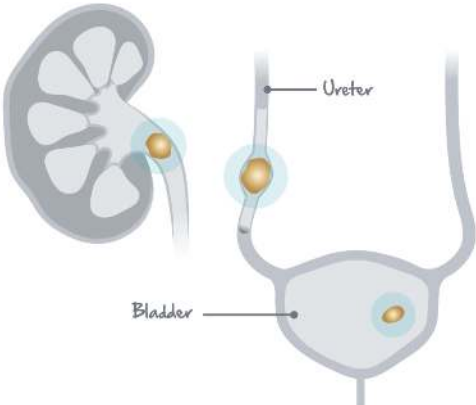
★Hearing loss

Disorder	Info
Conductive Hearing Loss	- Deafness which is caused by damage to the outer and middle ear - Results in reduced or muffled sound <u>Causes</u> - Malformed structure at birth - Aging - Noise strain - Injury - Infection - Blockage due to earwax, a tumour or foreign item
Sensorineural Hearing Loss	- Deafness which is caused by damage to the inner ear/nerve pathways from the inner ear to the brain - Results in faint or muffled sound <u>Causes</u> - Malformed inner ear at birth - Exposure to damaging noise can cause the hair cells in the cochlea to become fatigues/kill them - Head injury
Auditory Processing Disorder	- Not due to damage or defects in the structure of the ear - Results in Inability to detect the direction of a sound <u>Causes</u> - Processing problems in the auditory areas of the brain
Tinnitus	- When a sound is heard that isn't there - Results can include hearing a ringing in the ears: buzzing, hissing <u>Causes</u> - Exposure to loud sound - Injury

★Visual disorders

Refractive errors		
Myopia	Similarities	Hyperopia
- Causing the image to form in front of the retina - Refers to short-sightedness - Close objects are clear, but distant objects are blurry - The image is being focused in front of the retina <u>Causes</u> - The lens is too thick - Bends light too much - The eyeball is too long - The retina is further away	Results from when light is not being properly bent onto the retina so that vision is blurred <u>Causes</u> - Thickness of the lens - Shape of the eyeball	- Causing the image to form behind the retina - Refers to long-sightedness - Distant objects are clear, but close objects are blurry - The image is being focused behind the retina <u>Causes</u> - The lens is too thin - Doesn't bend light enough - The eyeball is too short

		
Retina Conditions		
Retinopathy	Macular degeneration	Colour blindness
• Damage to the retina, often caused by changes to the blood vessels surrounding this region  NORMAL RETINA  RETINOPATHY	• Caused by a distortion or absence of the fovea ◦ Central most sensitive part of the retina NORMAL MACULA  Macular region DRY MACULAR DEGENERATION  • Gradual loss of central vision WET MACULAR DEGENERATION  • Caused by abnormal growth of blood vessels under the retina • Leads to fluid leakage and sudden loss of vision	• Occurs when one of the three types of the colour detecting cone cells is absent or doesn't function properly  NORMAL EYE  COLOUR BLINDNESS
★ Loss of function in kidney		
Chronic Kidney disorders		
Glomerulonephritis	• Inflammation of the glomerulus • Often occurs after a bacterial infection  NORMAL KIDNEY  GLOMERULONEPHRITIS	
Diabetic Nephropathy	• Kidney damage caused by diabetes ◦ Unable to maintain healthy levels of sugar in blood ◦ Kidneys can become fatigued and damaged by extra strain • Less efficient removal of wastes • Larger substances can be filtered out of the blood when they shouldn't be able to	

	<table><tr><td data-bbox="457 191 801 478"></td><td data-bbox="801 191 2703 478"> NORMAL DIABETIC NEPHROPATHY </td></tr><tr><td data-bbox="457 478 801 932">Polycystic Kidney disease</td><td data-bbox="801 478 2703 932"> • Causes cysts to grow in the kidneys ◦ An abnormal, sac-like pocket which contains fluid, air or another substance • Inherited condition  NORMAL KIDNEY POLYCYSTIC KIDNEY DISEASE </td></tr><tr><td data-bbox="457 932 801 1593">Kidney stones</td><td data-bbox="801 932 2703 1593"> • Form when waste products crystallise into lumps • Can be incredibly painful and can cause blockages • Factors which increase a person's risk: ◦ Being dehydrated ◦ Consuming a poor diet ◦ Hereditary factors  Ureter Bladder </td></tr><tr><td colspan="2" data-bbox="457 1593 2703 1724">Effects of each of these kidney disorders: 1. Build-up of metabolic waste in the blood 2. Imbalanced amounts of salt and water in the body fluids </td></tr></table>		 NORMAL DIABETIC NEPHROPATHY	Polycystic Kidney disease	• Causes cysts to grow in the kidneys ◦ An abnormal, sac-like pocket which contains fluid, air or another substance • Inherited condition  NORMAL KIDNEY POLYCYSTIC KIDNEY DISEASE	Kidney stones	• Form when waste products crystallise into lumps • Can be incredibly painful and can cause blockages • Factors which increase a person's risk: ◦ Being dehydrated ◦ Consuming a poor diet ◦ Hereditary factors  Ureter Bladder	Effects of each of these kidney disorders: 1. Build-up of metabolic waste in the blood 2. Imbalanced amounts of salt and water in the body fluids	
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Effects of each of these kidney disorders: 1. Build-up of metabolic waste in the blood 2. Imbalanced amounts of salt and water in the body fluids									
8.5.2 investigate technologies that are used to assist with the effects of a disorder,	<table><tr><td data-bbox="457 1780 715 1822">★ Hearing loss treatment</td><td data-bbox="715 1780 2703 1822"></td></tr><tr><td data-bbox="457 1822 715 1885">Hearing Aids</td><td data-bbox="715 1822 2703 1885">A battery operated electronic device that fits into the hollow outside the ear canal</td></tr></table>	★ Hearing loss treatment		Hearing Aids	A battery operated electronic device that fits into the hollow outside the ear canal				
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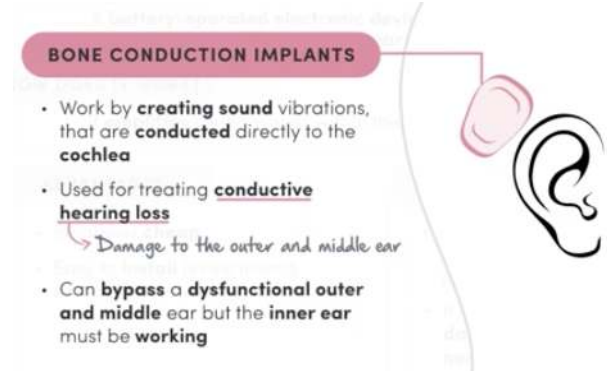
- including but not limited to:
- a. hearing loss: cochlear implants, bone conduction implants, hearing aids
 - b. visual disorders: spectacles, laser surgery
 - c. loss of kidney function: dialysis



- Amplifies sound waves within the ear canal
- The eardrum, ossicles and cochlea must still have an adequate level of function so that sound can be registered and converted into an electrochemical signal
- Can be used for people with conductive or sensorineural hearing loss

Bone conduction implants

Similar to hearing aids although implanted into the bone near the ear.

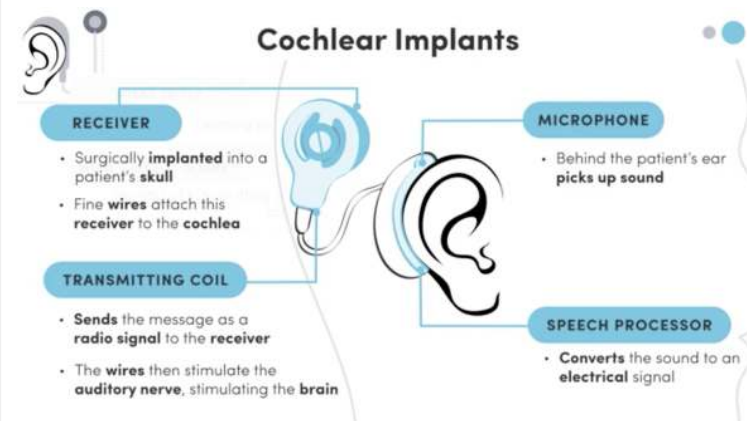


- Works by creating sound vibrations, that are conducted directly to the cochlea
- Used for treating conductive hearing loss
 - Damage to the outer and middle ear
- Can bypass a dysfunctional outer and middle ear but the inner must be working

Advantage	Disadvantage
• Relatively cheap • Easy to install (no surgery required)	• They amplify all sounds in the environment (including background noise) and can cause pain • It does not help people with severe damage to the inner ear/auditory nerve

Cochlear implants

Used by people with profound hearing loss (*profound damage to any region of the ear*)



- Instead of transmitting sound waves, it converts sound into electrical signals and directly stimulate the auditory nerve

Advantage	Disadvantage
• Provides hearing to profoundly deaf people	• Highly expensive • Operation side effects (e.g. infection, drooping face, numb tongue due to nerve damage) • The wearer must learn to interpret sounds again • Success of the implant system is limited if the auditory nerve or region of the brain is damaged

★ Visual disorder treatments

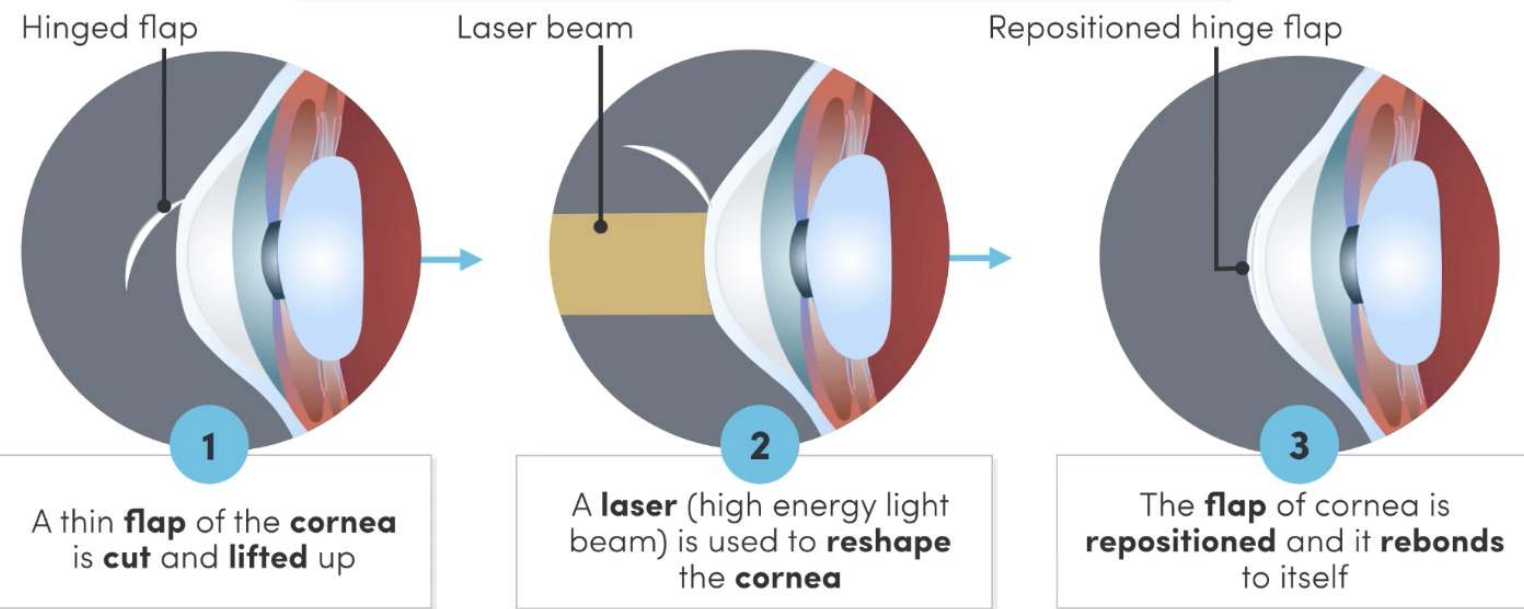
Glasses

- Glasses have lenses which are shaped in order to correct individual needs, by complementing for misshapen eye lenses or eyeballs
- Glasses cause light rays to bend before entering the eye so that the image focuses on the retina.
- There are two types of glasses

Concave	Convex
• Glasses with concave lenses are used to correct MYOPIA • Causes light rays to diverge so that the image forms further back, falling on the retina Without concave lens With concave lens	• Glasses with convex lenses are used to correct HYPEROPIA • Causes light rays to converge so that the image forms further forward Without convex lens With convex lens

Lasik

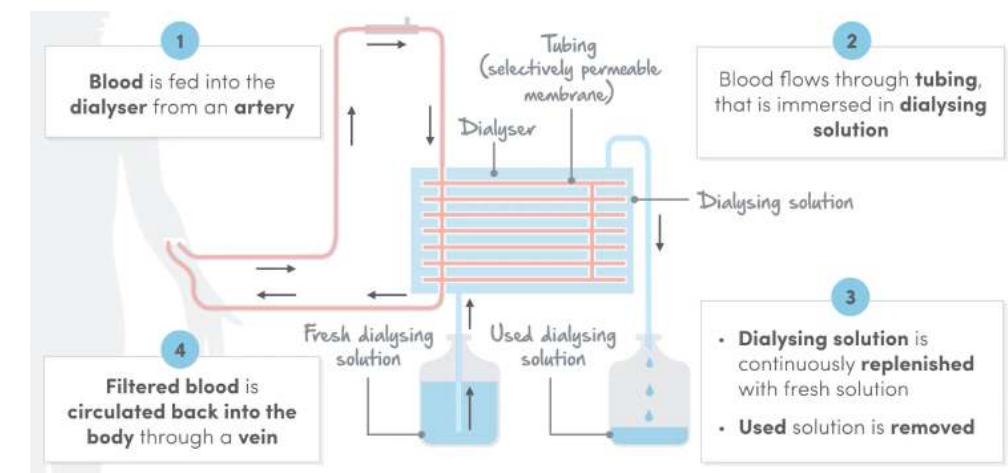
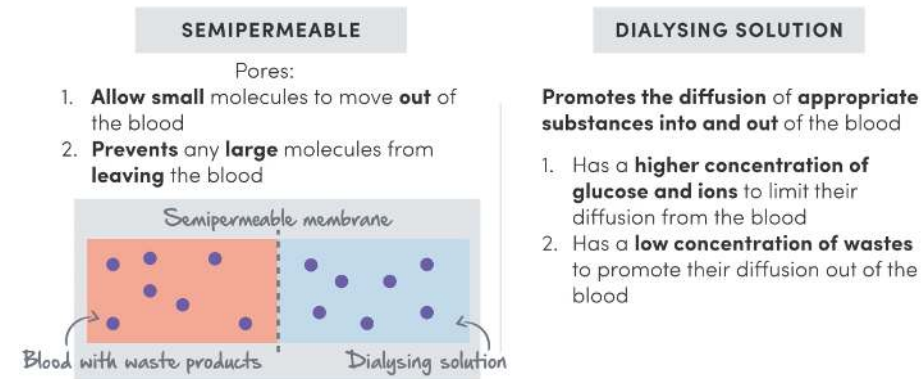
- Expensive and risky, but highly effective and fast



★ Kidney treatment

Dialysis

The process of removing waste products and excess fluid from blood by an external machine called a dialyser



Advantage

- Improved quality of life

Disadvantage

Time consuming

- Comes with a lot of equipment that is often only available at the hospital
- A person needs to be connected to the dialyser for at least two hours, two to three times a week

Blood issues

- Invasion of the individual's blood vessel can cause blood clotting, bleeding and infections

Kidney transplant

Replacing a damaged kidney with a healthy one from a matching donor

Advantage

Disadvantage

	<u>Improved quality of life</u> - When successful, can allow a person to live a longer more active life <u>Efficient Procedure</u> - Surgery normally takes 2-3 hrs and recovery only takes days <u>Kidney rejection</u> - Recipients immune system recognises that the kidney is foreign, and attacks it <u>Immunosuppressive drugs</u> - Transplant recipient is far more vulnerable to future infections and cancers <u>Long waiting lists</u> - Transplants require access to a donated organ
8.5.3 evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder	★<u>Evaluate technology that assists in hearing disorders</u> - Treatments in managing hearing loss can be highly effective, if the correct treatment is used for the specific disorder. There are two main types of treatments/technology that can improve one's hearing - hearing aids (+bone conduction implants) and cochlear implants. Hearing aids are used to amplify all sounds within the ear canal therefore, the individual using this form of treatment the eardrum, ossicles and cochlea must have an adequate level of function so that the sound being produced by the hearing aid can still be translated and registered. This form of treatment can be used by those who have disorders of conductive (damage to the middle or outer ear) and or sensorineural (damage to the inner ear). ★<u>Evaluate technology that assists in visual disorders</u> - Laser eye surgery or Lasik can be a highly effective treatment for those with eye disorders, as it means that the individual will be able to see without the need of glasses; but it is also expensive and risky. A thin flap of the cornea is cut and lifted up, and a laser is used to reshape the cornea. The flap of the cornea is repositioned and it rebinds to itself. The procedure is not highly invasive, and the patient is placed under general anaesthesia for the procedure. The results are effective and act fast. Therefore, for those where their sight is affected heavily due to their condition, Lasik eye surgery can be an option in order to aid their sight. ★<u>Evaluate technology that assists in kidney disorders</u> - Performed two to three times per week at a hospital or home. Helps drive salt and water balance to a normal level on a frequent basis to avoid accumulation of nitrogenous waste toxic to cells. Slow process requires several hours and takes time away from daily life, required to undergo dialysis on a regular basis as removal of nitrogenous waste occurs only when connected to the dialysis machine. Does not restore kidney function. Effective in acting as an artificial kidney. Allows the patient to continue to perform everyday functions, such as breathing normally, without experiencing many symptoms